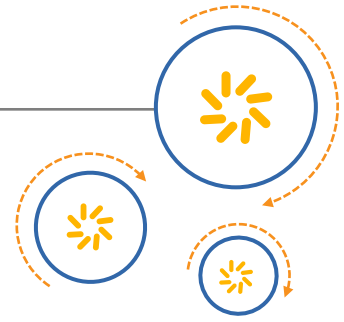




Qualcomm Technologies, Inc.



APQ8016E Processor Design Guidelines

September 2016

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Qualcomm Technologies, Inc.
5775 Morehouse Drive
San Diego, CA 92121
U.S.A.

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1 Overview

This document combines all layout guidelines available for the APQ8016E chipset and its individual ICs.

PCB designers can use this single document for mechanical and layout instructions rather than downloading all the chipset's design guidelines and searching for the layout material embedded within each one of them.

System and circuit designers are still encouraged to study the design guidelines for important information besides PCB details.

1.1 Chipset Block Diagrams

To improve the reader's understanding of the system, block diagrams for the system and detailed block diagrams for each of the major chips in the system are presented.

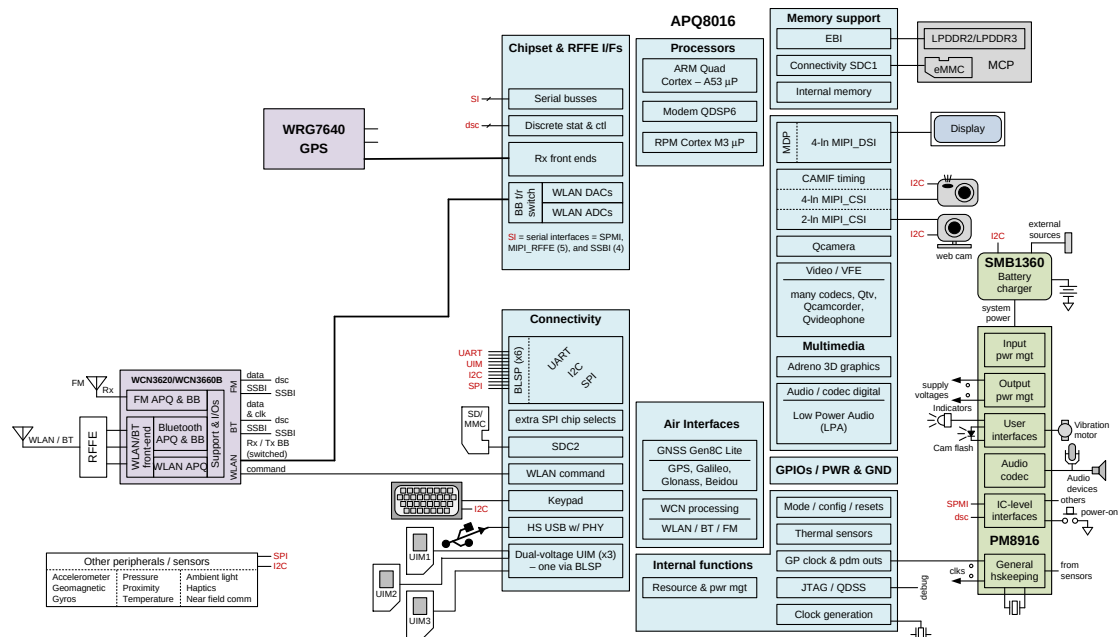


Figure 1-1 APQ8016E System Block Diagram

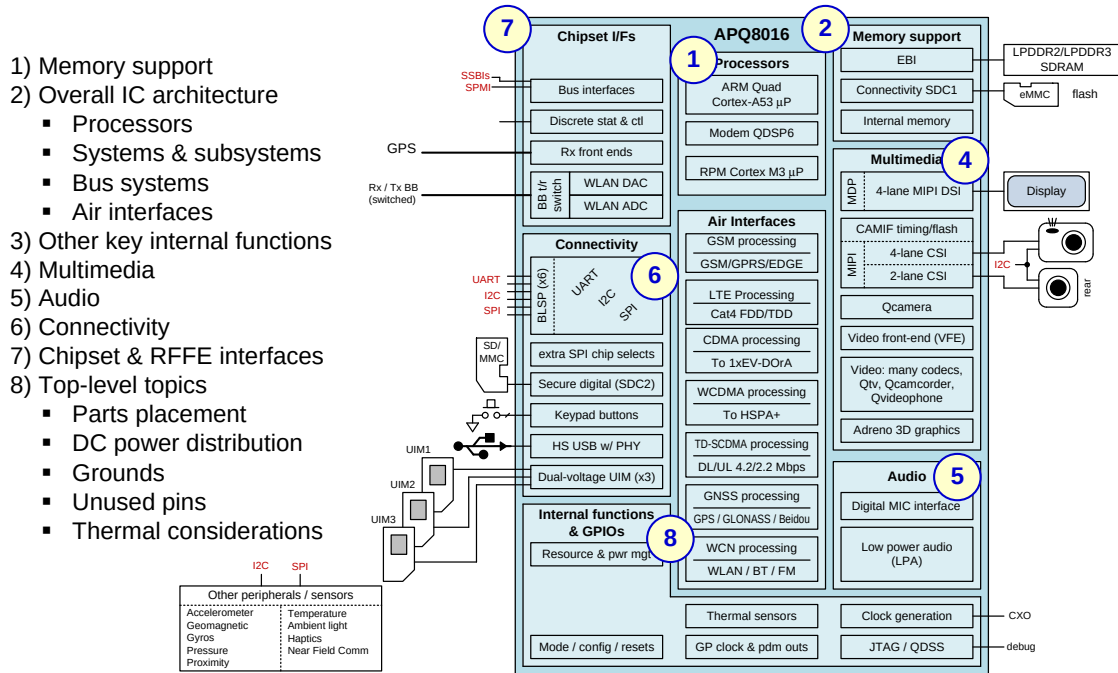


Figure 1-2 APQ8016E Processor - Major Functional Blocks

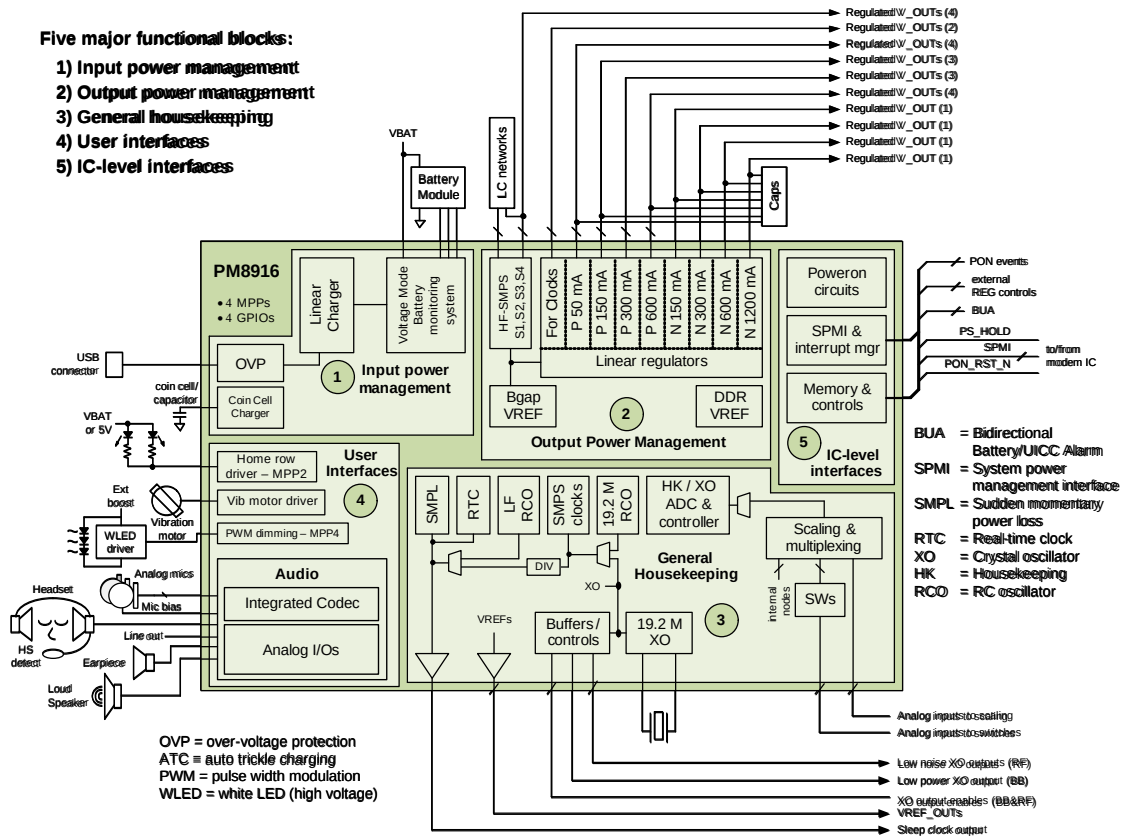


Figure 1-3. PM8916 Power Management – Major Functional Blocks

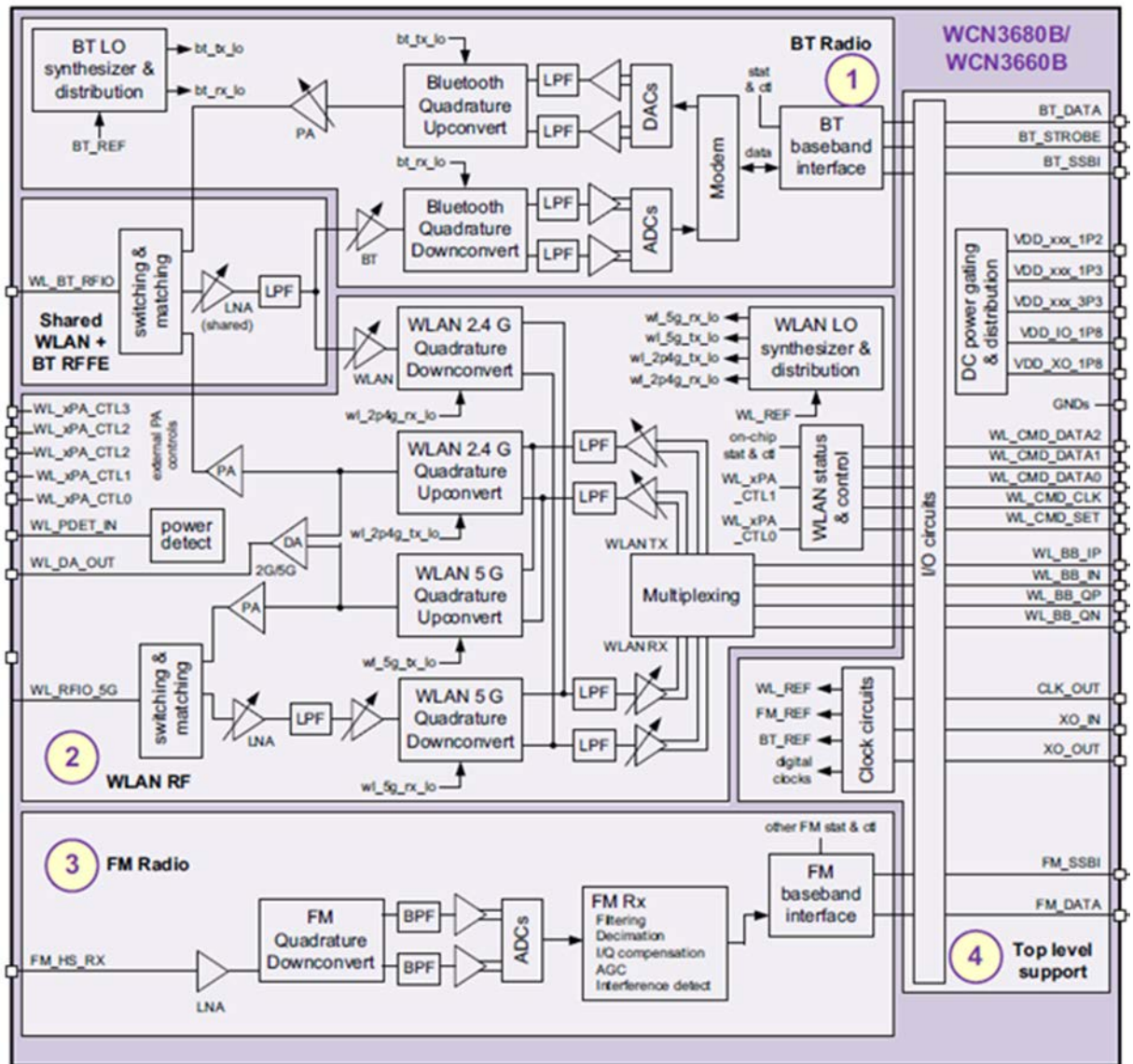


Figure 1-4 WCN3660B BT/WiFi/FM – Major Functional Blocks

Three major subsystems – WLAN, Bluetooth, and FM radio (split between two ICs) plus top-level support circuits

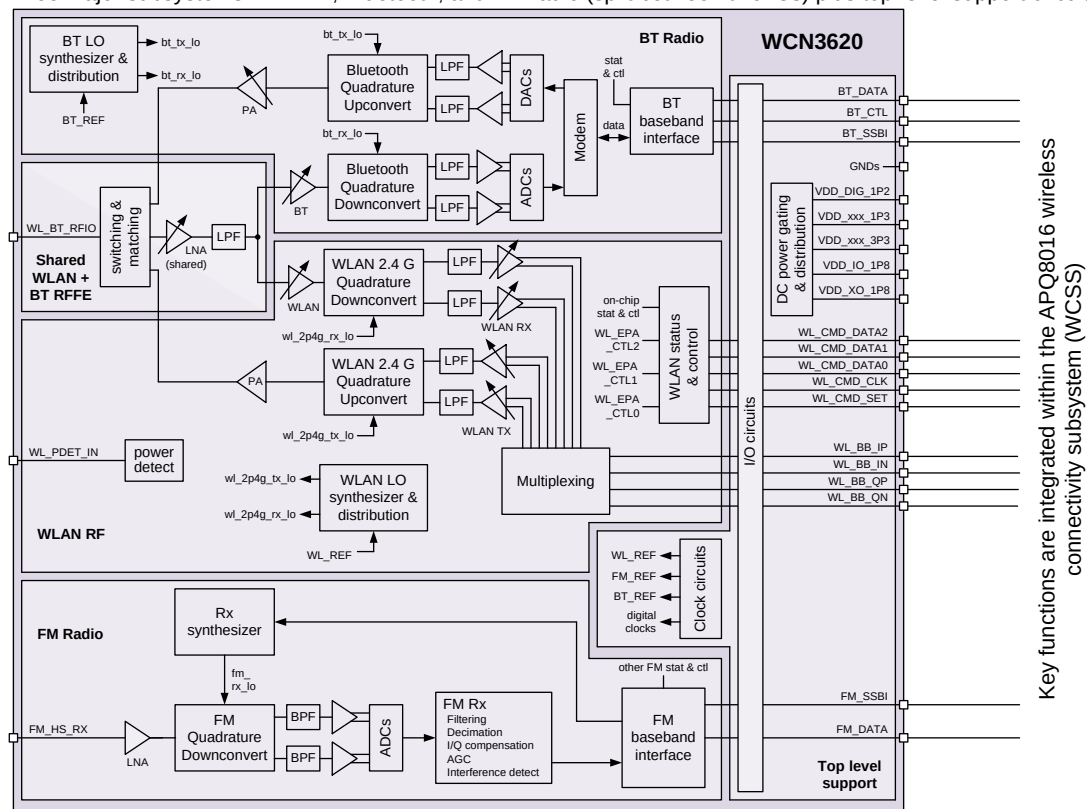


Figure 1-5. WCN3620 BT/WiFi/FM – Detailed Block Diagram

These document provides layout guidelines for the APQ8016E Chipset based on three known working designs. Other layouts and strategies are possible and it is up to the system designer to ensure that all of the requirements are met.

[Table 1-1](#) lists the three designs and highlights the strategies and key differences in each design.

Table 1-1 Layout strategies and differences

APQ8016E	3-4-3 Test Platform	2-6-2 Reference Design	2-4-2 Reference Design	Comments
Stack up	3-4-3	2-6-2	2-4-2	
PCB shape size	Rectangle full phone	C-shape	Rectangle half phone	
PCB single or double sided	Double sided	Double sided	Double sided	
DDR	Key signals on Layers 2 and 3	No GND fill on top layer; key signals on layer 2 only	Key signals on layers 3 and 5	See Section 3.4
PDN decoupling capacitor placement	Bottom layer	Bottom layer	Bottom layer	See Section 3.1

1.2 Acronyms and abbreviations

Table 1-2 Acronyms and abbreviations

Acronym	Definition
APC	Asynchronous Procedure Call
APPC	Advanced Peer-to-Peer Communications
ASM	Asynchronous State Machine
AVDD	Analog Supply - Power Side
BB	Baseband
BBRX	Baseband Receive
BGA	Ball Grid Array
BT	Bluetooth
CDMA	Code Division Multiple Access
CK	Cipher Key
CMD	Centralized Message Distribution
CP	Control Plane
CS	Chip Select
DAC	Digital-to-Analog Converter
DC	Direct Current
DDR	Dual Data Rate
DNC	Do Not Connect
DSI	Display Serial Interface
DVT	Digital Video Terminal
EBI	External Bus Interface
eMCP	Embedded Multi Chip Package
EMI	ElectroMagnetic Interference
eMMC	Embedded Multimedia Card
ESD	Electrostatic Discharge
EVM	Error Vector Magnitude
FM	Frequency Modulation
GND	Ground
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HPH	Head Phone
HS	High Speed
HS-USB	High Speed Universal Serial Bus
I2C	Inter-Integrated Circuit
IC	Integrated Circuit
IO	Input/Output
IR	Infra Red
IREF	Current Reference
LDO	Low Drop Out

Acronym	Definition
LNA	Low Noise Amplifier
LO	Local Oscillator
LPDDR	Low Power Double Data Rate
LTE	Long Term Evolution
MIC	Microphone
MID	Midpoint
MIPI	Mobile Industry Processor Interface
NFC	Near Field Communication
PCB	Printed Circuit Board
PCM	Pulse Code Modulation
PDM	Pulse Density Modulation
PDN	Power Distribution Network
PGND	Power Ground
PLL	Phase-Locked Loop
PMIC	Power Management Integrated Circuit
PWR	Power
QRD	Qualcomm Reference Design
QTI	Qualcomm Technologies, Inc.
RC	Resistor-Capacitor circuit
RF	Radio Frequency
SDC	Switched Digital Capability
SDIO	Secure Digital Input Output
SI	System Information
SMPS	Switched-Mode Power Supply
SPMI	System Power Management Interface
SSBI	Signal-Signal Beating Interference
SW	Software
SWP	Single Wire Protocol
TCXO	Temperature Compensated Crystal Oscillator
TIM	Thermal Interface Material
TSENS	Temperature Sensor
TX	Transmit
USB	Universal Serial Bus
USID	Unique System Identifier
VCO	Voltage Controlled Oscillator
WAN	Wide Area Network
WLAN	Wireless Local Area Network
XO	Crystal Oscillator

2 PCB Mechanical and Layout Highest Priorities

The Mechanical Layout of the system is critical to ensuring a successful design. This section describes some of the major factors to consider when making the initial design layout.

2.1 Priorities

2.1.1 PCB mechanical highest priorities

This sections contains the entire mechanical information needed to start a PCB design.

- Stack-up options and recommendations
- Thermal
- Design rules (via options and sizes, clearances, trace widths, etc.)
- Very high-level component placement and signal flow
- Shielding
- Antenna

2.1.2 Layout highest priorities

This section provides an introduction. Details are contained in later chapters.

- PDN and other critical power distribution and grounding
- DDR
- XO layout and distribution
- SMPS circuits
- WiFi, BT and analog baseband sensitive traces
- GPS circuits

2.2 Specific implementation details

2.2.1 PCB Stackup Concepts

In general PCBs can be built with multiple layers, and there are multiple types of interconnects between layers. The Figure below shows the concepts of some typical PCB Stackups and discusses the pros and cons of each option.

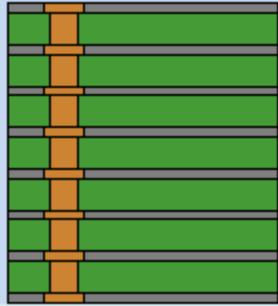
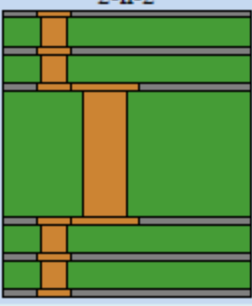
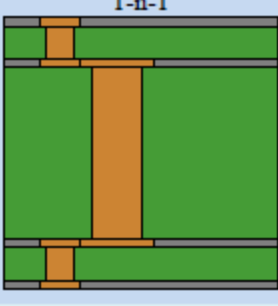
	Full Stacked Microvia	2-n-2	1-n-1
			
Routable?	Yes	Yes	Yes (assuming I/O usage is not 100%)
Breakout effort	Easiest Very few lines needed at min width & spacing	Moderate effort Many lines will have to be at min width & spacing	Challenging. Needs strict adherence to standard pattern. Most lines under the BGA will be min width & spacing.
Power Delivery Network	Relatively easy to meet impedance spec. (Note that DC resistance of certain full stacked processes can be quite high, and need to be taken into account)	Will require careful design of power planes and decoupling network to achieve required impedance spec	Will require careful design of power planes and decoupling network to achieve required impedance spec
Signal Integrity	Relatively easy to manage	Needs more effort	Needs more effort
Relative Cost	\$\$\$	\$\$	\$ (Least expensive option)

Figure 2-1 PCB Stackup Concepts

A 1-N-1 PCB stack up presents many difficulties for connecting the entire system. All signal layers should reference a solid reference plane (power or GND) in order to allow for good signal return paths. If a signal changes reference planes, remember to place stitching vias nearby, or if the reference planes are at different potentials, place the bypass capacitors nearby. Additionally, power and GND layers are adjacent in order to facilitate minimal spacing between power and GND planes, and to minimize parasitic inductance in the power delivery network (PDN).

The 2-N-2 PCB stack up eases many of the design challenges associated with the 1-4-1 PCB stack up. Firstly, it provides two extra layers of routing resources. Secondly, it allows for all of the APQ signals to immediately breakout as stripline transmission lines with solid reference planes nearby for good impedance control. It maintains the configuration of having GND planes near all power layers in order to reduce loop inductance of PDNs.

Fully stacked micro-vias provides the best routing and the best signal integrity, easing the system design. The disadvantage of this stackup is the higher relative cost. Manufacturer and designer should consider this option if their volumes are low and if they want the lowest risk design with the highest chances for first pass success.

Table 2-1 APQ8016E chipset stack up summary

Circuit – IC types	Supported stack ups	Comments
Digital BB and PDN – APQ8016E	1-N-1, 2-N-2	
Memory – LPDDR2 and LPDDR3	1-N-1, 2-N-2	1-N-1 is theoretically possible (has not been implemented and tested)
Power management PM8916 and SMB1360	1-N-1	SMB1360 is not required in all designs.

2.2.2 Design rules

Table 2-2 shows the minimum design rules needed to breakout the APQ8016E package.

Via/Via Pad Clearance Rules			General	Min
Via	Via Diameter	Laser-Via Drill	0.1	0.1
		Through-Via Drill	0.25	0.25
		Laser-Via Pad (Outer)	0.25	0.25
	Via Pad Diameter	Laser-Via Pad (Outer)	0.25	0.25
		Laser-Via Pad (Outer)	0.45	0.4
		Through-Via and Through-Via	0.45	0.4
	Distance (Drill center to Drill Center) (Different Net)	Through-Via and Through-Via	0.55	0.5
		Through-Via and Buried-Via	0.55	0.5
		Laser-Via and Laser-Via	0.35	0.35
	Distance (Drill center to Drill Center) (Same Net)	Through-Via and Through-Via	0.45	0.4
		Through-Via and Buried-Via	0.45	0.4
		Laser-Via and Laser-Via	0.25	0.225
Trace / Via / Copper Clearance Rules			General	Min
Conductor	Trace Width	Build up layer	0.075	0.075
		Core outer layer	0.075	0.075
		Core inner layer	0.075	0.075
	Space of trace to Trace	Build up layer	0.075	0.075
		Core outer layer	0.075	0.075
		Core inner layer	0.075	0.075
	Space of trace to pad	Build up layer	0.1	0.075
		Core outer layer	0.1	0.075
		Core inner layer	0.1	0.075
	Space of trace to copper	Build up layer	0.1	0.1
		Core layer	0.1	0.1
	Space of PAD to PAD	Build up layer	0.1	0.1
		Core layer	0.1	0.1
	Space of copper to Board edge	Build up layer	0.3	
		Core layer	0.4	
Table Continued Next Page				

Component Clearance Rules				
Components	Space of component to component or shield	Components body from Board Edge		0.5
		Components from Panel Edge		0.5
		(0402 & less) Component body to component body		0.25
		BGA chip outline to other component body		0.3
		Mechanical Component body to other Component body		0.4
		Shield body to component body		0.4
		Shield body to Shield body		0.5
		Shield body to Board edge		0.3

Table 2-2 APQ8016E design rules

2.2.3 Placement

2.2.3.1 General placement guidelines

The general guidelines are:

- Place the core chipset relative to the APQ pinout as shown in [Figure 2-2](#)
- Digital devices and traces should not be placed near sensitive signals like RF and clock
- Locate power management device centrally to service all of its loads
- Locate receive section away from the antenna feed points
- Shield WGR and processor under separate cavities
- Keep SPKR and MIC away from sensitive RF lines

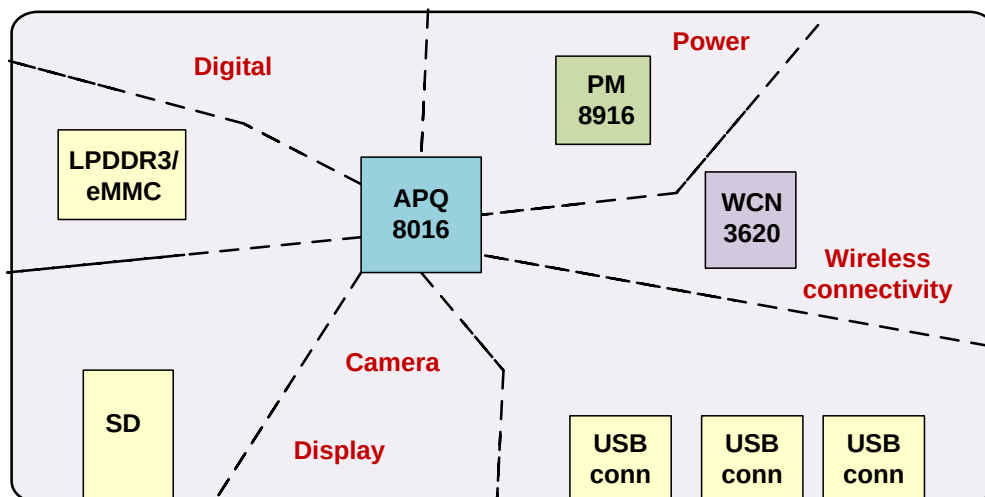


Figure 2-2 Example placement (DragonBoard 410c)

2.2.4 Thermal

Separate highest power consumption ICs.

- Do not allow overlap on opposite sides of the PCB
- Place connectors on opposite sides of key ICs, wherever possible

Use Thermal Interface Material (TIM).

- Eliminate air gaps between the top of key ICs and heat spreaders; use TIM under compression and thermal “grease” for better thermal conduction

Use heat spreaders.

- Connect TIM to graphite sheets, metal shields, and metal battery case
- Use large surface areas with high thermal conductivity

Use air gaps.

- Balance the heat flow between the front and back of the PCB
- Insulate hot spots on the device skin from hot areas below

Use thermistors and internal TSENS.

- Place thermistors close to XO, WLAN, camera, and the charger to better control IC temperatures
- Use dedicated thermistors to better control maximum skin temperature
- Use internal TSENS to control APQ Tj

2.2.5 Shielding

- All circuits susceptible to noise or interference, such as WiFi/BT front-end and GPS LNA input, should be shielded.
- All strong noise and interference generating circuits, such as APQ, SMPS, and memory, should be shielded.
- Ensure that if the shield is a two-piece can, that the lid makes proper contact to the soldered-down frame (dimples), or radiated sensitivity can be degraded.
- Ensure that **all** shield ground pads are connected to the inner layer main ground with as many ground vias as possible.

3 Digital Baseband

3.1 PDN – Power Delivery Network

A properly designed power delivery network (PDN) provides a sufficiently low impedance path for the transient currents that the APQ8016E devices demand. The resulting voltage excursions caused by the interaction of these currents with the PDN impedances of the PCB must remain within the operating voltage range of the APQ8016E device specification to help prevent system crashes.

This section is intended for engineers who are designing with devices that operate with clock frequencies of approximately 800 MHz and higher. This section contains important information about how to design the power delivery network (PDN) to minimize impedance over the frequency range of interest. Proper PDN design prevents violations of V_{min} or V_{max} during voltage transients that occur under normal operating conditions.

3.1.1 Key Concepts

What is included in the PDN?

- Voltage regulators residing in the power management IC (PMIC)
 - Switched mode power supply (SMPS)
 - Low-dropout regulator (LDO)
- Regulator output inductor (SMPS only) + bulk capacitor(s)
- All passive components and their connections to the processor power grid
- Copper traces connecting the regulator output to the processor power pins
- Copper plane connecting processor ground pins to regulator ground pins

A properly designed PDN helps ensure supply voltage compliance to the required operating conditions of processors and other integrated circuitry.

Proper PDN design ensures that $V_{min} \leq V(t) \leq V_{max}$ during all di/dt events.

- $V(t)$ is the voltage measured at the processor power pins as a function of time.
- V_{min} is the minimum voltage allowed (DC + transient) at the processor power supply pins to guarantee proper operation over all variations of process and temperature as listed in the device specification.
- V_{max} is the maximum voltage allowed (DC + transient) at the processor power supply pins to guarantee proper operation over all variations of process and temperature as listed in the device specification.

3.1.2 Power Delivery Concepts

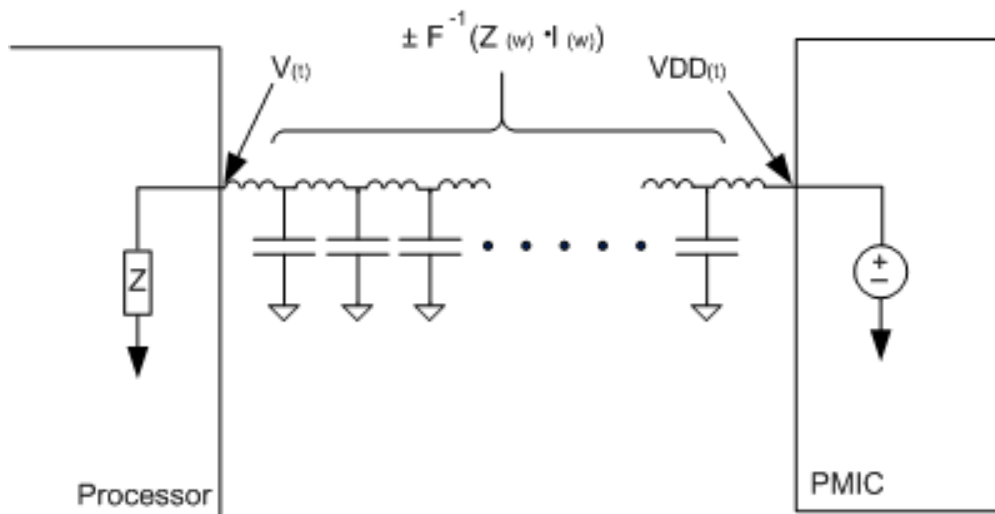


Figure 3-1 Representative Lumped Model PDN Schematic

Key formula: $V(t) = VDD(t) \pm F^{-1}\{Z(w)I(w)\}$

- $VDD(t)$ is the instantaneous voltage at the PMIC output.
- $I(w)$ is the load current as a function of frequency.
- $Z(w)$ is the impedance of the PDN determined from the load (processor power supply pins), back toward the PMIC and is dominated by:
 - PCB metal power planes (DC to ~3 kHz)
 - PMIC regulator response and external bulk capacitors (~3 kHz to ~300 kHz)
 - PCB metal planes and local decoupling capacitor values (~300 kHz to ~10 MHz)
 - Local decoupling capacitor connections to processor power supply pins and distance between power and ground planes (~10 MHz onwards)

Impedance: Resistor $Z = R$

Inductor $Z = j\omega L$

Capacitor $Z = 1/(j\omega C)$

On a log/log scale, these impedances graph as lines of constant slope vs w .

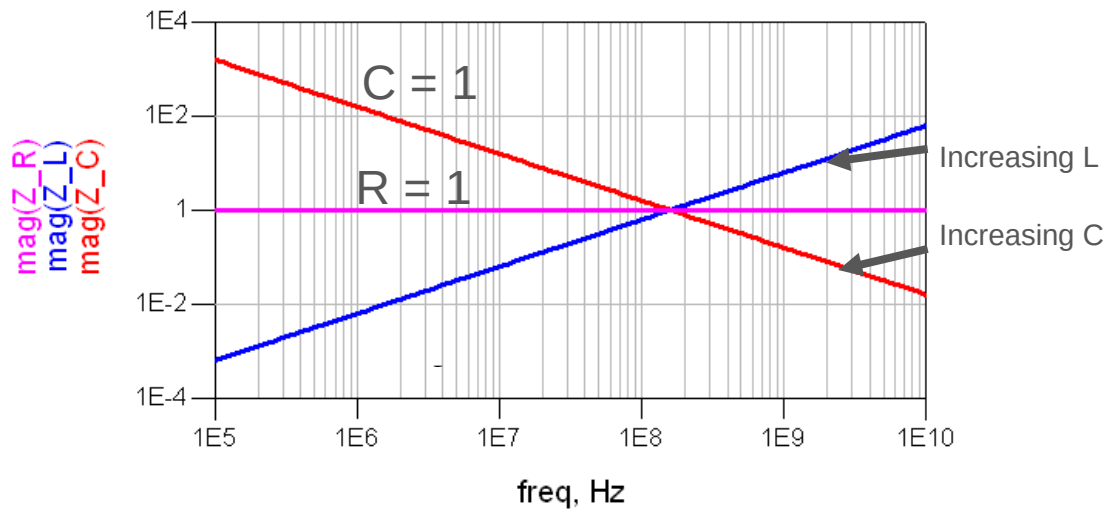
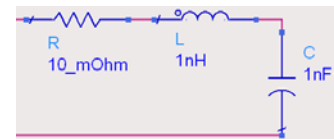
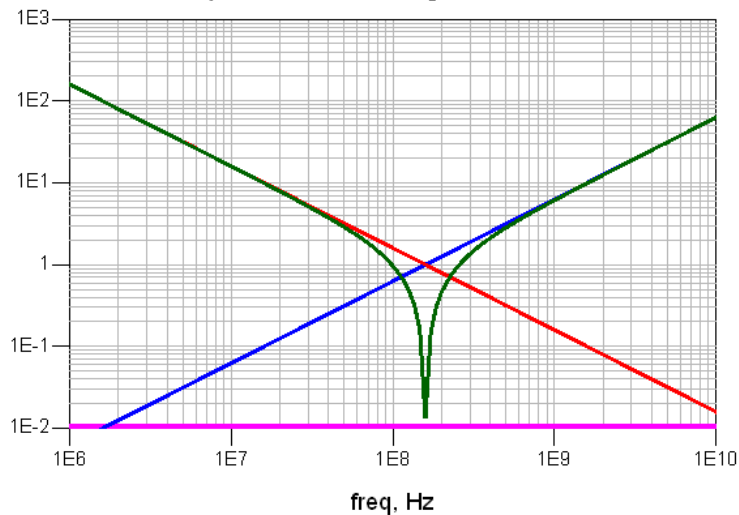


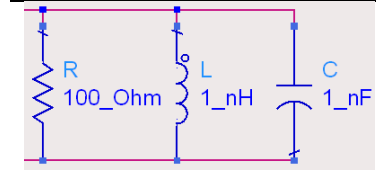
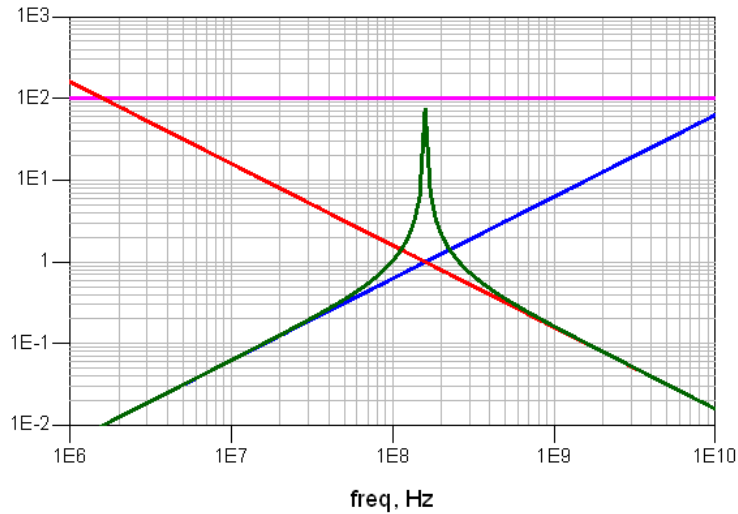
Figure 3-2 R, L, C Impedances Over Frequency

When R, L and C components are arranged in a series or parallel circuit as shown below, the combined impedance can be very low or very high. These concepts are used to model the PDN and help simulate a proper solution. Using these concepts an example of optimizing one branch of the Power Delivery Network will be presented.



$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Figure 3-3 Series RLC resonance



$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Figure 3-4 Parallel RLC resonance

The PDN impedance (Z) is modeled by combinations of capacitance, inductance, and resistance.

- Inductor: $Z = j\omega L$
- Capacitor: $Z = 1/(j\omega C)$
- Resistor: $Z = R$

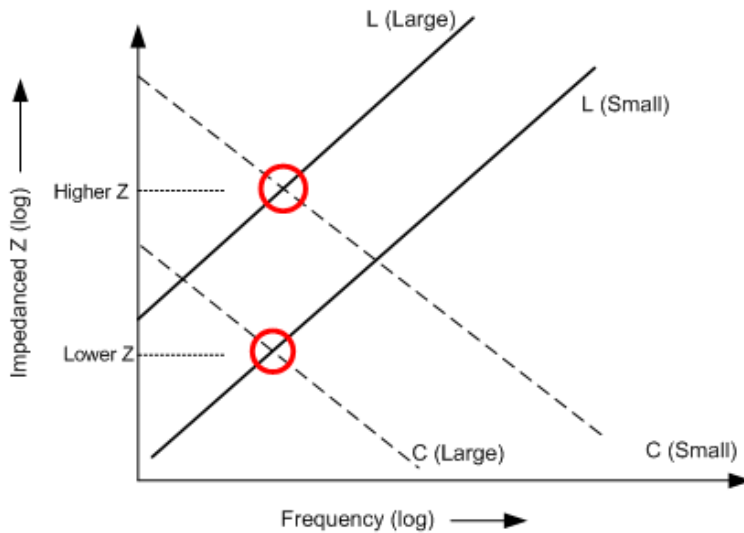


Figure 3-5 Changing Impedance (Z) by Changing Component Values

- Combinations of larger inductance and smaller capacitance result in a higher impedance.
- Combinations of smaller inductance and larger capacitance result in a lower impedance.

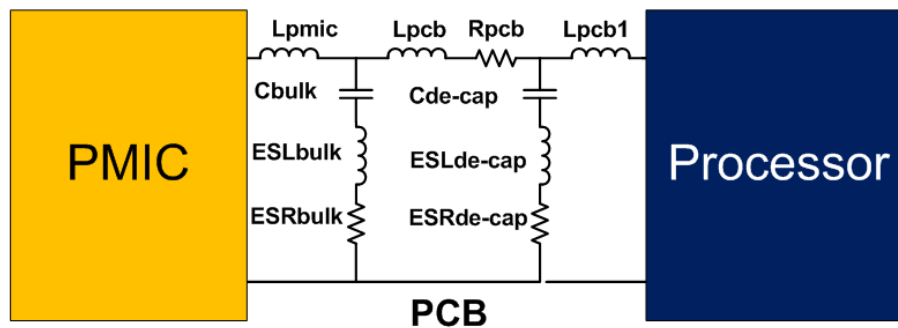


Figure 3-6 Simplified PDN Model

Key Concept – Loop Inductance

The amount of loop inductance associated with the PDN routing on a PCB is primarily determined by:

- The physical dimensions of the current carrying conductors (power net) and the closeness of the return path (ground net).
- The physical dimensions and placement of the passive components connected to the PDN (primarily decoupling capacitors and possibly resistors).

Loop inductance is directly proportional to the area encompassed by a power net and its return path. A large area implies a large loop inductance.

- Therefore, key steps to reducing loop inductance are:
 - Reducing the lengths (x) of both power and ground nets.
 - Reducing the vertical distance (y) between power and ground (return) nets.
 - See [Table 3-1](#).
 - Routing multiple current loops in parallel. $L_{total} = L1 \times L2 / (L1 + L2)$. This is because the parallel loops can support more current for the same voltage drop.
 - Wider power and ground (return) traces with broadside (over-under) coupling effectively provide more loops in parallel to reduce inductance.
 - Optimally placing and connecting decoupling capacitors. This reduces effective loop area because the capacitors electrically close the parallel inductive branch loops; good placement + connection = smaller loops in parallel.

NOTE: See Section [3.1.3.3](#), ‘Capacitor Placements.’

At higher frequencies (not shown in figure), the PDN impedance is a function of package RLC and board loop inductance.

The PDN response in a given frequency range can be resistive, capacitive, or inductive. It is composed of both series and parallel resonances as shown below.

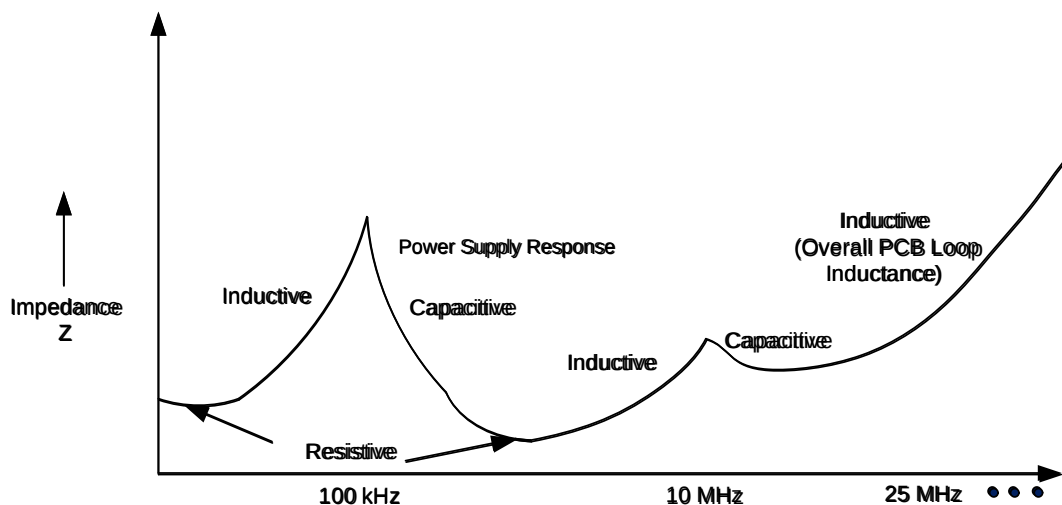


Figure 3-7 Impedance over Frequency In a More Complex System

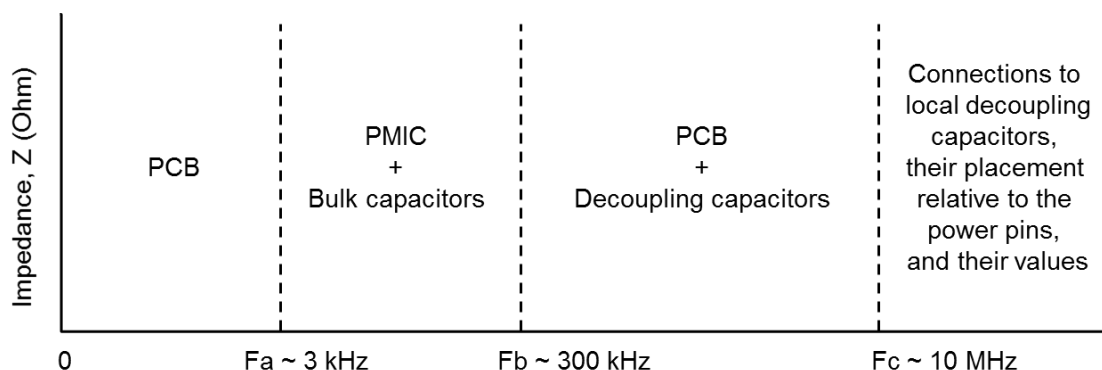


Figure 3-8 Dominant Effects On Impedance (Z) Over Frequency Range

NOTE: The exact frequencies F_a , F_b , and F_c depend on the PMIC regulator used, bulk capacitors, inductor (SMPS only), board impedance, decoupling capacitors, etc.

When designing a PDN the following guidelines are used to optimize the PDB in certain frequency bands.

DC–3 kHz → PCB metal routing to meet the DC resistance specification

- First, address thermal design requirements relative to the placement of the PMIC and processor
- Maximize metal thickness ($\frac{1}{2}$ vs. $\frac{1}{3}$ oz. Cu)
- Use sufficiently wide power traces
- Use multiple parallel power and ground vias

3 kHz–300 kHz → PMIC regulator response and bulk capacitors

- Use supplied PMIC settings and reference schematic values
- Follow design guidelines (later in this document) for optimal placement of the bulk capacitors

- Maximize metal thickness ($\frac{1}{2}$ vs. $\frac{1}{3}$ oz. Cu); use wider power traces
- Use multiple parallel power and ground vias

300 kHz–10 MHz → PCB layout and decoupling capacitors

- Engineer capacitor values, body sizes, and placement to meet impedance specification

10 MHz onwards → Capacitors and PCB connection

- Current always takes the path of least impedance. At these high frequencies, PDN loop inductance, value, and placement of capacitors attached to power pins dominates impedance.
- Local decoupling capacitors should be placed as close as possible to the processor power and ground pins. Use back-side capacitors if possible. Each capacitor should have its own via directly to the ground plane and power plane layer.
- Power and ground vias and planes should be as close together as possible.

See [Table 3-1](#) and [Figure 3-9](#).

3.1.3 Factors Affecting Impedance

The Following Sections give guidance on designing the Power Delivery Network for a single example supply. It is necessary to repeat this process for each of the power rails on the APQ8016E.

The use of a PCB CAD package that can simulate resistances, and impedances is **required** in order to successfully design a system based on the APQ8016E chipset.

3.1.3.1 Static DC IR Drop

Calculate the resistance of power trace between PMIC and APQ8016E using:

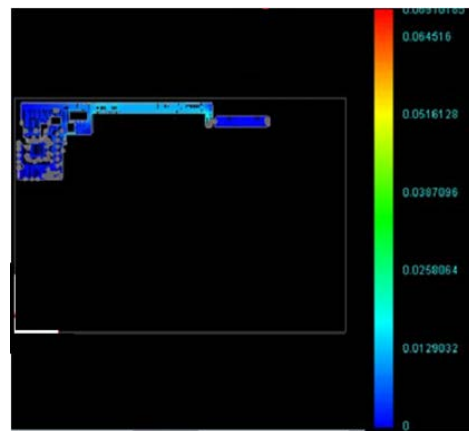
- $R = \rho (L/A)$ where
- ρ = Resistivity of copper
- L = Length of trace

A = Cross-section area = Width of trace $\times$ thickness of trace

Calculate the minimum number of vias required to carry the current. Always use more vias than the minimum required.

- Minimum vias $\geq$ Total current divided by the Current carrying capacity of each via
- Copper paste filled vias are used in certain PCBs. Copper paste is electrically and thermally less conductive than copper. More vias are required when using copper paste vias.

Static IR drop analysis shows current densities at different locations in the layout.



3.1.3.2 PCB Stackup

Power and ground should be located on adjacent layers separated by as thin a dielectric as possible to reduce loop inductance as much as possible.

Table 3-1. PCB Stackup for Minimum Loop Inductance

No	Layer	Thickness (μm)	Scenario I	Scenario II
1	Cu	25		
	Dielectric	50		
2	Cu	20		
	Dielectric	50		
3	Cu	20	Gnd	Signal
	Dielectric	65		
4	Cu	20	Signal	Gnd
	Dielectric	65		
5	Cu	17	Pwr	Pwr
	CORE	100		
6	Cu	17	Gnd	Signal
	Dielectric	65		
7	Cu	20	Signal	Gnd
	Dielectric	65		
8	Cu	20		
	Dielectric	50		
9	Cu	20		
	Dielectric	50		
10	Cu	25		

Better (Dielectric is thinner)

Worse (Dielectric is thicker)

3.1.3.3 Capacitor Placement

Lower power-ground loop area reduces loop inductance thereby improving the effectiveness of a capacitor at high frequencies.

Local decoupling capacitors must be placed as close to the APQ8016E pins as possible.

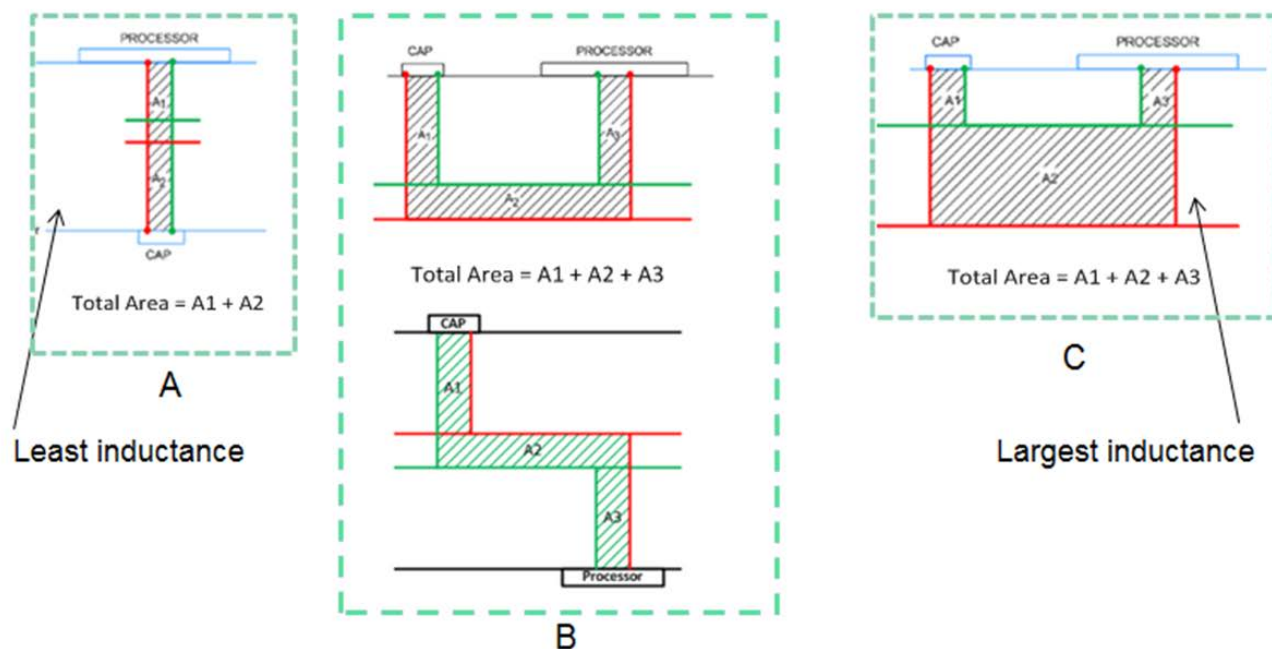


Figure 3-9. Capacitor Placement and Routing Options

NOTE: Geometries are not to scale.

3.1.3.4 Decoupling Capacitor Selection

Capacitor effective loop inductance is geometry based.

- Body size influences self-inductance; capacitors with smaller body sizes have lower ESL.
- Locate power-ground vias as close to each other as possible; small loops have lower inductance.
- Locate vias close to the capacitor terminals; small loops have lower inductance.
- Multiple parallel power-ground via pair connections reduce mounting inductance; parallel loops have lower inductance.
- Avoid sharing vias with adjacent capacitors; parallel loops have lower inductance, series loops have higher inductance.
- Three-terminal capacitors such as X2Y devices can offer lower inductance than two-terminal devices.
 - With good layout the three-terminal devices are connected with small loops and parallel loops.
 - The multiple terminal design also provides small loops and parallel loops inside the device for lower ESL.

3.1.3.5 Decoupling Capacitors Surface Routing

The routing of the tracks joining the decoupling capacitors and the vias carrying the current to the reference planes has a large effect on the effectivity of the capacitor. Figure 3-10 shows examples of the surface tracks and via placements that can improve the effectivity of capacitors.

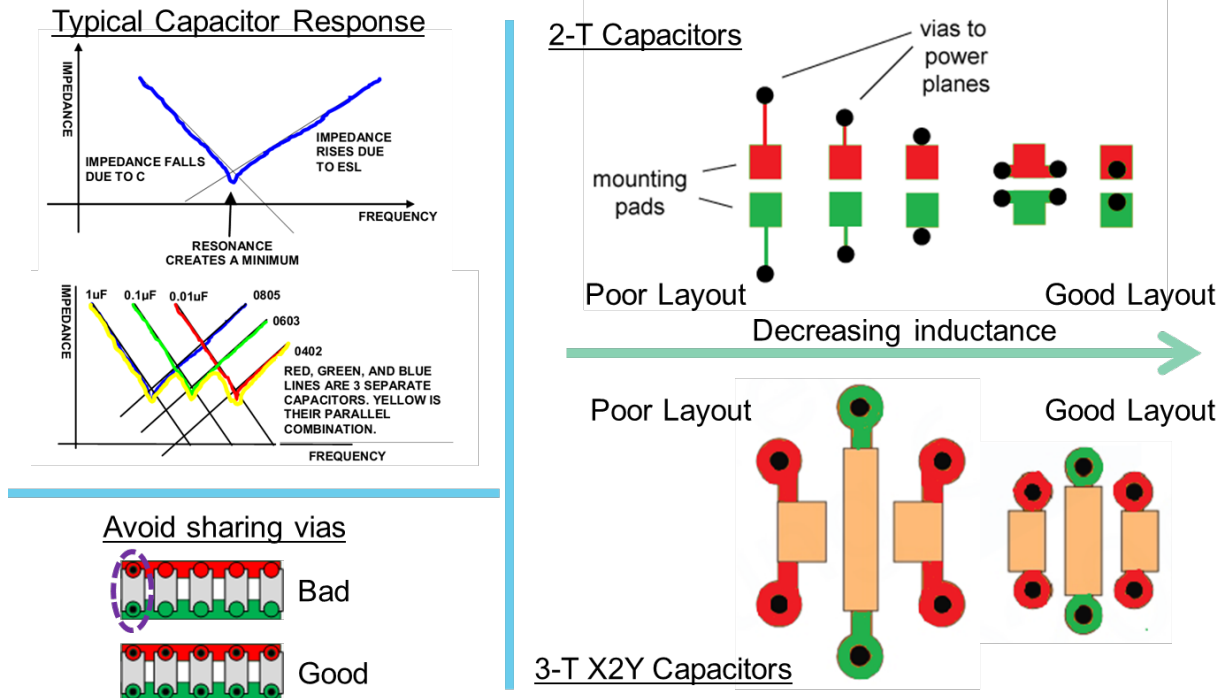


Figure 3-10 Decoupling Capacitor Surface Routing and Vias

3.2 PDN Design Example - Simulation

This section shows the step by step process to optimize one section of the system PDN, this process must be repeated for exact power delivery network in the system. Simulation is required.

The following definitions are used throughout this design example:

- Effective capacitance: The total shunt capacitance over the PDN between the source and the load
- Effective inductance: The loop inductance from the processor power pins to the closest “local” bypass capacitors

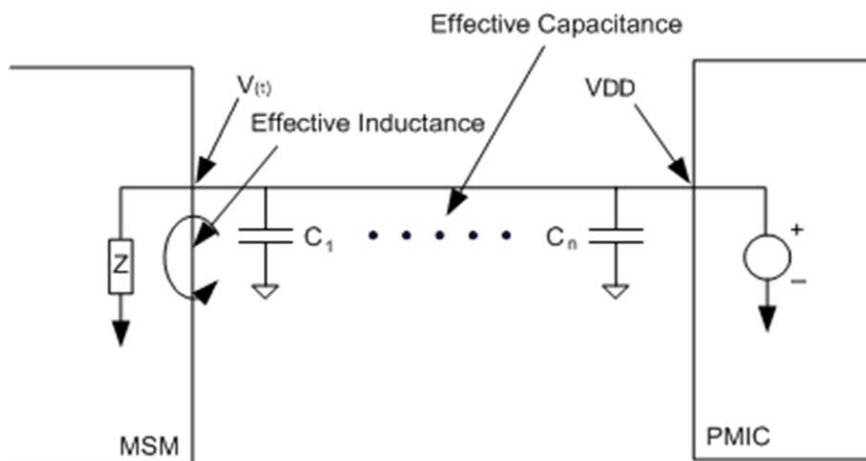


Figure 3-11 Basic PDN

This PDN impedance reduction example uses a device similar to the APQ8016E and is used to show the concepts of PDN optimization. This example is not from an APQ8016E system.

NOTE: All impedance graphs shown in the following slides are simulated results that correlate accurately to measured results.

- Bulk capacitor (at the PMIC output) ([modification case 1](#))
- Wide traces/copper fill ([modification case 2](#) and [case 3](#))
- Local decoupling capacitors (at the APQ8016E input)
 - Placement (minimize the distance to the processor power supply pins) ([modification case 4, step 1](#) and [case 5, step 1](#))
 - Value ([modification case 5, step 2](#))
- Layer thickness ([modification case 5, step 3](#))
- Distance between PMIC and the processor power supply pins (General – Via Routing)
- Distance between power and ground planes (General – PCB Area and BOM Count)

Via ([General – Trace Width](#))

Proper PDN design ensures that $V_{min} \leq V(t) \leq V_{max}$ during all di/dt events.

- $V(t)$ is the voltage measured at the APQ8016E power pins as a function of time.
- V_{min} is the minimum voltage allowed (DC + transient) at the processor power supply pins to guarantee proper operation over all variations of process and temperature as listed in the device specification.
- V_{max} is the maximum voltage allowed (DC + transient) at the processor power supply pins to guarantee proper operation over all variations of process and temperature as listed in the device specification.

3.2.1.1 Initial Layout and Conditions

In this example L402 is the inductor for the switching power supply. R504 is a current sense resistor intended for power measurement. The initial circuit layout met the following conditions:

DC resistance (by PowerDC):

- Power path:
 - From APQ8016E to R504: 1 mΩ
 - R504: 4 mΩ
 - From R504 to L402: 6.5 mΩ (11.5 mΩ total)
- Ground path:
 - From U201 to GND node near L402: 1.7 mΩ
- Total is 11.5 mΩ + 1.7 mΩ = 13.2 mΩ – This does not meet the 10 mΩ specification at DC.

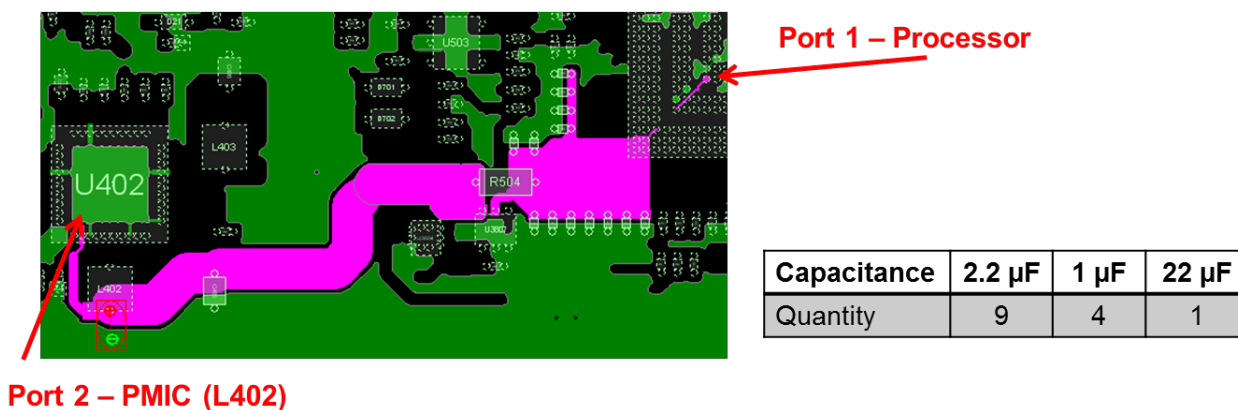


Figure 3-12 Initial Design Layout and Conditions

3.2.1.2 Modification – Case 1

Additional bulk capacitors are placed at the PMIC side to improve the low-frequency response.

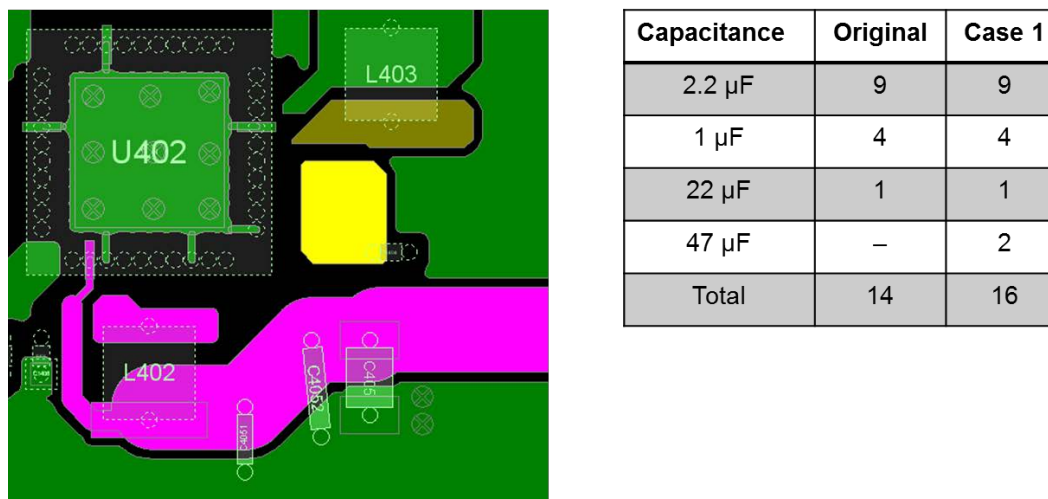


Figure 3-13. Additional Bulk Decoupling Capacitors – Case 1

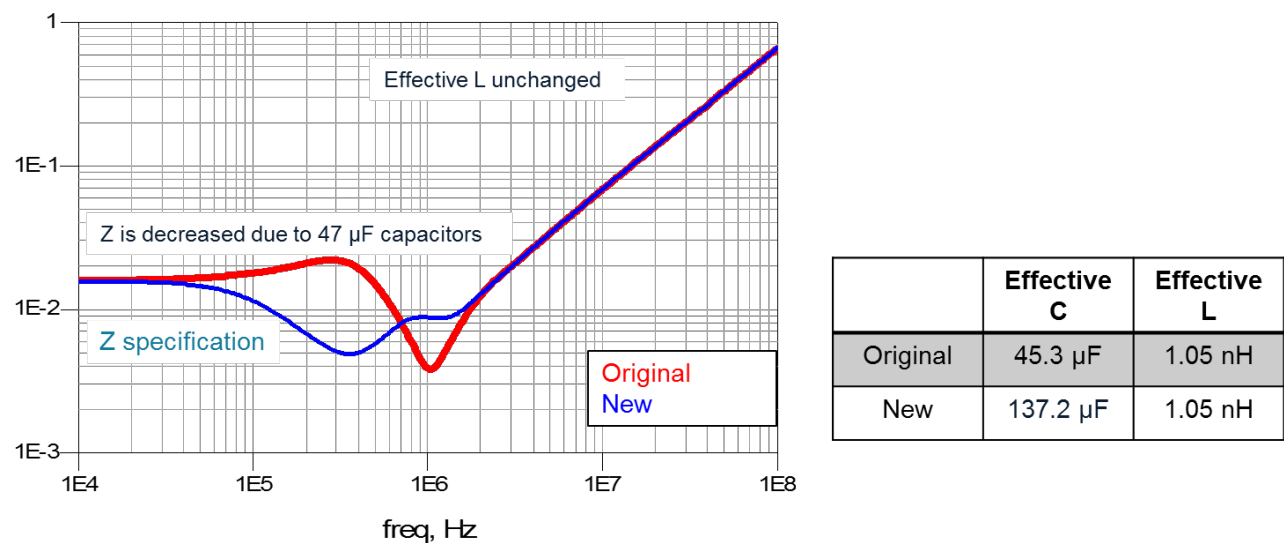


Figure 3-14. Simulated Impedance – Case 1

Observation: Large bulk capacitors improve PDN impedance as shown, but even more important is their role in reducing PMIC effective impedance below 100 kHz (not shown in these simulations). Follow the PMIC data sheet and reference schematic for proper values.

3.2.1.3 Modification – Case 2

Changing the values of the capacitors can improve the overall impedance. Note the capacitor changes, some are made larger, and some are made smaller.

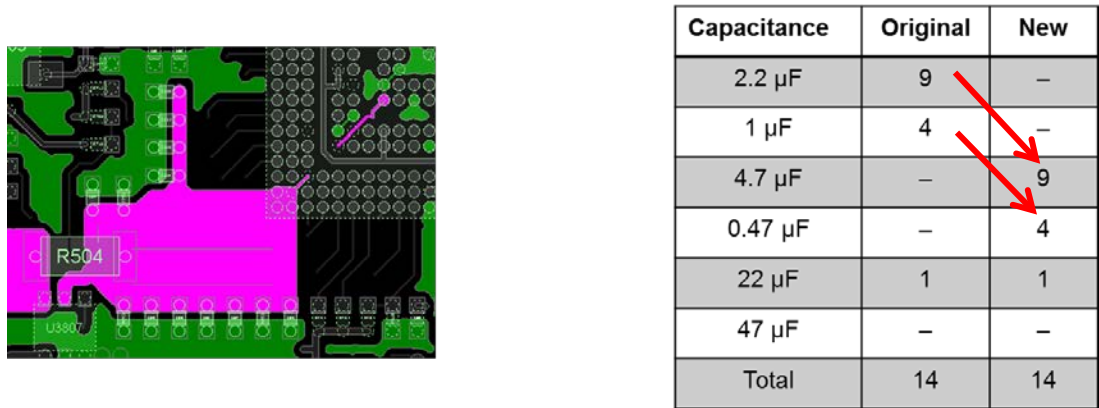
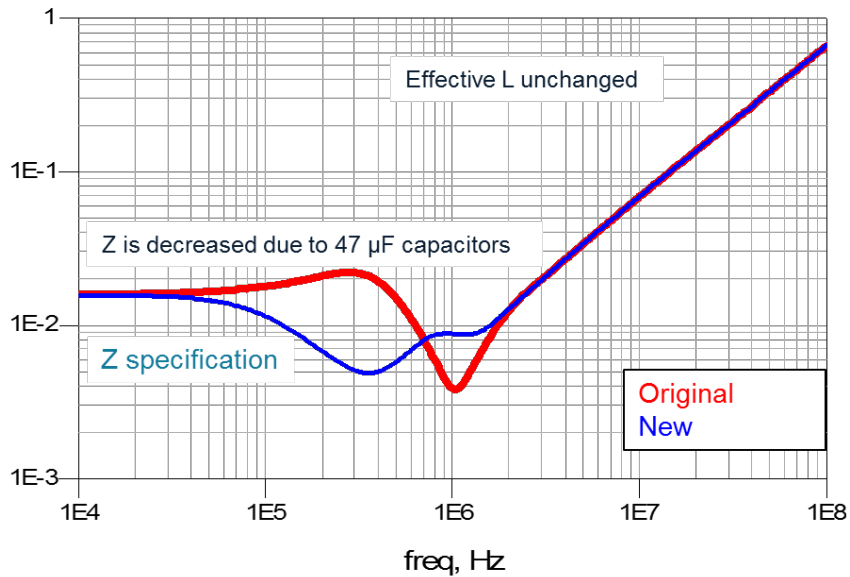


Figure 3-15. Changing Capacitor values – Case 2

- No layout/capacitor placement changes
- Replaced only the 2.2 μF capacitor with 4.7 μF capacitors (larger) and replaced the 1 μF capacitors with 0.47 μF capacitors (smaller)



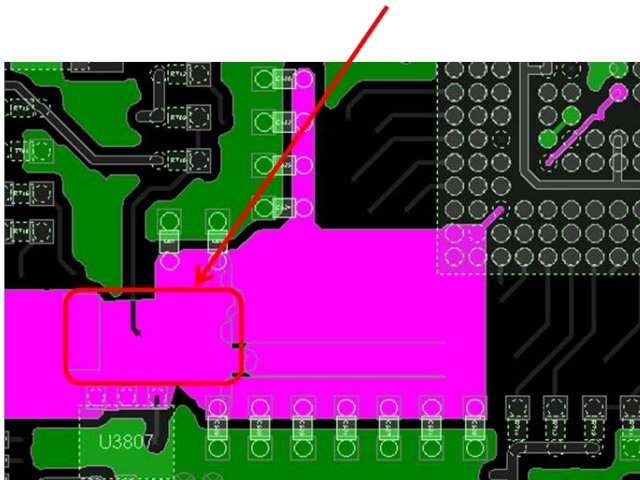
	Effective C	Effective L
Original	45.3 μF	1.05 nH
New	137.2 μF	1.05 nH

Figure 3-16. Change in PDN Impedance – Case 2

Observation: Increasing the value of local capacitors without changing the PDN routing did not help reduce the effective inductance. Effective inductance is dominated by the current path from the chip pins to the local decoupling capacitors, not the capacitor values themselves.

3.2.1.4 Modification – Case 3

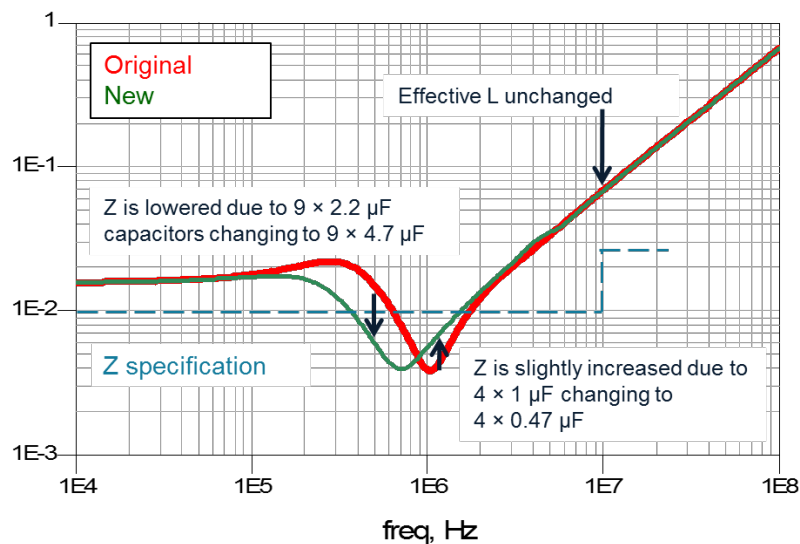
Take out the current measurement resistor and fill in the shape.



Capacitance	Original	New
2.2 μF	9	9
1 μF	4	4
22 μF	1	1
Total	14	14

- No changes in the bypass capacitor arrangements

Figure 3-17 Removal of current measurement resistor – Case 3



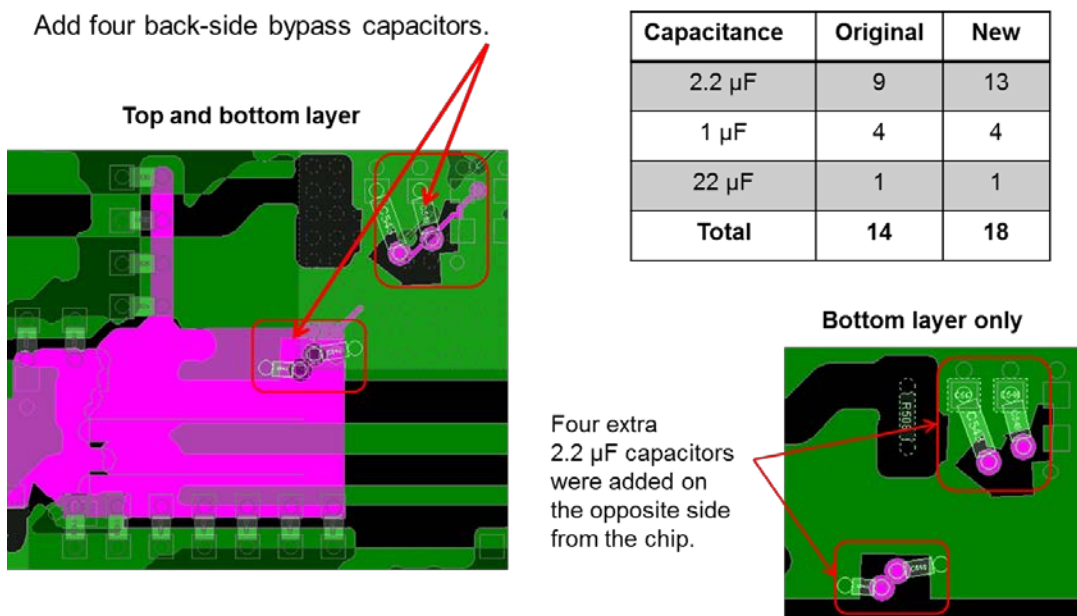
PMIC = ideal voltage source

	Effective C	Effective L
Original	45.3 μ F	1.05 nH
New	66.3 μ F	1.05 nH

Figure 3-18 Change in PDN Impedance – Case 3

Observation: Removing the series R showed a 4 m Ω improvement in low--frequency impedance from the APQ8016E chip to the PM8916 (see Original Layout/Conditions).

3.2.1.5 Modification – Case 4

**Figure 3-19 Addition of backside capacitors – Case 4**

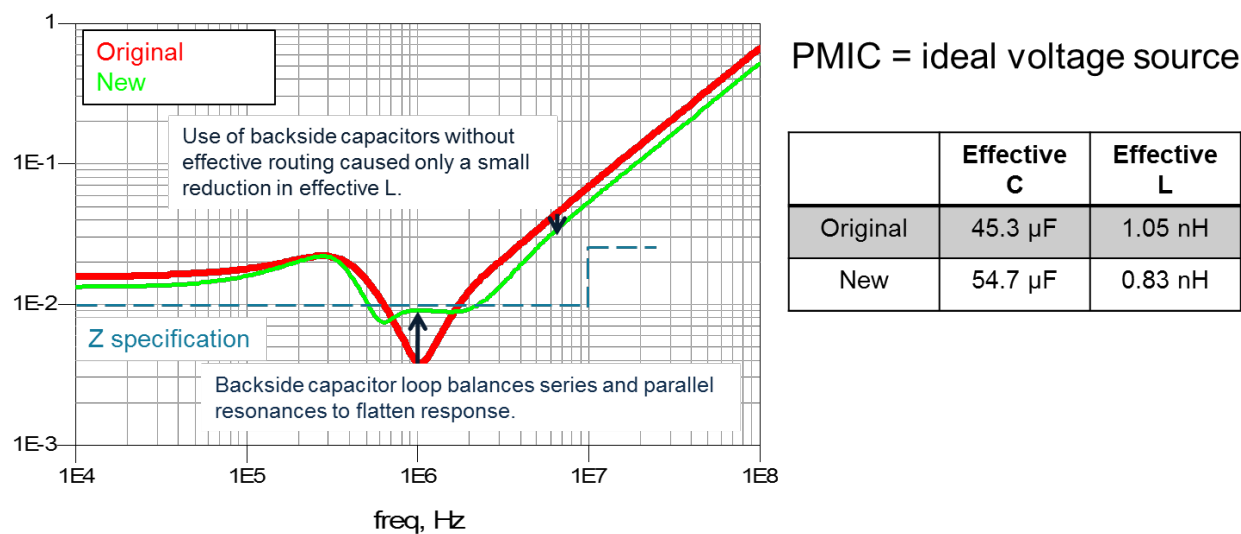


Figure 3-20 Change in PDN Impedance – Case 4

Observation: Adding back-side capacitors without optimizing the capacitor routing only marginally reduced the PDN inductance. The next case shows that re-engineering the routing of those capacitors was key.

3.2.1.6 Modification – Case 5 Step 1

Use back-side capacitors and reduce the size of other capacitors to place them as close to the processor power pins as possible.

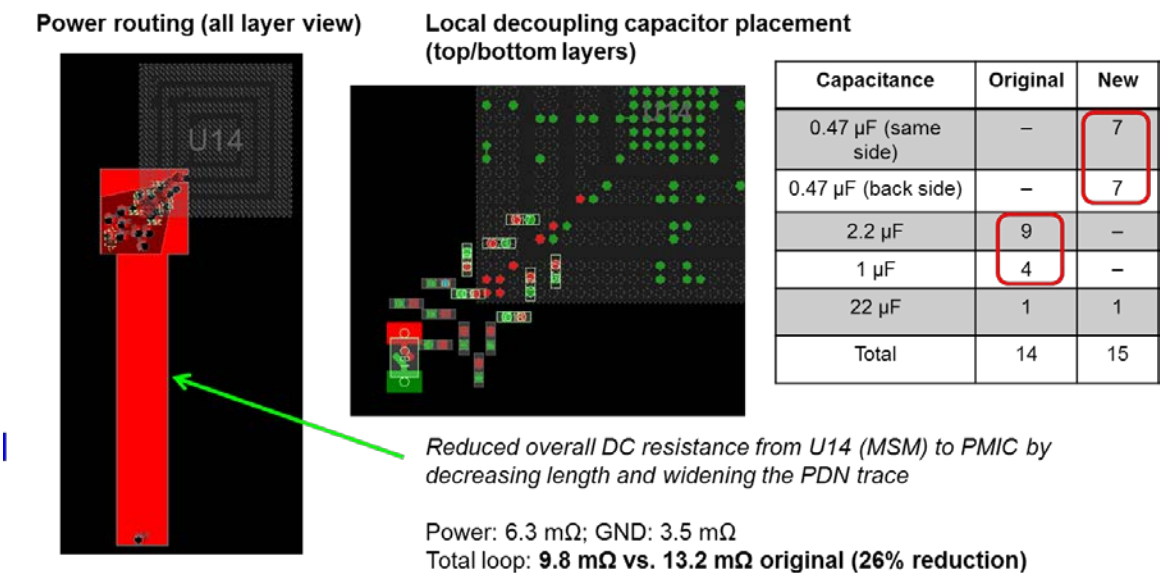


Figure 3-21 Moving capacitors to backside – Case 5 Step 1

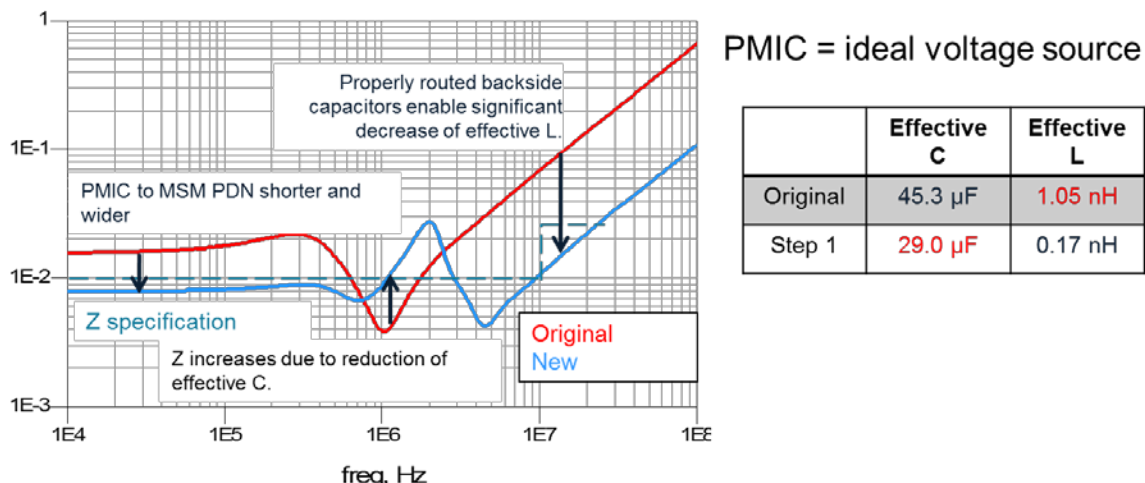


Figure 3-22 Change of PDN Impedance – Case 5 Step 1

Observation: With an improved breakout strategy and re-engineering of the via patterns, the PDN impedance is significantly reduced at both low and high frequency.

Back-side capacitors with close **placement** and short **routing** of the traces to the processor power pins and ground plane reduce effective L. However, reduction of effective C has caused an increase in impedance at mid-frequency.

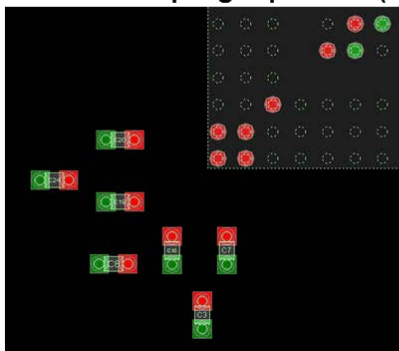
3.2.1.7 Modification – Case 5 Step 2

- To reduce the LC resonance peak ~ 1.5 MHz, all of the same-side decoupling capacitors were changed from $0.47 \mu\text{F}$ 0201 to $2.2 \mu\text{F}$ 0402. The pad size and PDN routing stayed the same; only the capacitance and ESL value in the circuit definition were changed.
- The impedance beyond 10 MHz is dominated by the effective inductance of the back-side capacitors and their routing to the processor power supply pins.

NOTE: There was not enough room for the required power/ground vias if all back-side 0201 capacitors were replaced by 0402.

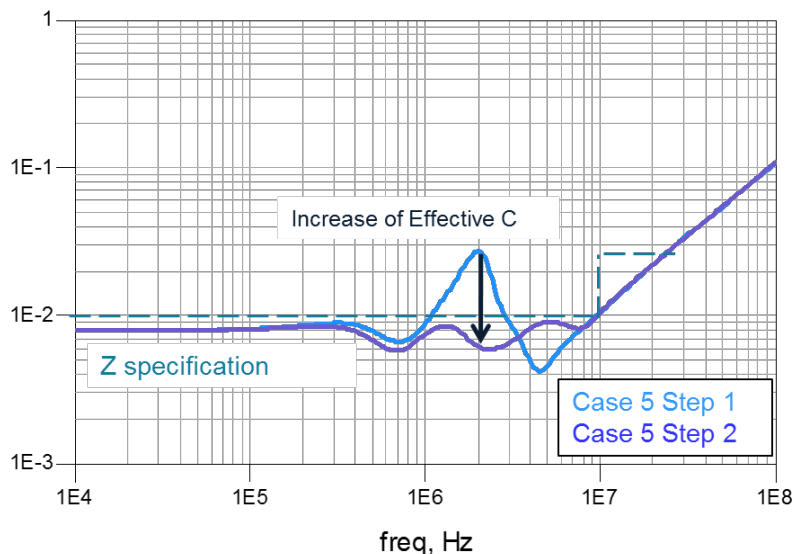
Improve the LC resonance region.

Local decoupling capacitors (top side)



Capacitance	Quantities
2.2 μF (top side)	7
0.47 μF (back side)	7
22 μF	1
Total	15

Figure 3-23 Local Top Side Decoupling Capacitors – Case 5 Step 2



PMIC = ideal voltage source

	Effective C	Effective L
Case 5 Step1	29.0 μF	0.17 nH
Case 5 Step 2	41.2 μF	0.17 nH

Figure 3-24 Change of PDN Impedance Case 5 Step 2

Observation: Replacing all of the same-side 0.47 μF capacitors with the 2.2 μF capacitors helped reduce the mid-frequency LC resonance peak. The high-frequency response (> 10 MHz) stays the same as Case 5 Step 1 because there were no routing changes. (This effective inductance is due to the back-side capacitors and their routing to the processor power supply pins.)

3.2.1.8 Modification – Case 5 Step 3

Change in board stack-up

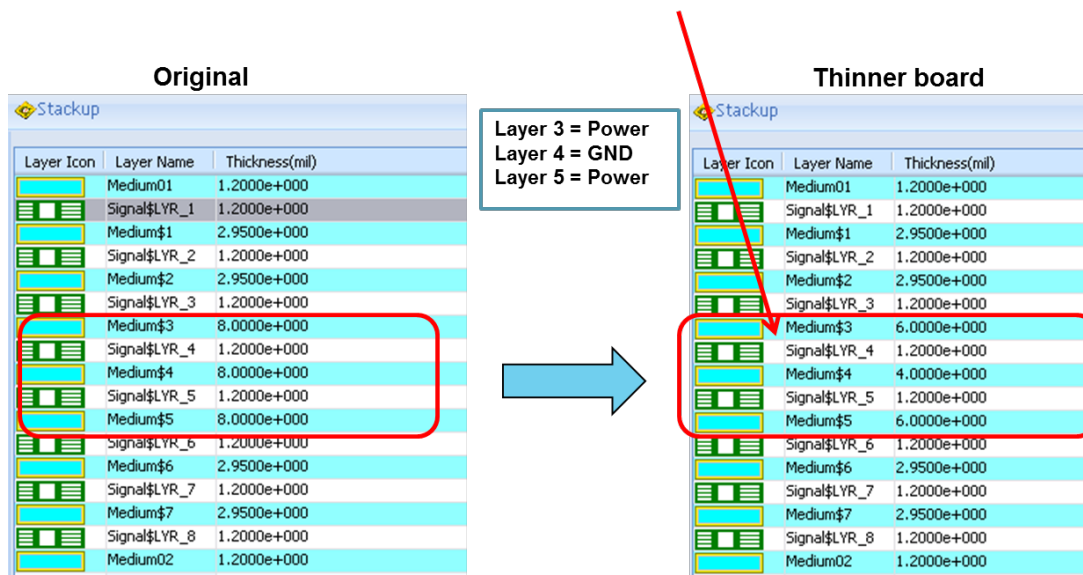
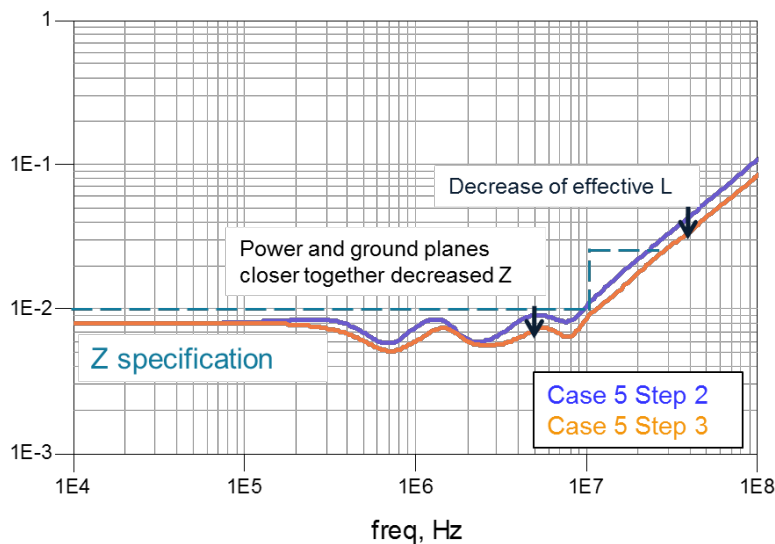


Figure 3-25 Change of Dielectric layer thickness – Case 5 Step 3

- The dielectric thickness between core layers was reduced as shown above.
- The total board thickness was reduced by 200 μm .



PMIC = ideal voltage source

	Effective C	Effective L
Case 5 Step 2	41.2 μ F	0.17 nH
Case 5 Step 3	41.2 μ F	0.135 nH

Figure 3-26 Change in ePDN Impedance – Case 5 Step 3

Observation: When the board thickness (distance between power and ground planes) was reduced, the vias and current loop associated with the back-side capacitors became smaller, reducing the effective inductance.

3.2.1.9 Summary

The PDN for this branch has been significantly improved by making multiple small improvements. This example demonstrates some of the basic techniques used to improve the system PDN, and shows that small changes can make significant improvements.

Table 3-2 Changes in Capacitor Selections

Capacitance	Original	Case 5 Step 1	Case 5 Step 2
2.2 μ F (same side)	9	–	7
1 μ F (same side)	4	–	–
0.47 μ F (same side)	–	7	–
0.47 μ F (back side)	–	7	7
22 μ F (PMIC load capacitor)	1	1	1
Total	14	15	15

Table 3-3 PDN Improvement Summary

Design changes	Modifications	Z_{PDN} DC–100 kHz	Z_{PDN} 100 kHz–10 MHz	Z_{PDN} (> 10 MHz)
Original		13.2 m Ω	–	164.85 m Ω
Modification Case 5 Step 1	- Layout changes - Decaps selections/placements	9.8 m Ω	22 m Ω	26.69 m Ω
Modification Case 5 Step 2	Larger capacitance/dimension change (0201 → 0402)	9.8 m Ω	8 m Ω	26.69 m Ω
Modification Case 5 Step 3	Board thickness reduced by 200 μm	9.8 m Ω	8 m Ω	21.19 m Ω

PDN impedance specification achieved.



3.3 Experimental Verification of PDN Design

When the final product design and manufacturing is completed experimental verification for each node of the PDN is required to ensure that the design matches the simulations. Without design verification possible errors in the simulation will go undiscovered. The results will be unstable systems and frequent unexplained crashes.

3.3.1.1 Stress Test

- Customers must run stress tests on the final system and measure the effectiveness of the PDN. Additionally Customers should also run their own high concurrency tests at temperature extremes.
- These stress tests can only be used only on a physical board, not in simulation.
- Designs must meet the PDN specifications (listed in the device specifications) in simulation before manufacturing. Physical stress test is a supplement to, not a replacement for, meeting the PDN specifications in simulation.

3.3.1.2 Recommended Lab Setup

- See the next slide for probe setup.
- Use an oscilloscope with Bandwidth greater than 500 MHz, DC; couple the probes, ensure there is no low-pass filter, 50 Ohm input impedance, and infinite persistence mode to capture peaks. Use 50 Ohm input because it is high-impedance to the PDN and has the advantage of allowing a very stable mechanical connection to the PDN. This allows repeatable measurements without picking up RF noise.
- Connect a 50 Ohm coaxial cable to the scope (not a high impedance scope probe).

PCB: Solder the center conductor of a semi-rigid coaxial shield to the capacitor closest to the APQ8016E. Connect the shield of the coaxial cable to the ground closest to the above mentioned capacitor. The semi-rigid coaxial shield should have an SMA-type connector to connect to the 50 Ohm coaxial cable to the scope.

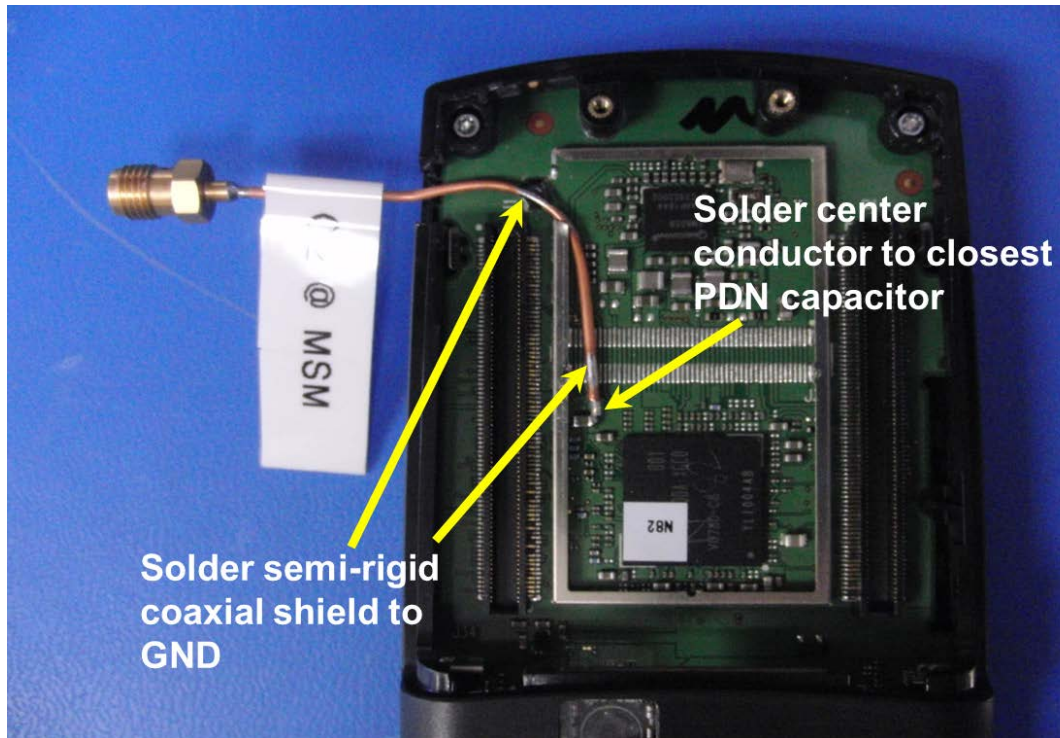


Figure 3-27 Example Verification Test Probe Setup

3.3.1.3 Experimental Results

Scope capture of the supply voltage on a typical system

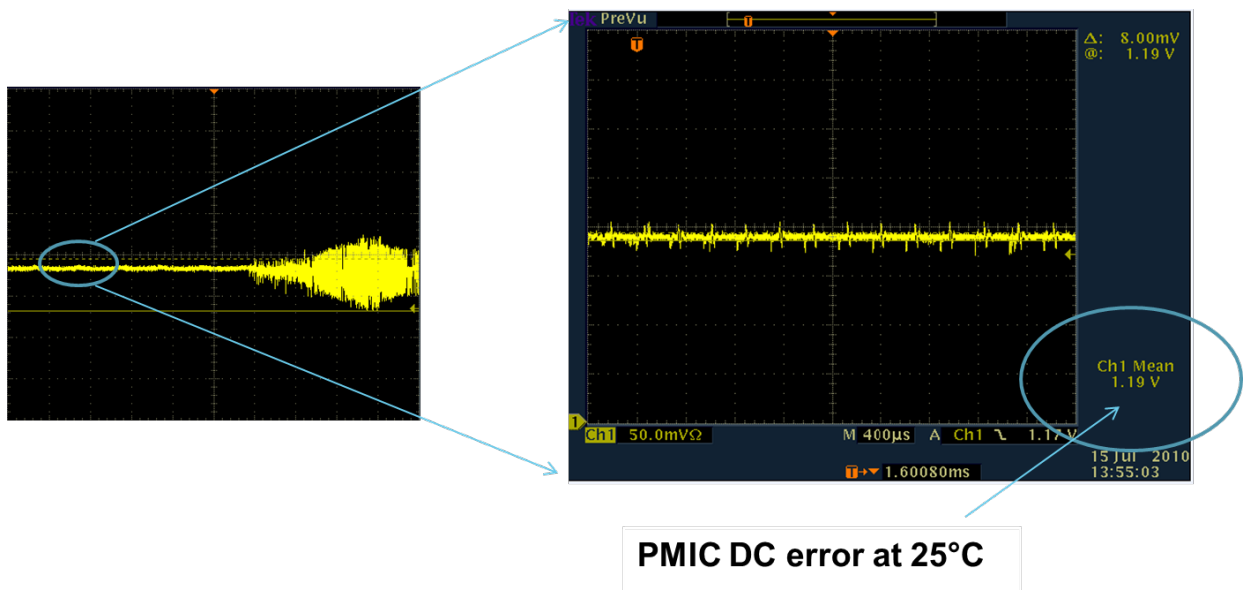


Figure 3-28 Measured Supply Noise

3.3.1.4 Interpreting the observed measurement

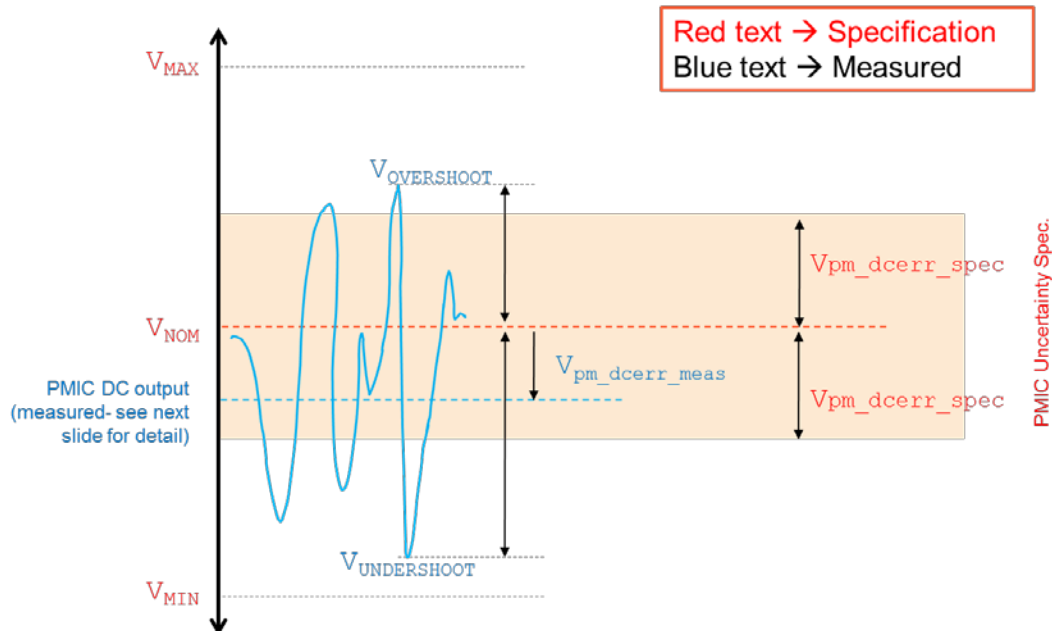


Figure 3-29 Experimental Results Terms and Definition

- V_{nom} = The PMIC set point (example: 1.2 V for the APQ8016E device).
- V_{min} and V_{max} = Refer to the specification document.
- $V_{undershoot}$ = The absolute value of measured undershoot relative to V_{nom} (when running the AC1 stress test).
- $V_{overshoot}$ = The absolute value of measured overshoot relative to V_{nom} (when running the AC1 stress test).
- $V_{pm_dcerr_meas}$ = The measured DC error of the PMIC relative to V_{nom} . This will be negative if the actual DC voltage is below V_{nom} and positive if it is above V_{nom} .
- $V_{pm_dcerr_spec}$ = The error specified in the PMIC datasheet for PMIC output voltage relative to V_{nom} . Typically, it is expressed as a percentage of output voltage; it is converted to mV.

Example: For PM8916 with APQ8016E, it was $\pm 1\%$ of V_{nom} . Hence, $V_{pm_dcerr_spec} = 0.01 * 1.2 \text{ V} = 12 \text{ mV}$.

3.3.1.5 Calculations

- $V_{margin_l} = V_{nom} - V_{min} - V_{undershoot} - (V_{pm_dcerr_meas} + V_{pm_dcerr_spec})$
- $V_{margin_h} = V_{max} - V_{nom} - V_{overshoot} - (-V_{pm_dcerr_meas} + V_{pm_dcerr_spec})$
- Positive V_{margin_l} and V_{margin_h} ensures that the device V_{min} and V_{max} specifications are not violated.

3.3.1.6 Further measurements

- Repeat the test setup and PCB solder connection as outlined above in Lab Setup and repeat the measurement and verification process for every supply to the APQ8016E.
- Repeat the preceding steps over temperature (-30°C, 25°C, and 60°C).

3.4 DC considerations for PDN design

The DC resistance path from the buck regulator sense point or from the bulk capacitor of a buck regulator to the APQ8016E balls must meet the PDN DCR specification. For linear low drop out regulators (LDOs) the DC resistance path is from the PMIC LDO pin to the APQ8016E balls. Both the resistance from the VDD side of the regulator to the APQ8016E VDD balls and from the regulator VSS side to the APQ8016E VSS balls together must meet the PDN DCR specification.

3.4.1 PDN Impedance Targets – APQ8016E

Table 3-4 PDN system specification (PCB+ baseband IC)

Power domain	Required PCB routing inductance (caps shorted)	Max impedance		
		DC to 10 Hz	10 Hz to 10 MHz	10 MHz to 500 MHz
VDD_MEM	500 pH	11 mΩ	104 mΩ	250 mΩ
VDD_CORE	250 pH	5 mΩ	79 mΩ	160 mΩ
VDD_APC	200 pH	4 mΩ	45 mΩ	140 mΩ

Power domain	Maximum impedance		Pin number of positive ports	Pin number of negative ports
	DC to 10 Hz	10 Hz to 25 MHz		
VDD_MEM	11 mΩ	78 mΩ	All VDD_MEM pins	All GND pins
VDD_CORE	5 mΩ	78 mΩ	All VDD_CORE pins	All GND pins
VDD_APC	4 mΩ	45 mΩ	All VDD_APC pins	All GND pins
VDD_P1	35 mΩ	141 mΩ	J10, J14, K11	G10, G12, G14, G16, K9, J16
		141 mΩ	J20, J22	G18, G20, G24, J18, K21
		141 mΩ	J24, K23	G24, K21
		141 mΩ	K15, K19	K17, K21
		141 mΩ	J30	G28, G30, G32, J32

It is recommend to do PDN simulation up to 500 MHz for VDD_APC, VDD_CORE and VDD_MEM power rails and up to 25 MHz for VDD_P1.

3.4.2 PDN simulation and optimization

Customers should start with the PDN optimization result of 410c DragonBoard reference design, *MSM8X16 + PM8916 Preliminary Reference Schematic* (80-NK807-42), as the starting point of their PCB board PDN optimization.

- Use the following 410c design as a starting point, then after simulations and testing it may be possible to remove some capacitors, which are not required.
- Spread capacitors evenly along the APQ8016E periphery – instead of lumping in a few corners.
- The preferred path is to place these on the backside of the board underneath the APQ8016E chipset for optimal impedance profile.

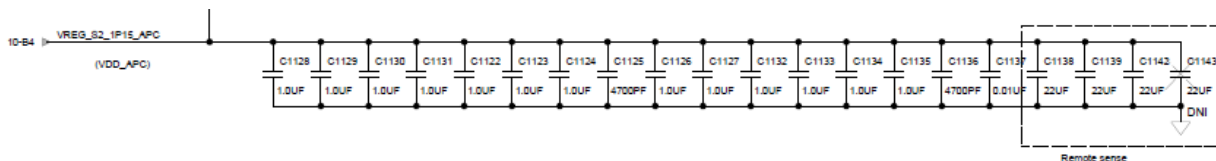


Figure 3-30 VDD_APC

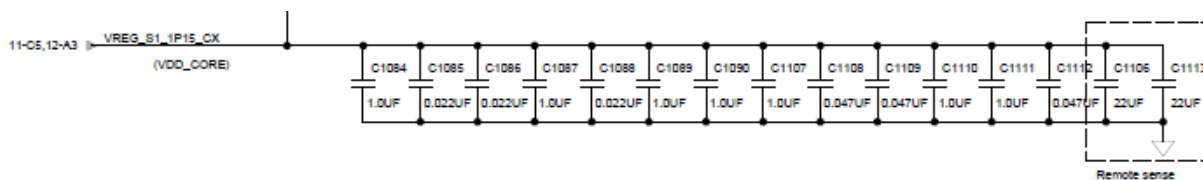


Figure 3-31 VDD_CORE

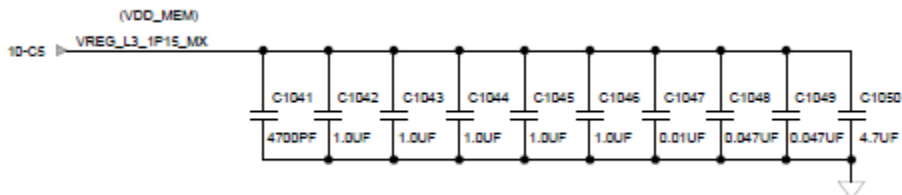


Figure 3-32 VDD_MEM

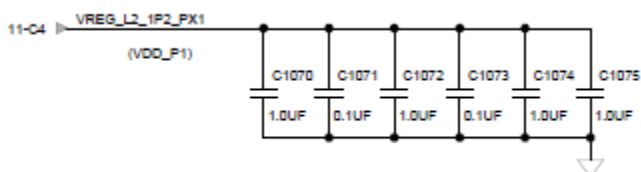


Figure 3-33 VDD_P1

3.5 APQ8016E Routing Guidelines

This Section gives guidelines for the signal breakout and routing rules for each major group of signals.

3.5.1 APQ8016E pin breakout

The key routing challenge is the initial signal breakout from under the APQ8016E, in order to make this possible, it will be necessary to use tracks that are smaller than the typical design rules in this area of the PCB.

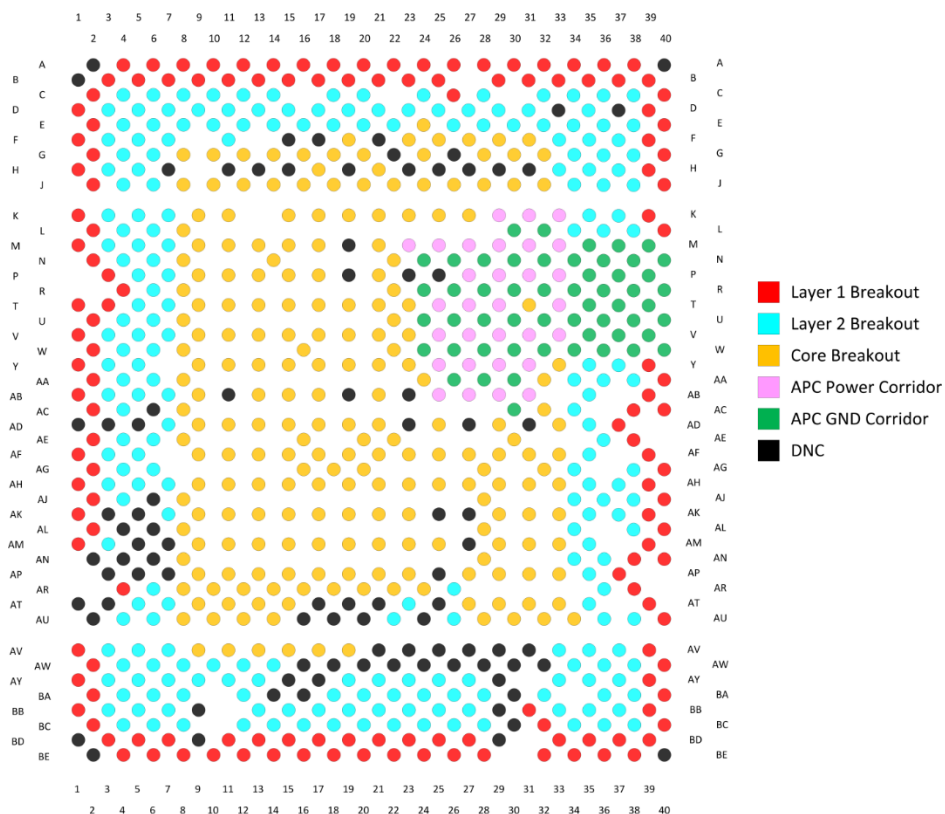


Figure 3-34 APQ pin breakout

After routing away from the APQ device, trace widths and spacing can be increased to 100 μm to ease manufacturing.

3.5.2 Ground and Power Routing

3.5.2.1 Ground

It is important to have a solid ground plane throughout the board; this will be the primary reference plane for most signals. This serves as a signal return for most signals and as reference plane for impedance controlled traces on the board. Having a solid ground plane makes it much easier to manage signal integrity and to reduce radiated emissions.

Layer 1 should include GND fill to provide return path for signals on L2.

- The GND plane on L1 will not be continuous, since there are components on it. Return path of L2 should be focused on L3.
- L1 should be filled with GND for EMI purposes only when GND fill have enough via connecting to main GND.

- If allowed, signals on L1 should get to L2 as soon as possible.

3.5.2.2 Power

The most important power rails within the core of the BGA are: VDD_APC, VDD_CORE and VDD_MEM.

There should be at least one core via for every two pins of the above mentioned rails attaching it to an associated power fill on an inner layer. Routing of high-current DC supplies require sufficient trace widths, fill areas, power planes, and several vias between layers to minimize the IR drop between the supply source (such as the PM8916 device) and the APQ8016E device power pins. A straightforward calculation gives an estimate of the required minimum trace width:

1. Find the maximum current (I_{MAX}) conducted by the trace – the sum of maximum currents expected for all its loads.
2. Define the regulator's operating output voltage (V_{REG}).
3. Calculate the maximum tolerable trace resistance (R_{MAX}) assuming a 1% IR drop:

$$R_{MAX} = 0.01 \times V_{REG}/I_{MAX}$$

4. Estimate total trace length (L) based on the preliminary layout.
5. Determine the copper thickness (T): 1 ounce copper foil thickness is 1.34 mil – scale as needed for thicknesses other than 1 ounce.
6. Calculate the minimum trace width (W_{MIN}) allowed:

$$W_{MIN} = \rho \times L/(R_{MAX} \times T) \text{ where } \rho = \text{copper resistivity } (1.7 \times 10^{-8} \Omega\text{-m})$$

3.5.2.3 APC power and ground

A power corridor through the periphery IO pins allows a low inductance connection to HF decoupling capacitors. Capacitors should be placed as close to the APQ8016E chipset as possible. The ground shape on Layer 1 should flood the decoupling capacitor pads if possible as opposed to using thermal ties.

The power corridor has been designed in such a way that no core vias should be needed to make the connection between the APQ8016E pins and the local HF capacitors.

- Layer 1 should be GND (green), and layer 2 VDD_APC (pink) as shown in [Figure 3-35](#).
- Make sure there are multiple branches connecting VDD_APPC pins to capacitors.

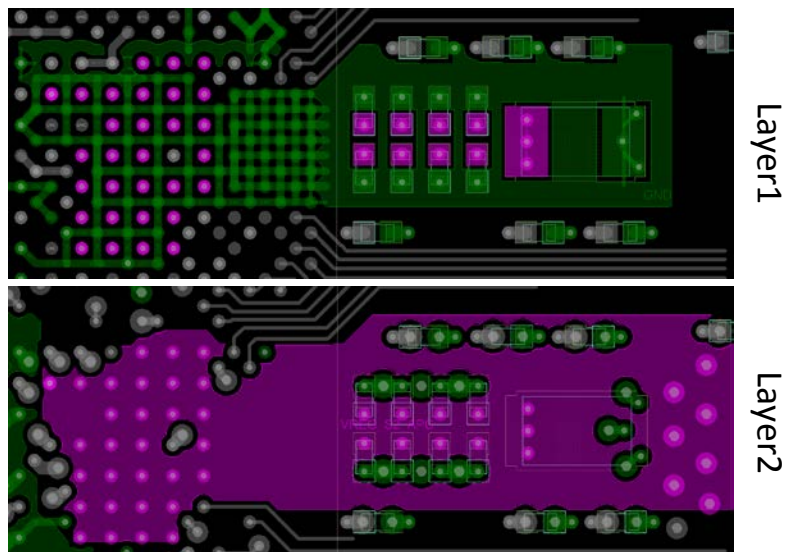


Figure 3-35 Power corridor

3.5.2.4 VDD_CORE

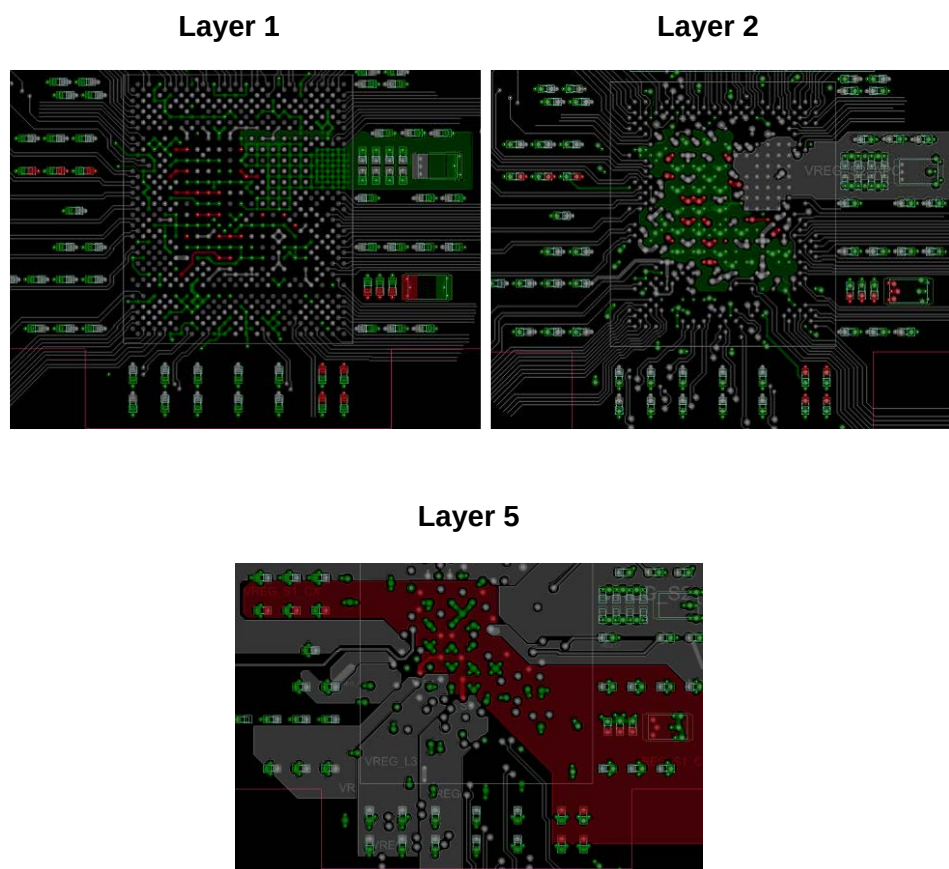


Figure 3-36 Core power

VDD_CORE is highlighted in red.

Make sure there are multiple branches connecting VDD_CORE pins to capacitors.

3.5.2.5 VDD_MEM

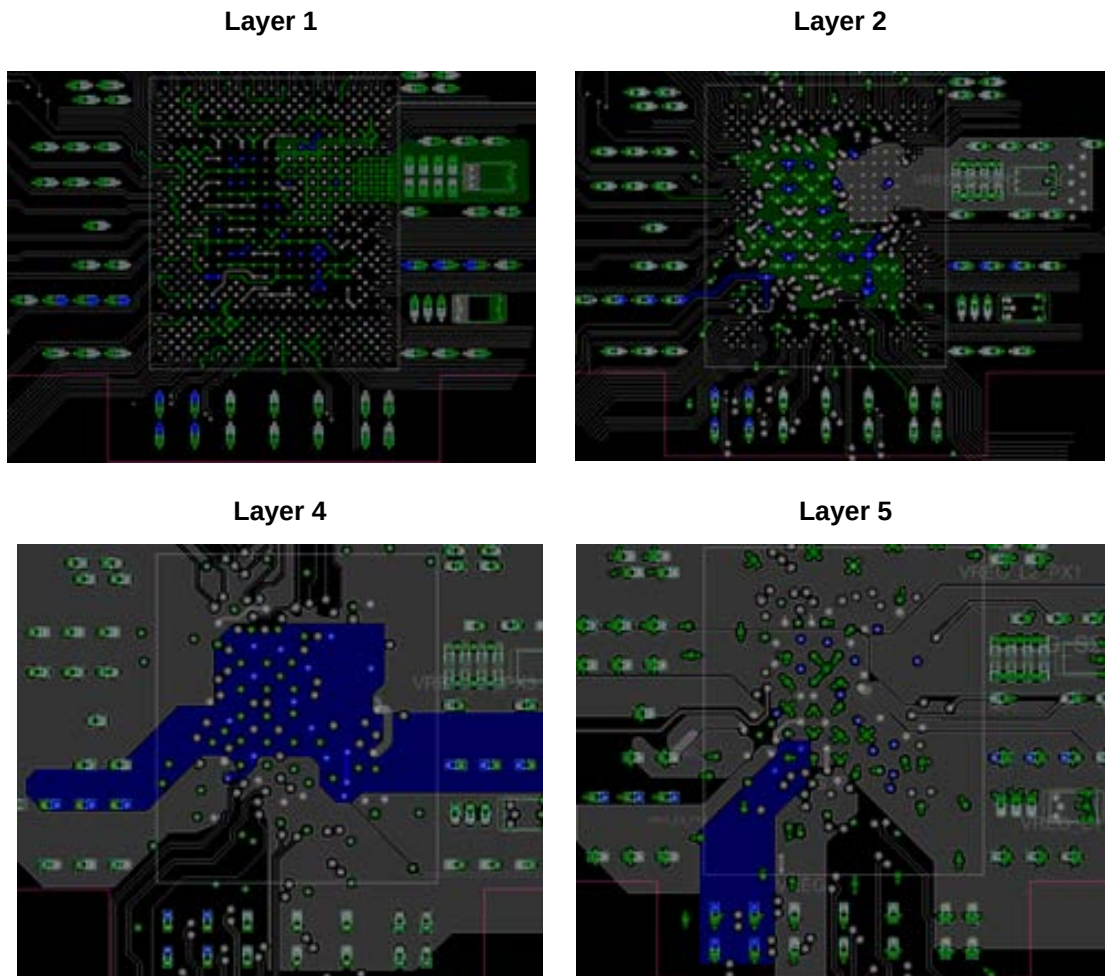


Figure 3-37 VDD_MEM

VDD_MEM is highlighted in blue.

Make sure there are multiple branches connecting VDD_MEM pins to capacitors.

3.5.3 DDR Routing

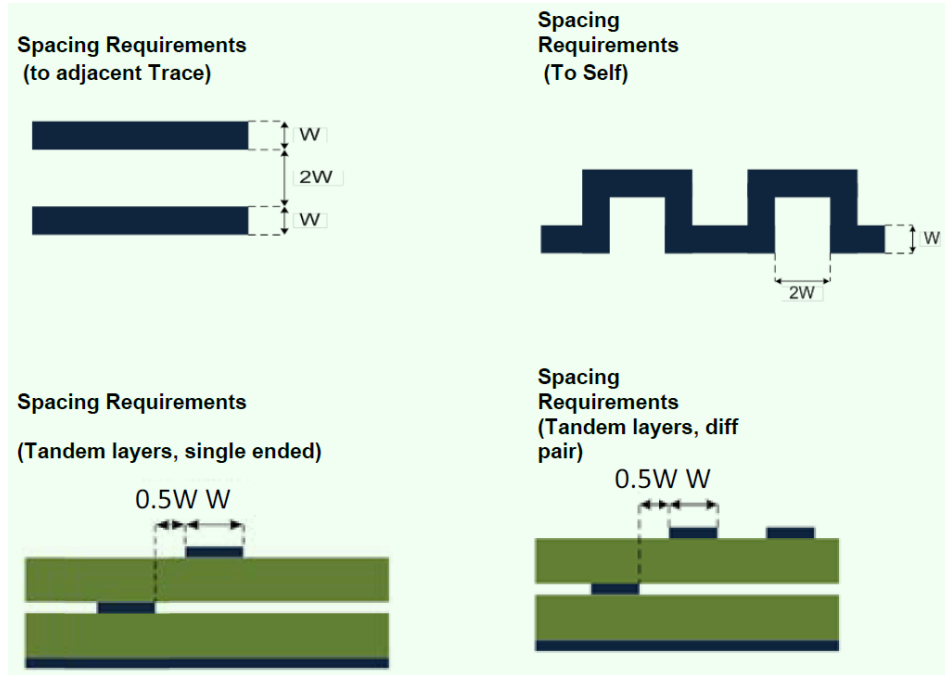


Figure 3-38 DDR

DDR memory should be placed close and directly adjacent to APQ8016E EBI pins, while allowing space for bypass capacitors. EBI signals should be routed adjacent to a solid power fill and ground fill areas to provide proper return paths and maximize shielding. Signals may either reference GND or power. It is important that the power shape on Layer 6 be solid. Maintain constant thickness of the EBI traces to ensure that there are no impedance discontinuities.

NOTE: The important thing is to pick an impedance, and then make sure the impedance on the line varies as little as possible.

Routing is very tight in this region and care should be taken that broad side coupling is minimized. DDR memory area is very constrained, requiring specific routing guidelines.

- Unless there are PCB/placement/stack-up constraints, EBI signals are recommended to be routed in inner layers. Advantage of routing EBI signals in inner layers is, it can reduce EMI radiations.
- When routing is done in external layers, ensure that extra care is taken in shielding the signals.

3.5.4 Digital clocks

APQ8016E has two separate balls that connect to the PM8916 BBCLK1 driver. APQ8016E pin AL34 (CXO) and pin AD35 (USB2_HS_SYSCLOCK) are both clock inputs. These pins have been located physically close to one another so that they can be routed in a daisy-chain topology. [Figure 3-39](#) shows the APQ8016E-Breakout-Study (3-4-3 Test Platform) board CXO clock signal routing colored red on layer 2.

- The route between AD35 and AL34 should be done on Layer 2 (the same layer that AD35 breaks out on) and should be made as short as possible.
- USB_SYSCLK should be the first load, and CXO the final load.

Many other clock signals besides CXO are considered aggressors to sensitive analog signals such as BBRX, TXDAC, WLAN, and GNSS, so those should be isolated as much as possible. Signals crossing on adjacent layers should cross at 90° to minimize coupling.

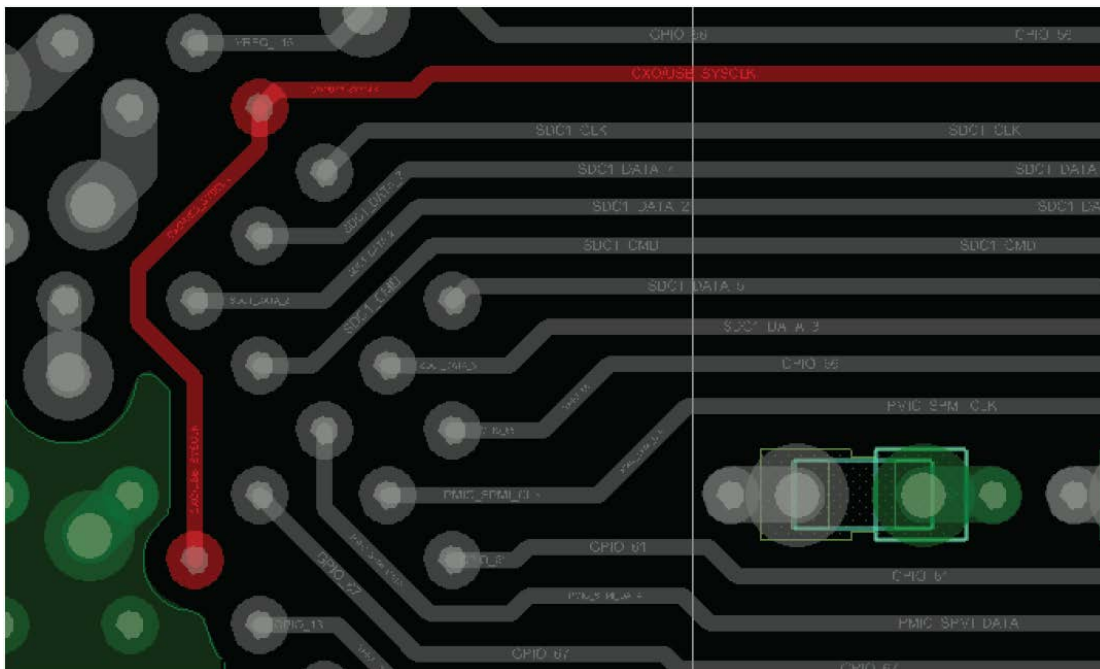


Figure 3-39 Digital clocks

3.5.5 TXDAC

The TXDAC routing for APQ8016E should be very similar to past platforms. TXDAC I/Q nets are both victims and aggressors, so those should be isolated from adjacent signals.

- These signals are sensitive to noise and should be isolated from any aggressor traces that could run both parallel and perpendicular.
- Noise/spurs on the I and Q signals can result in failing Rx-band noise specifications.

The IREF signal is very sensitive and should be isolated from all other signals including the TXDAC I and Q signals. Improperly routed or decoupled IREF could result in TXDAC performance degradation.

3.5.6 BBRX

The BBRX routing for APQ8016E has specific isolation targets between I/Q lines of the same channel as well as channel to channel isolation and isolation from other digital signals such as GPIOs. BBRX traces should be routed as low capacitance and low resistance and should be closely matched. When routing these traces it is important to avoid routing both parallel and perpendicular to other interfaces whenever possible.

3.5.7 MIPI signal

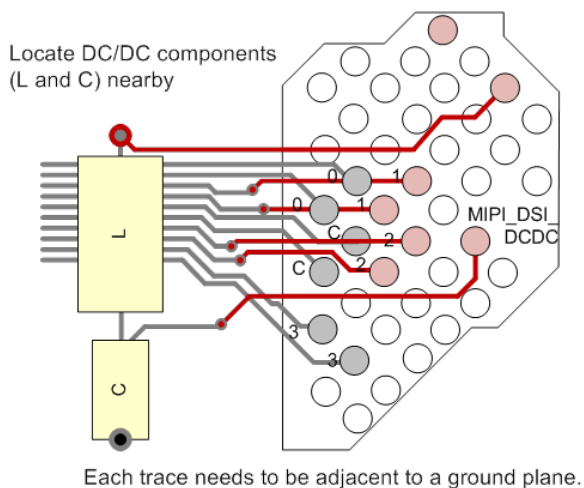
3.5.7.1 MIPI DSI

Protect CSI signals from noisy signals (clocks, SMPS, etc.)

- 750 MHz clock rate; 1.5 GHz data rate
- Differential pairs, 100 Ω nominal, $\pm 10\%$
- Total routing length < 305 mm
- Intra-pair length matching < 5 ps (0.67 mm)
- Inter-pair length matching < 10 ps (1.3 mm)
- Lane-to-lane trace spacing = 3x line width
- Spacing to all other signals = 4x line width
- Calibration voltage
- Maintain a solid ground reference for clocks to provide a low-impedance path for return currents.
- Each trace needs to be next to a ground plane.

VDD_MIPI_DSI_0P4

- DC resistance < 10 m Ω
- PCB trace loop inductance < 1.2 nH



Broadside coupling occurs when traces on adjacent layers overlap or nearly overlap

- This is unavoidable near MSM during breakout
- But once routed away, force traces for each pair to separate from others to minimize coupling

Figure 3-40 MIPI DSI placements

3.5.7.2 MIPI CSI

Protect CSI signals from noisy signals (clocks, SMPS, etc.). Other comments and guidelines:

- 750 MHz clock rate; 1.5 GHz data rate
- Differential pairs, 100 Ω nominal, $\pm 10\%$
- Total routing length < 305 mm
- Intra-pair length matching < 5 ps (0.67 mm)
- Inter-pair length matching < 10 ps (1.3 mm)
- Lane-to-lane trace spacing = 3x line width
- Spacing to all other signals = 4x line width

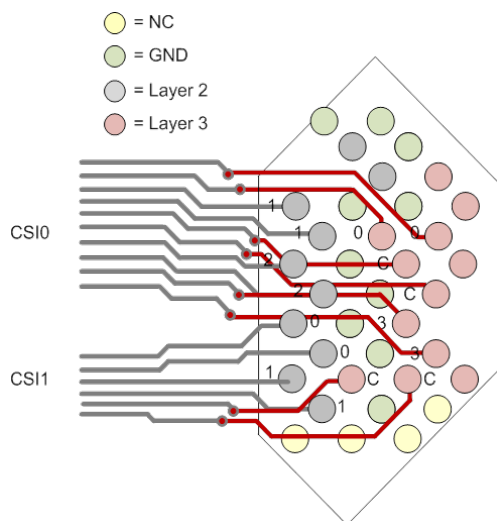


Figure 3-41 MIPI CSI placements

3.5.8 USB signal

HS-USB guidelines:

- Up to 480 Mbps data rate
- 90 Ω differential, $\pm 10\%$ trace impedance
- Trace delay < 4 ns
- Data jitter = 60 ps
- Differential data pair matching < 6.6 mm (50 ps)

Other comments and guidelines:

- External components should be located near the USB connector.
- Relatively fast edge rates, so they should be routed away from sensitive circuits and signals (RF, audio, and 19.2 MHz XO).
- If USB connector is used as the charger input:
 - USB_VBUS node must be routed to the PMIC using extremely wide traces or sub-planes.
 - Detailed recommendations are provided in the PMIC training.

- Even if the USB connector is not used for charging, USB_VBUS can be used as the power bus for the USB. This trace width must be sized depending on the length of VBUS and the expected current.
- USB peripheral currents will be ~200 mA.
- Loading on the USB DP/DM lines could cause USB receiver sensitivity issues. Perform USB electrical tests (eye diagram and receiver sensitivity) to ensure proper USB functionality.
- Refer to Application Note: Tuning the 28 nm USB Phy Eye Diagram and Receiver Sensitivity (80-NA648-1) for information regarding eye diagram tuning by software.
- It is not recommended to install series switches on the USB lines.

3.5.9 SDC signal

SDC1 and SDC2 signals are very high-speed signals.

- Protect other sensitive signals/circuits from SDC corruption.
- Protect SDC signals from noisy signals (clocks, SMPS, etc.).

Other comments and guidelines:

- Up to 200 MHz clock rate;
- 50 Ω nominal, $\pm 10\%$ trace impedance
- CLK to DATA/CMD length matching < 1 mm
- 30 – 35 Ω termination resistor on clock lines near the APQ8016E device
- Total routing length < 50 mm recommended
- Routing distance from the APQ8016E device clock
- Pin to termination resistor < 1 mm
- Spacing to all other signals = 2x line width
- Bus capacitance < 15 pF
- VDD_P7 (SDC1 pad power) and VDD_P2 (SDC2 pad power) loop inductance < 3 nH

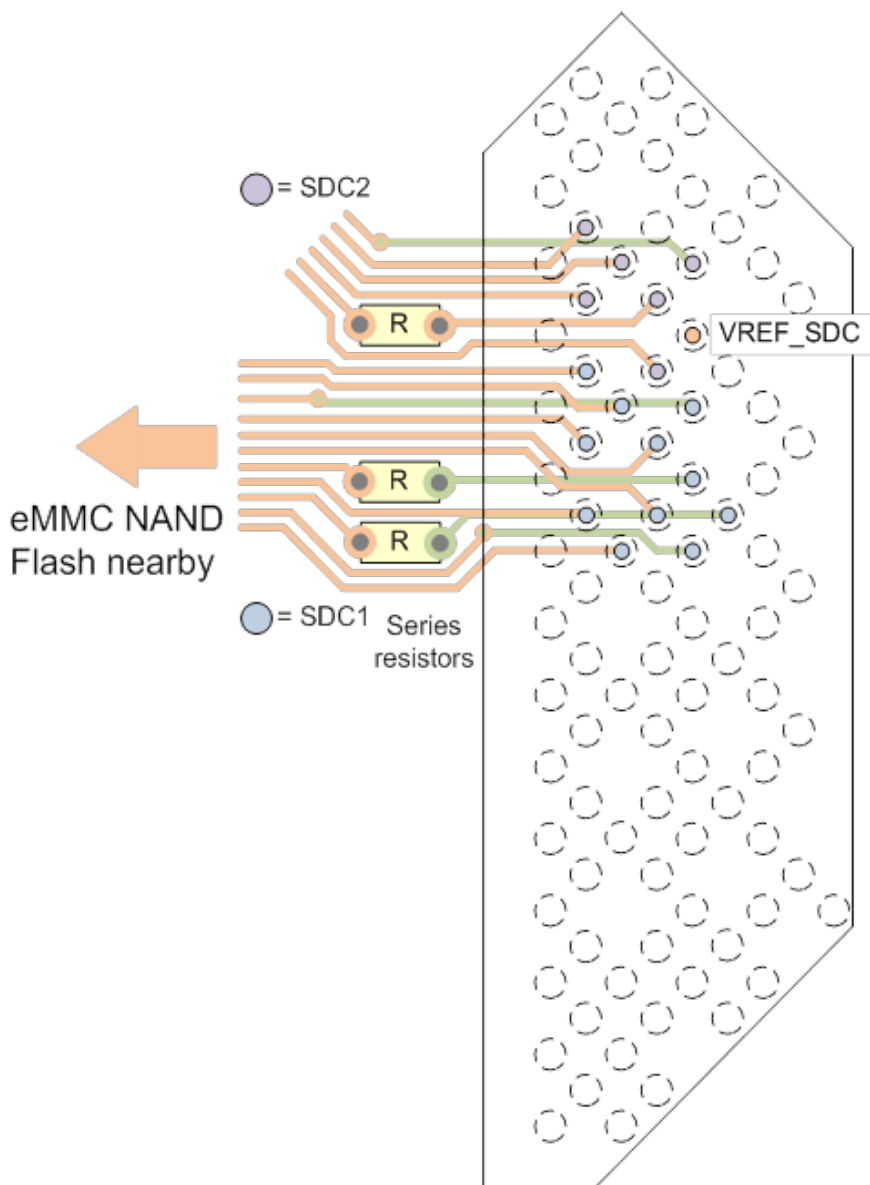


Figure 3-42 SDC placement

3.5.10 SPMI signal

- SPMI data and SPMI clock traces should be controlled impedance = 50 Ω .
- SPMI data and SPMI clock trace spacing should be a minimum of 2x dielectric thickness.
- Spacing between SPMI data/clock and other traces should be a minimum of 2x dielectric thickness.
- The capacitors should be placed as close as possible to the APQ8016E with trace parameters:
 $L \leq 1.65$ nH, $R \leq 70$ m Ω .
- Routing order: APQ8016E $\leftrightarrow$ capacitor $\leftrightarrow$ PMIC (capacitor should be placed between the APQ8016E and the PMIC).

- SPMI data trace length should be within ± 2 mm of the SPMI clock trace length.
- The total trace length of SPMI data and SPMI clock lines should be within 10 mm to 40 mm.

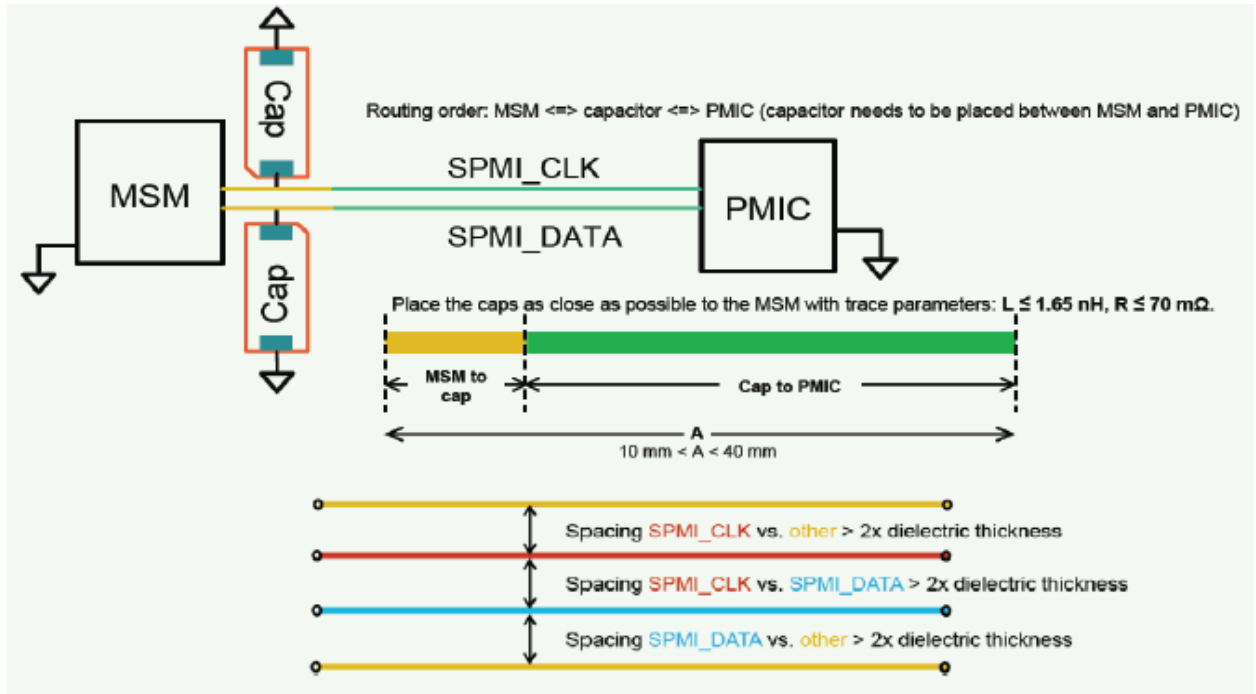


Figure 3-43 SPMI placement

3.6 DDR Memory and PCB Stackup Strategies

Three layout strategies have been tried and tested on APQ8016E designs (3-4-3 Test Platform, 2-6-2 Reference Design, and a 2-4-2 Reference Design). All three strategies are guided and verified by DDR SI simulation, and test results indicate that the different approaches work. The following content provides detailed description for each strategy.

3.6.1 3-4-3 Test Platform strategy

The 3-4-3 Test Platform fully incorporates APQ8016E-Breakout-Study strategy, which is single-side board and chooses G-S-S-G-S-P-x-x layer assignment for DDR signal routing. Note that Layer 1 and 4 are assigned to GND, and all DQs are routed on Layer 2 and 3 in order to maintain good impedance control of these lines. All CA/CK signals are routed on Layer 5. As these signals may reference either.

Layer 4 GND or Layer 6 Power it is important that the power shape on Layer 6 be solid. Routing is very tight in this region and care should be taken that broad side coupling is minimized. DDR signal Layer 1 is filled with GND for EMI shielding purpose, both Layer 2 and Layer 3 take Layer 1 and 4 as reference return GND, and Layer 2 and 3 follow the tandem layers spacing rule.

3.6.2 2-6-2 Reference Platform strategy

The 2-6-2 Reference Platform reference design is a dual-side C-shape board, since APQ8016E/eMCP/PM8916 has to be placed on the narrow 'I' area of C-shape, big vias must be

minimized as much as possible and DDR signals have to be routed as many as possible on Layer 2 with small vias after breakout. So the DDR layer assignment is S-S-G-S-P/G-G-x-x-x

Note there is no GND fill on the top layer. This is mainly because of impedance control. If layer 2 signals refer to layer 1 GND and layer 3 GND for return path, the impedance is less than 40 Ω which is out of the $45 \pm 10\%$. The same concern of possible crosstalk between tandem layers exist, so there are no tandem layers routing. PM8916 is in a separate shielding chamber from APQ8016E + LPDDR + eMMC.

3.6.3 2-4-2 Reference Platform strategy

The 2-4-2 Reference Platform reference design is a dual-side half board. Considering the difficulty to have a solid GND fill on Layer 1, the 2-4-2 Reference Platform chooses G-G-S-G-S-P/G-x layer assignment for DDR signal routing. All DQ signals are routed on Layer 3 and all DQS are on Layer 5, both Layer 3 and Layer 5 have solid reference return GND which permits the desired impedance. There are no tandem layers routing for DDR signals to avoid possible crosstalk between tandem layers. PM8916 is in the same shielding chamber as APQ8016E+LPDDR+eMMC. DVT RF test data show no any RF sensitivity issue related to DDR signals.

3.7 Comparisons of strategies

Table 3-5 Comparison of strategies

	Strategy 1		Strategy 2		Strategy 3	
Layer	3-4-3 Test Platform		2-6-2 533M Reference Platform		2-4-2 533M Reference Platform	
1	Breakout	DDR	Breakout	DDR	Breakout	DDR
2	GND	GND	Signal		Signal	No GND fill
3	Signal	DQ/DQS/DM	GND	GND	Signal	Most DQ/DQS/DM/CA
4	Signal	DQ/DQS/DM	Signal	DQ/CA/DM	GND	GND
5	GND	GND	GND	GND	Signal	CA/CK, main GND
6	Signal	CA/CLK/CKE/CS	Signal	DQS/CA/CK/DM	GND	VREL_L2/GND
7	Power	VREG_L2	PWR/GND	VREG_L2 /GND	PWR/GND	GND
8	Signal	Not DDR related	Signal	Not DDR related	Signal	Not DDR related
9	Signal	Not DDR related	Signal	Not DDR related		
10	Signal	Not DDR related	Signal	Not DDR related		

NOTE: The requirement of maximum length of DQ/DQS/CA/CK is not critical, but the length matching of DQ to DQS and CA to CK are critical.

Type of guidance	Guideline		Requirement	
			3-4-3 Test Platform strategy	2-4-2 Reference Platform strategy 2-6-2 Reference Platform strategy
General	CLK frequency		533 MHz	533 MHz
	Single-ended impedance		50~65 Ω	45 Ω $\pm$ 10%
	Differential impedance		90 Ω $\pm$ 10%	90 Ω $\pm$ 10%
	EBI to all other signal spacing		4x W	4xW
DQ , DQS, and CA	DQ to DQS matching DM to DQS matching CA to CK matching		2.5 mm	2.5 mm
	DQS to DQSB matching CK to CKB matching		< 0.5 mm	< 0.5 mm
	DQS to CK matching		TBD	TBD
	DQ to DQ spacing within byte DQ to DQS spacing within byte CA to CA spacing CK/CKB to all others spacing	Same layer:	2x W	1.5x W
		Tandem layer:	> 0.5x W	N/A
	Byte-to-byte spacing		2x W	1.5x W
	DQ max length DQS/DQSB maximum length		TBD TBD	TBD TBD
	CA maximum length CK/CKB maximum length		TBD TBD	TBD TBD



4 Power Management and Codec Routing

4.1 SMB1360 power management

4.1.1 SMB1360 component placement

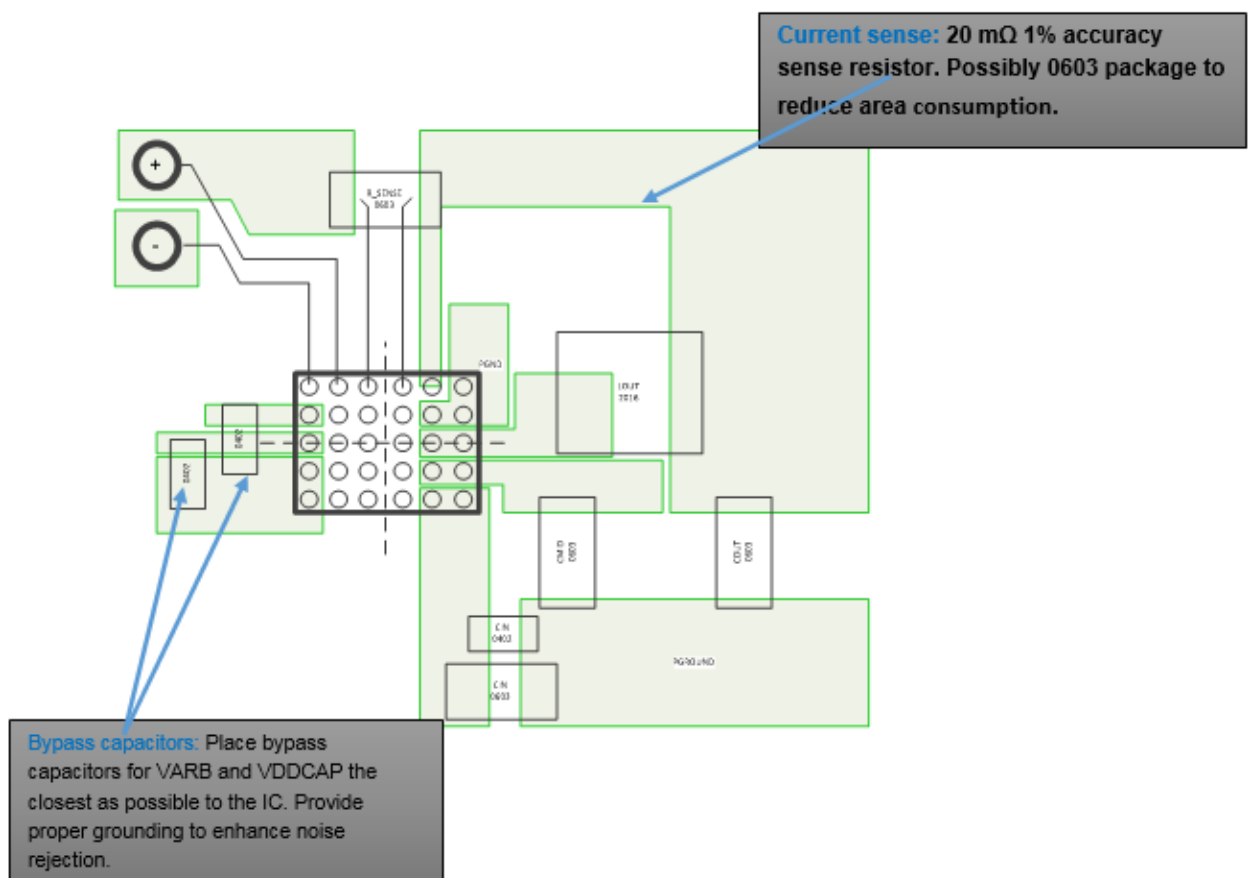


Figure 4-1 SMB1360 component placement

4.1.2 SM1360 layout (Optional for charging enable devices)

- Connection from USB power source to DCIN should be at least of 200 mils wide.
- Input bypass capacitors should be near IC, especially the HF filtering 0.1 μF capacitor.
- Bypass capacitor grounds should use larger than usual ground pads, with at least a couple of vias to connect to the inner ground layer.

Switching node

- Output inductor should be near the switching (SW) nodes (pins 17 and 18).
- Trace widths from the SW nodes to the output inductor should be at least 200 mils.

Power ground (PGND)

- Power ground plane is on an inner layer; for proper connection, the IC power ground must use big ground pads on the component layer, and then connect to the inner layer through an appropriate number of vias.
- The same surface ground pad should be used as the MID bypass capacitor ground connection.

Analog ground (AGND)

- AGND pins should be connected to inner ground through a properly sized ground pad and an appropriate amount of vias.
- The same ground pad should be used as VDDCAP and VARB bypass capacitors' ground.

Midpoint capacitors (MID)

- MID pins connect to power ground through a bypass capacitor; keep traces from the MID pins to the capacitor short, and at least 200 mils wide.

Battery current sensing (CS_P, CS_N)

- Place the sense resistor as close to the IC as possible.
- Route traces from CS_P and CS_N to their respective resistor nodes as a differential pair.
- Use 200 mil widths or larger to/from both sides of the current sense resistor.

Battery voltage sensing (BATT_P, BATT_N)

- The CS_P and CS_N guidelines generally apply to BATT_P and BATT_N.
- The battery negative node should connect to power ground near the PCB negative connector.

Other general layout recommendations

- Use as much copper as possible for power paths, including the input (DCIN), the switch node (SW), and GND. This will greatly improve thermal performance. Unused inputs can help. For example, if AC/USB5 is register controlled, then AC/USB5 (D4) is not used. The GND plane from the GND balls (D1, D2, and D3) can be extended by routing through D4 with solid copper.
- Use the thickest copper possible for the top layer to improve thermals.
- Ensure that the impedance from the GND balls to the GND plane is as small as possible by placing filled vias below each GND ball to connect the GND balls directly to the internal GND plane.
- For best current sensing, keep these impedances as low as possible:
 - From input capacitors to input balls
 - From MID capacitors to MID balls

4.2 PM8916 power management

4.2.1 PM8916 component placement

Figure 4-2 shows the top-level placement only; critical layout details are distributed throughout this document where needed. 2520 inductors and 0805 capacitors (above 10 mF) are used to keep the cost low.

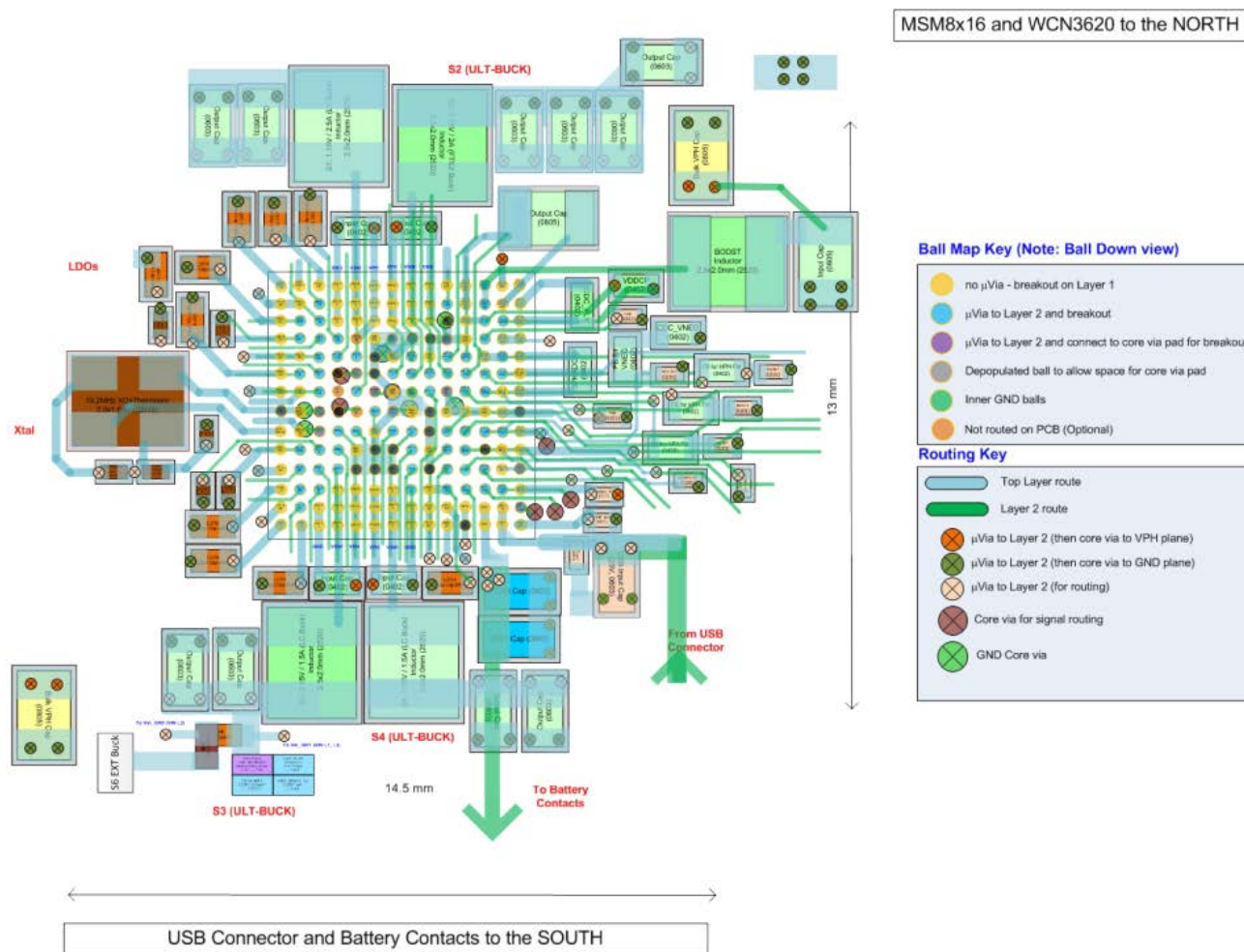


Figure 4-2 PM8916 component placement

4.2.2 PM8916 input power

- High-current paths: Use extremely wide traces/fill areas or sub-planes
- USB connector to pins N13 and P13 (USB_IN)
- Use thick traces or fill areas for all high-current paths – distributing VBAT to PMIC regulators, PA and any other external high current load

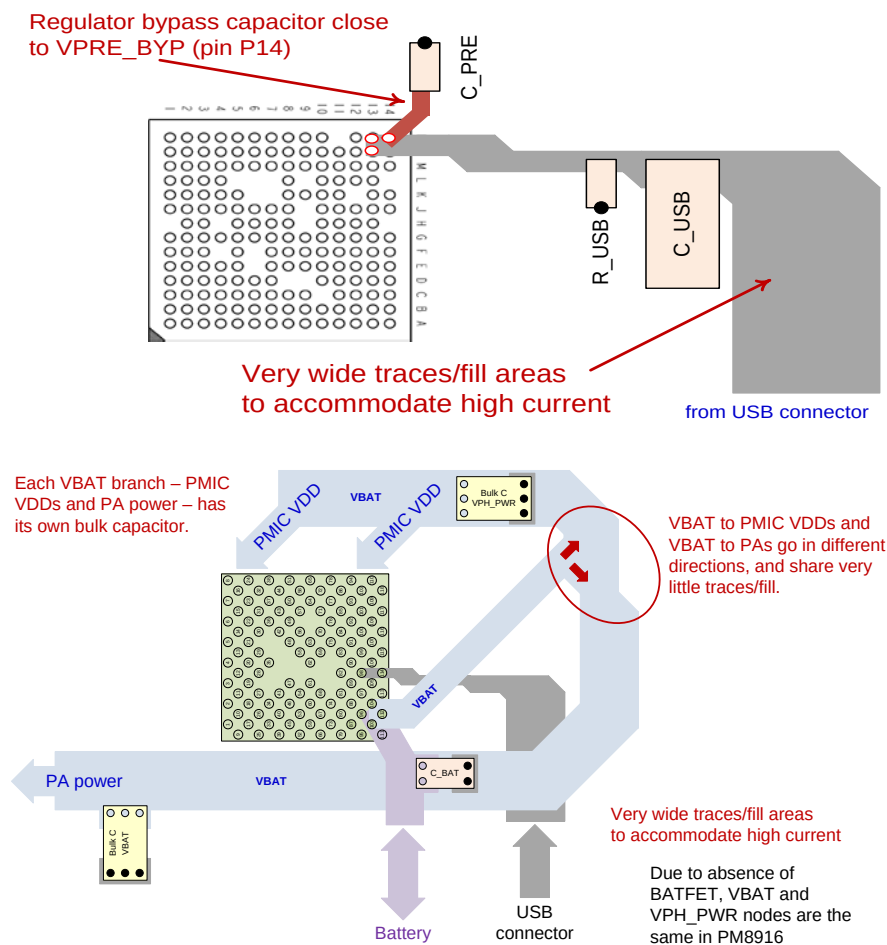


Figure 4-3 PM8916 input power

4.2.3 PM8916 other layout guidelines

SMPS

- Buck routes VSW_Sx (from PMIC pin to inductor) and GND_Sx (from PMIC pin to input capacitor, including ground vias) should be able to electrically and thermally handle expected peak currents.
- GND_Sx must connect directly to buck input capacitor, and then input capacitor should via directly to main ground. Input-capacitor loop inductance should be less than 5 nH.
- For local sensing: GND side of output capacitor should have a dedicated GND trace back to the input capacitor and GND_Sx. Do not add GND vias at the output capacitor.
- VREG_Sx (feedback) always connects to the output capacitor, regardless of local or remote sensing.

Remote sensing

- Place output capacitor close to APQ8016E load; it should via directly to main ground.

- Main ground plane should be continuous (not fragmented) between output capacitor and PMIC.

LDO

- Pseudo-capless LDOs do not need a local capacitor at the PMIC if a suitable capacitor is at the load. However, the maximum trace inductance depends on the capacitor value and peak current. If this inductance requirement cannot be met, a local capacitor is needed.

Charger

- Connection from USB connector to USB_IN pin should be wide and capable of handling 1.4 A current.

Clocks

- GND_XOADC must be routed out as a thin trace that connects to main ground at a point where temperature transients are not expected. If it cannot break out from under the PMIC, it should be connected to main ground using a dedicated via.
- Connect GND_XO to ground terminal of VREG_XO capacitor and then connect to main ground using dedicated via right underneath the capacitor. Do not connect to any other ground. Do not connect to XTAL ground.
- Connect GND_RF to ground terminal of VREG_RFCLK capacitor and then connect to main ground using dedicated via right underneath the capacitor. Do not connect to any other ground.
- XTAL should be placed 1 cm away from the PMIC, with in/out traces about 1 cm in length. If the desired placement is not possible, try to keep in/out traces about 1 cm long.
- Clear out all metal on one layer below the XTAL to minimize heat transfer to the XTAL.
- XTAL_IN and XTAL_OUT should be kept away from all noisy signals, especially 19.2 MHz clock output signals (to avoid frequency pulling). Ground shielding should be used wherever available.
- The dedicated clock LDO load capacitors should each connect directly to the main ground using dedicated vias. This applies to VREG_XO and VREG_RFCLK.
- When the PMIC's 19.2 MHz clock signals (BB_CLK1 and RF_CLK1) are branched during routing:
 - Make the common route a 50 Ω trace, and each individual segment a 100 Ohm trace.
 - Match the branch lengths; even a 5 mm difference can cause problems.
 - Keep branches as short as possible.
 - If these routing guidelines cannot be met, add a buffer at the BB_CLK1 branch point to USB.

General

- GND_REF and REF_BYP must connect directly to the capacitor using a thin trace. The capacitor should have a dedicated via directly to main ground, with no connections to intermediate ground fills.
- The AVDD_BYP capacitor should use a dedicated via directly to main ground.

Vibration motor

- The vibration motor's ground (GND_DRV) should connect to main ground.

4.3 PM8916 codec

4.3.1 Codec related capacitors and inductors – layout and placement

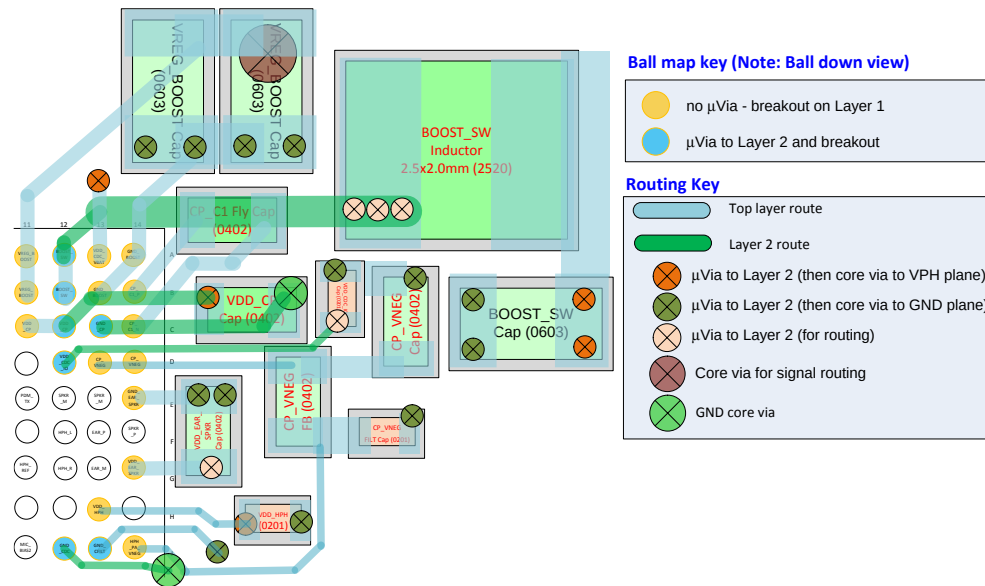


Figure 4-4 PM8916 codec related layout and placement (1)

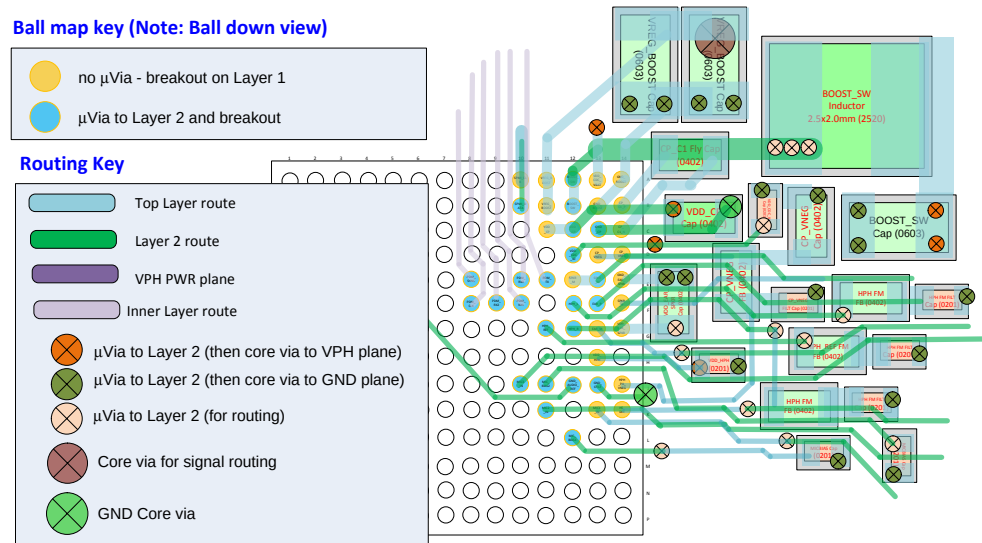


Figure 4-5 PM8916 codec related layout and placement (2)

Charge pump capacitors

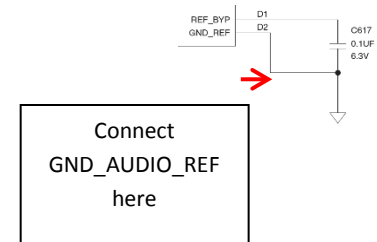
- Minimize trace inductance.
- Minimum number of via in bypass capacitor loop.
- Tie capacitors negative to GND_CP or directly to GND plane.

Speaker driver capacitors

- Place capacitor as close to VDD_EAR_SPKR pin as possible.
- Minimize loop inductance on VDD_EAR_SPKR and GND_EAR_SPKR trace.
 - Can use adjacent layers if they are directly on top of each other; otherwise, use top layer routing.
- Maximum two layers routing.
- Route GND_EAR_SPKR to capacitors negative.
- Tie capacitors directly to GND plane.

Boost capacitors and inductor

- Place capacitor and inductors as close to the corresponding pins as possible.
- Minimum number of via and trace inductance.
- Route GND_BOOST to capacitors negative.
- Tie capacitors negative directly to GND plane.



Codec ground GND_AUDIO_REF (GND_CDC)

- Must be connected to either:
 - GND_REF (PM8916 pin D2) or
 - The ground of the bandgap reference bypass capacitor connected to GND_REF.
- Trace from GND_AUDIO_REF to band gap reference ground carries minimum current. It is acceptable to use 5 mils, 1 inch trace. This trace needs to be isolated from interference.
- GND_AUDIO_REF must not be connected to other circuits.

Analog input

- 4 – 5 mil trace widths; 4 – 5 mil spacing between traces.
- Differential route for MIC1 and MIC3 with MIC GND; ground MIC GNDs in the same ground as GND_CFILT near the PMIC.
- Coplanar ground fill on both sides; Sandwiched between ground planes – grounds above and below.
- Isolate from noise sources, such as antenna, RF signals, EBI, SMPS, clocks, and other digital signals with fast transients.

Analog output

- Coplanar ground fill on both sides (of traces or pair as appropriate); Sandwiched between ground planes – grounds above and below.
- Isolate from noise sources such as antenna, RF signals, EBI, SMPS, clocks, and other digital signals with fast transients.
- EAR output signal – route as differential pair with 10 mil trace widths.
- SPKR output signals – route as differential pair with below trace widths:
 - $VDD_EAR_SPKR = V_{bat}$, 8 Ω load: 20 mils OR $VDD_EAR_SPKR = V_{bat}$, 4 Ω load: 25 mils.
 - $VDD_EAR_SPKR = 5\text{ V}$, 8 Ω load: 25 mils OR $VDD_EAR_SPKR = 5\text{ V}$, 4 Ω load: 30 mils.
- HPH output signals – not a differential pair; 10 mil trace widths for HPH_L and HPH_R; 15 mil trace widths for HPH_REF.
- Connect HPH_REF to the ground pin of the jack connector and route HPH_REF in between HPH_L and HPH_R for best crosstalk minimization.

4.3.2 PDM interface

- Digital signals that carry audio clock, sync, and Tx and Rx data between the APQ8016E codec digital and PM8916 codec analog.
- Signals switching at 9.6 MHz or 12.288 MHz, depending on the codec clock rate.
- Codec PDM signal routing guidelines:
 - Do not add any test point or stub to these signal traces.
 - Route these signals with less than 5 mm length mismatch and with matching impedance (e.g., 50 Ω) on an inner layer shielded between the two ground layers.
 - Isolate these signals from all sensitive traces, such as RF, BT, WLAN, power, etc.

5 Wireless and Connectivity Routing

PA 3 V and 1.3 V should be wide enough (especially the Wi-Fi PA V_{DD} trace) to withstand 1 A current. It should have adequate ground via under the ground pad. All bypass capacitor ground pads should be connected to ground with a blind via and a buried via. The traces and vias of bypass capacitors should be far from the Tx trace. Keep all other traces away from the CLK trace. The power trace must be placed far from clocks, e.g., 19.2 MHz, RTC, SDIO clock, PCM clock, etc.

5.1 Signal layer recommendation for 2-N-2 stack-up

- Layer 1 – Parts placement; components, RF microstrip, traces to pins on the outer rows
- Layer 2 – WCN RF ground; WCN ground pins and bypass capacitor grounds
- Layer 3 – WCN signals; digital I/Os, analog baseband, power buses, TCXO, bypass capacitors to WCN pins on inner rows
- Layer 4 – Main PCB ground plane

5.2 WCN3620 placement



Figure 5-1 WCN3620 placement

- RF matching components close to WCN pins

- Filter matching components near the filter (**critical**)
- Low-value VDD bypass capacitors close to their WCN pins (though RF matching components have higher priority)
- WCN and its discrete components in a dedicated shield area
- External coupler output (if used) must be isolated from the 2.4 GHz trace

5.3 WCN3620 layout

5.3.1 Keep-out area

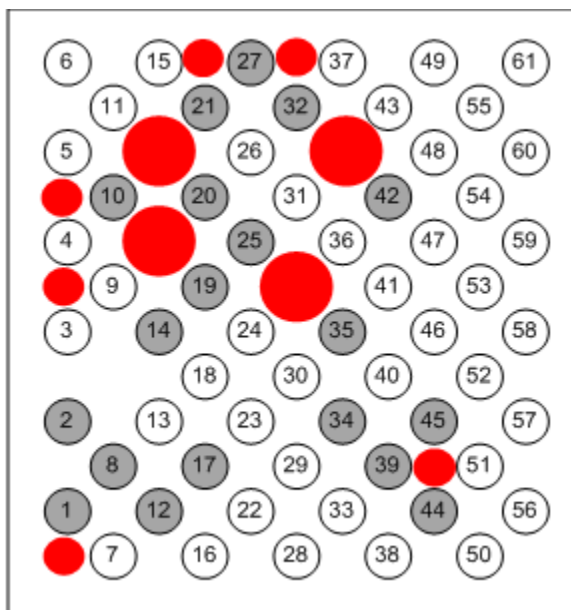


Figure 5-2 WCN3620 layer 1 mandatory keep-out areas

5.3.2 Power

5.3.2.1 3.3 V power supply

- The total resistance between PMIC and the WCN pins must be less than 100 mΩ (**critical**).
- Place C1 away from pin 5 (WL_BT_RFIO), preferably close to the C4 (**critical**).
- Star route 3.3 V VDD traces from the shared capacitor to pin 4 (VDD_BT_DA_3P3) and pin11 (VDD_WL_2GPA_3P3) with a minimum routing distance between pins of > 5 mm (**critical**).
- Instability in the 2.4 GHz Tx output may occur without this recommended star routing.

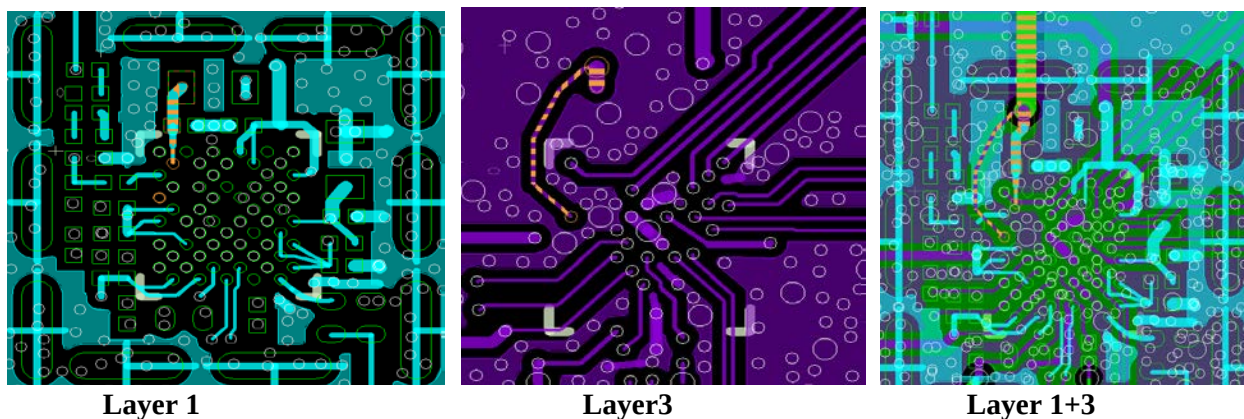


Figure 5-3 Power supply examples

5.3.2.2 1.3 V power supply

- The total resistance from PMIC to the WCN pins must be less than 100 m Ω (**critical**).
- It is recommended to have minimum 20 mil trace width with shortest length between PMIC to WCN.

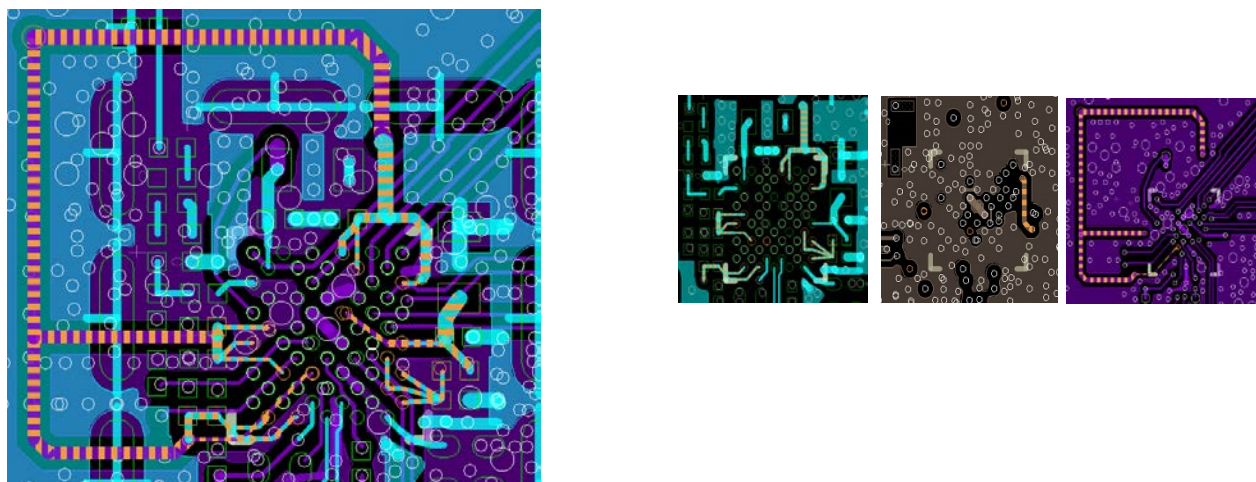


Figure 5-4 1.3 V power supply

5.3.2.3 Power distribution routing

- 3.3 V and 1.3 V high current traces should maintain a length-to-width ratio of less than 10 to maintain a maximum 0.1 Ω IR drop from the PMIC to the WCN pins. (**extremely critical**).
 - Failure to maintain this ratio can cause Tx EVM degradation.
- Route supply voltage rails from the PMIC to the inner layers.
- Keep all supply traces away from the RF pin 5 (WL_BT_RFIO) and RF traces.
- Isolate the 1.3 V and 3.3 V traces from each other; do not route in parallel.
- Do not run supply traces from one side of the WCN3620 through the WCN3620 to the other side; it is recommended to run the supply trace around the WCN3620 and then have a short trace from the outside directly to the WCN3620 supply pins.

- Route to bypass capacitors first and then continue route to the WCN pin.
- It is highly recommended to have good isolation between the RFIO and 3.3 V WLAN PA supply (**extremely critical**).

Good isolation between RFIO and 3.3 V WLAN PA supply

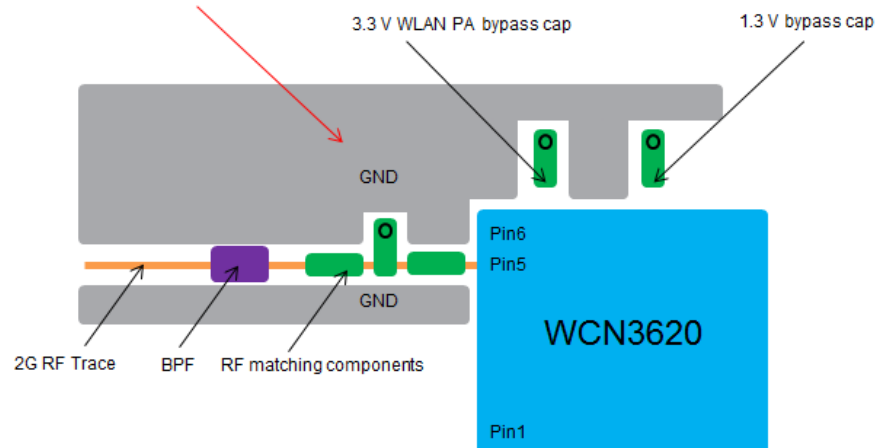


Figure 5-5 Isolation between RFIO and 3.3 V power supply

5.3.2.4 Ground

- Pin 25 and pin 35 are the analog ground pins and they need to be isolated from the other digital ground pins.
- Follow the mandatory keep-out areas on layer 1 under the IC.
- There should be no layer 1 ground pour below the IC.
- Provide a solid, continuous ground flood (WCN RF ground) on layer 2 below the IC.
- Connect the IC ground pins and bypass capacitors' ground pads directly to the WCN RF ground on layer 2 using micro-vias at each pin or pad (**critical**).
- Using the lower layer (main PCB ground plane) for ground return increases the loop inductance and might make bypassing less effective.

5.3.3 Signal

5.3.3.1 19.2 MHz clock

- Route the 19.2 MHz system clock with isolated inner-layer traces all the way from PMIC to WCN pin (**critical**).
- Note: PMIC 19.2 MHz clock is the only clock source for WCN3620 and this signal needs to be well isolated/protected).
- Keep clock traces away from any supply, I/Q, and RF traces.
- Keep 10 mil keep out from GND copper pour.

- 19.2 MHz clock trace should be routed away from FM RF trace to avoid FM desense at 96 MHz.

5.3.3.2 Analog baseband I/Q signals

- Add GND with GND vias between two signal pairs every 75 mil from WCN to APQ8016E.
- Keep the I/Q trace pairs equal length, symmetric, and well isolated. Maintain equal lengths for all signals within 40 mil.
- The resistance and capacitance on each pair should be equal; total capacitance should be < 10 pF.
- Crosstalk should be < 60 dB at 50 MHz.
- Keep the I/Q signals away from the RF routing area, high-speed digital, and clock signals.
- Use the same number of vias for each set of differential lines.

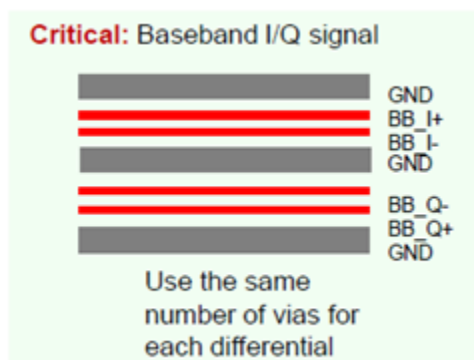


Figure 5-6 Baseband I/Q signal

5.3.3.3 High-speed digital signal

- Keep the high-speed digital signals of equal length or within 100 mil.
- Keep the high-speed digital signals away from any supply, I/Q, and RF traces.
- Keep 10 mil keep out from the GND copper pour.

5.3.3.4 Bandpass filter

- Requires GND cutout on layer 2 if ACPF9002 is used.

5.3.3.5 Wi-Fi antenna routing area

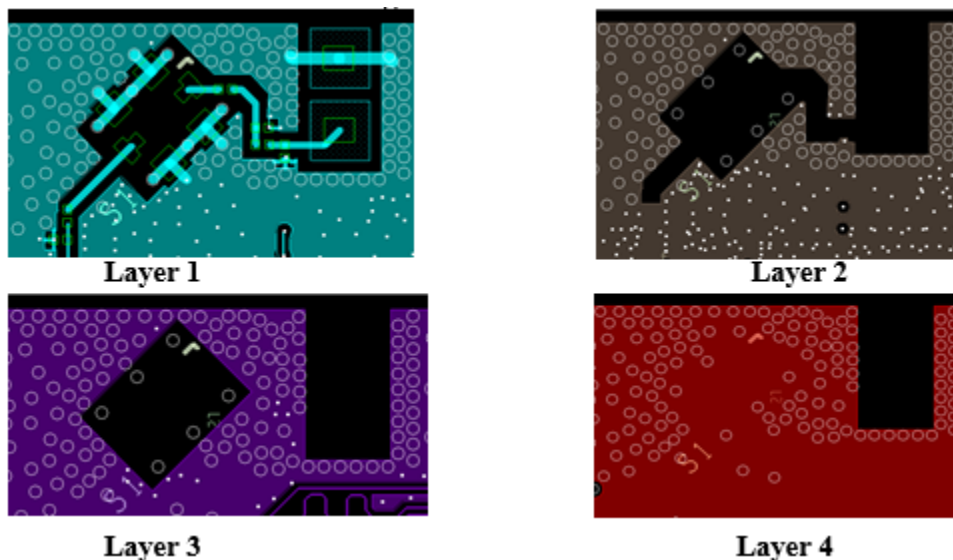


Figure 5-7 Wi-Fi antenna routing areas

5.3.3.6 WLAN/BT RF trace

- Keep the RF trace at 50 Ω (critical).
- Use layer 2 as the reference ground from the WCN to the RF connector.
- Use layer 3 as the reference ground from the RF connector to the antenna.

5.3.3.7 FM RF trace

- The FM matching/WAN filtering components should be placed close to the chip.
- Refer to *WCN36x0 FM Design Guidelines* (80-WL300-7) for FM general guideline.
- For a Murata connector, there should be no ground fill layers 1, 2, and 3.

5.4 FM Design

The FM design includes the standalone analog WCN FM, APQ8016E baseband, PMIC, display, and peripherals. The FM band operates close to the clock and signal data frequencies of other technologies; therefore, the concurrency impact on FM is higher than with other radios, which operate at higher frequencies.

As the analog FM radio operates concurrently with other digital subsystems, they will be the source of both conducted and radiated noise in the FM band. Good FM radiated performance is determined not only by FM circuit design, but also by good integration of the entire chipset. This document helps address these dependencies.

5.4.1 Challenges of FM antenna design and radiated performance

Typically, the higher-frequency RF antennas within smart phones are smaller and more efficient, and are also easier to match to 50 Ω because of their shorter wavelength. However, FM radios must have the ability to use any compatible *headset cable* as an antenna. Headset cables are neither efficient as an antenna, nor a good 50 Ω match. This is because the wavelength at FM frequencies (76 – 108 MHz) is too long to design a small, efficient antenna. A quarter-wave monopole antenna, for example, has dimensions of 3/4 meter at 100 MHz.

The four wires in the headset (audio left/right, microphone, and ground) may have different electrical terminations, so there is no FM standard for headsets. FM radiated performance may vary with different headsets as a result; therefore, a good consistent headset must be used to evaluate FM radiated performance.

Note that the headset audio ground wire is widely used as the FM antenna. However, FM on audio left or right could also be used as the antenna, instead of the ground wire. Reference schematics are provided for both types later in this document.

Good FM conducted performance does not guarantee good radiated performance, but is still essential for validation.

CAUTION: Good layout placement, routing, and shielding guidelines must be followed for all digital components that can be noise or spur aggressors for FM.

Maintaining low-inductance ground return paths for all other chips and peripherals requires careful attention to PCB layout. This prevents noise and spur harmonics in the FM band, and desensitizing the FM radio. Aggressor examples include harmonics of PMIC switching frequency at 1.6/3.2 MHz and 9.6 MHz, codec clocks of 9.6 MHz or 12.288 MHz, and system clock of 19.2 MHz.

5.5 WCN36x0 Guidelines

Use the correct keep out areas for FM RF trace and RF components:

- Connecting the ground pads of L2 and C64 to the clean ground flood at the inner layer through vias is recommended.
- Place these components close to the WCN36x0(A) FM_RX_HS pin (WCN3660(A)/WCN3680 is used as an example in the diagram).

The FM 1.3 V power supply uses star routing:

- Use star routing from C17 to VDD_FM_PLL_1P3 and VDD_FM_VCO_1P3, and from C33 to VDD_FM_RXBB_1P3 and VDD_FM_RXFE_1P3, respectively.
- Connect C17 and C33 ground pads to the clean ground flood at the inner layer through vias.
- FM ground pins should follow the WCN ground-pin routing requirement; they must be connected to the main reference ground directly through the dedicated vias. Do not connect these pins to the noisy ground flood.

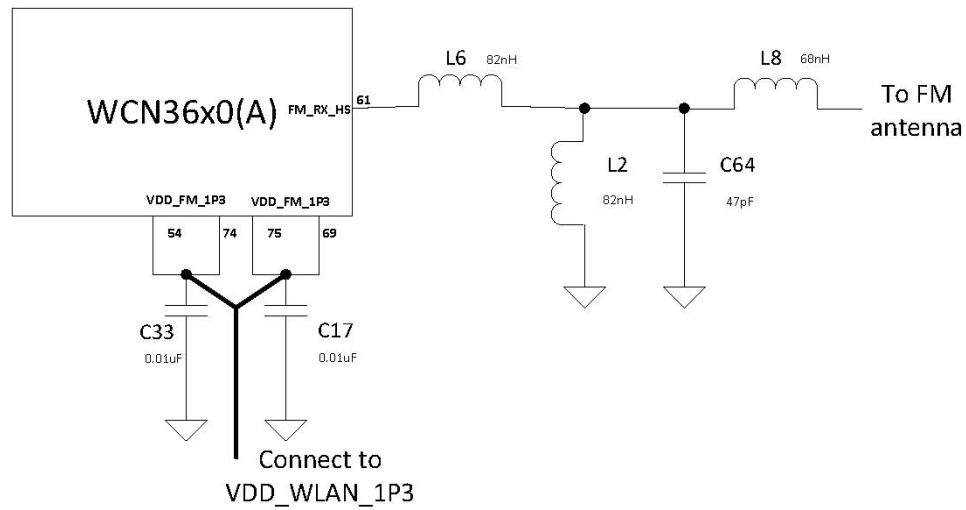


Figure 5-8 FM design

m5 freq= 76.00MHz S(5,5)=0.492 / 31.822 impedance = $Z_0 * (1.867 + j1.278)$	m6 freq= 106.0MHz S(5,5)=0.748 / 55.410 impedance = $Z_0 * (0.620 + j1.734)$
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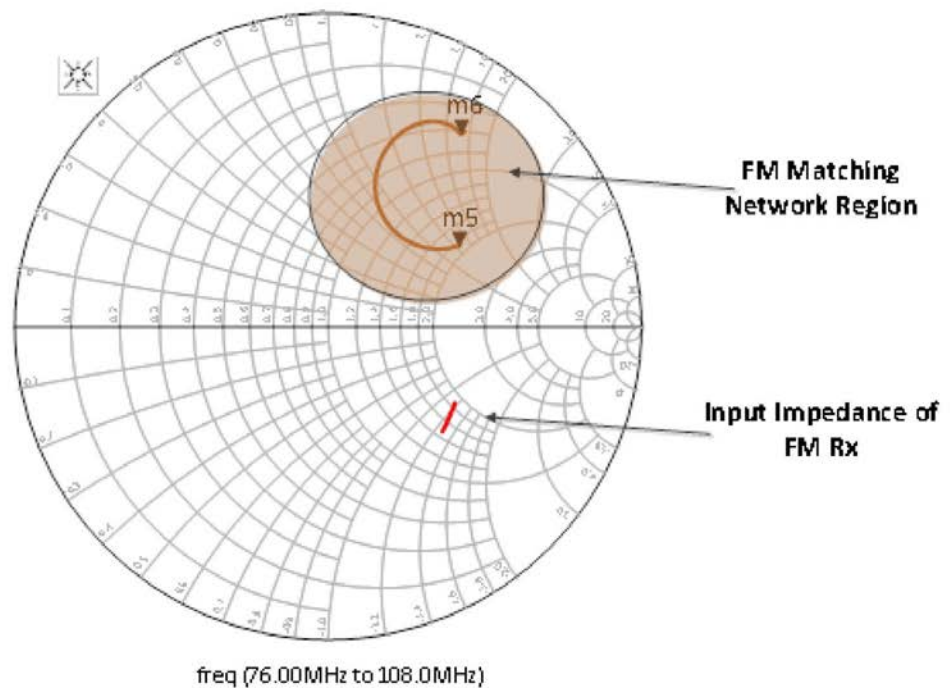


Figure 5-9 Target impedance when looking from chip LNA input toward FM ANT

5.6 Connectivity Check Lists

5.6.1 PMIC Checklist

Well-grounded shield walls are critical. The PMIC must be well shielded with good connections between shield walls and board ground. Use the following guidelines:

PMIC bypass capacitor:

- The PMIC buck supplies can radiate switching harmonics of 3.2 MHz and 1.6 MHz into the FM band. This requires good bypass capacitance filtering and ground return.
- Add a high-frequency bypass capacitor to PMIC buck supplies that are physically close to the WCN36x0 and FM headset connector. This capacitor reduces the buck-supplied harmonic levels in the FM band.

The PMIC buck supply ground must have local GND and main GND:

- Incorrect PMIC GND and lack of shielding will cause radiated and conducted interference to the FM and, potentially, to other technologies.
- The PMIC is platform dependent. See the relevant PMIC design documents.

5.6.2 PMIC Buck Supply Ground

The PMIC buck supply ground must have both local GND and main GND. Refer to [Figure 5-10](#).

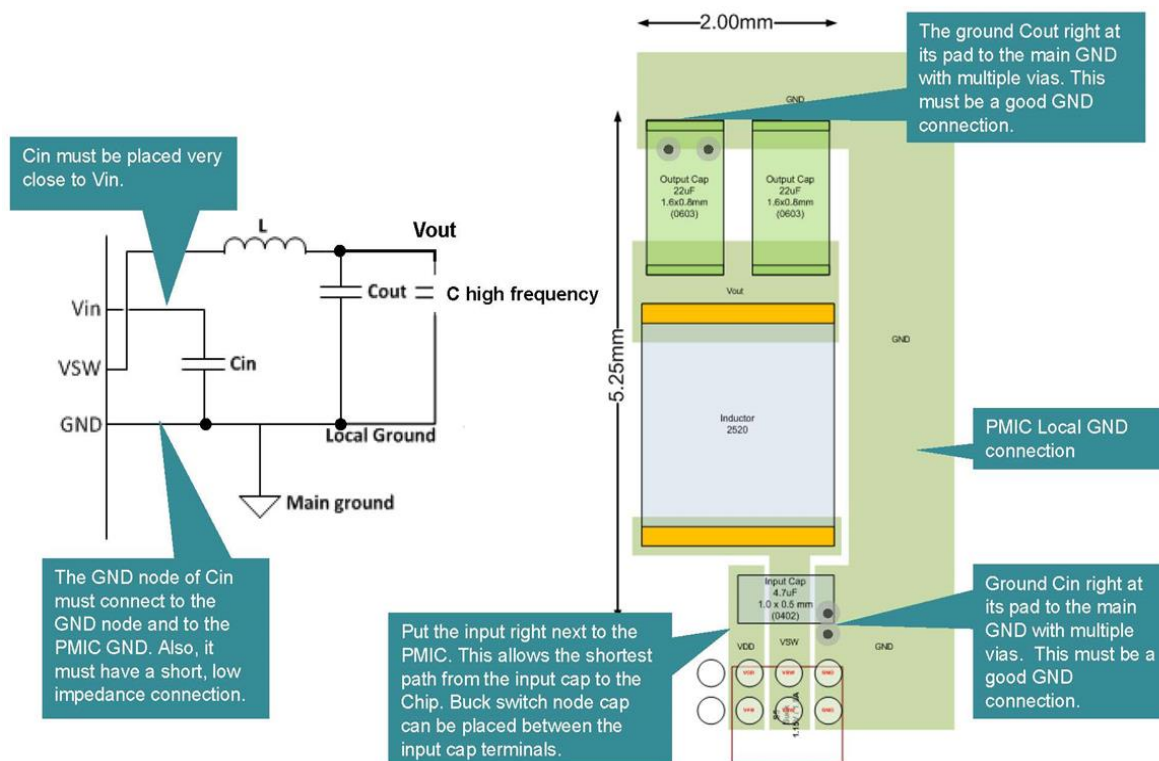


Figure 5-10 PMIC buck ground supply

5.6.3 Codec Checklist

Use the following guidelines for the codec:

- Place series ferrite beads (471 Ω @ 100 MHz) for AUDIO_L/R, and 0 Ω for HPH_REF close to the codec.
- Place series ferrite beads (1000 Ω @ 100 MHz) and shunt capacitors (470 pF) for AUDIO_L/R and HPH_REF close to the headset jack.
- Connect all the shunt components to the dedicated **clean** and **quiet** audio jack ground (layer 2 is recommended), and connect the dedicated audio jack ground to the main reference with low impedance (0.1 Ω or less).
- Use AUDIO_L/R and HPH_REF traces for good isolation, all the way from the codec to the headset.

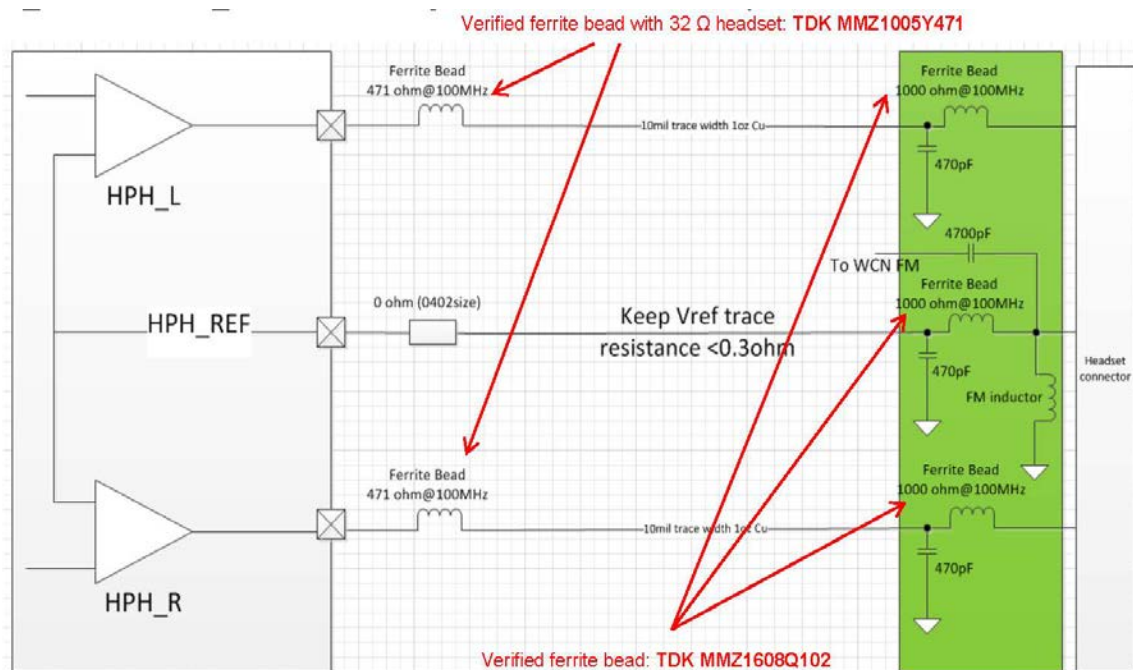


Figure 5-11 Codec design

NOTE: Ferrite bead selection should meet the FM and audio THD requirements.

- Use an RC filter for codec MCLK, if MCLK is from baseband (APQ8016E) or PMIC.
- Include an RC filter placeholder option, in case harmonics of the MCLK need to be filtered.
- Use the AUDIO_MCLK trace for good isolation all the way from the baseband/PMIC to codec.
- Place an RC filter close to the MCLK source (baseband or PMIC), and keep it under the baseband or PMIC shield.

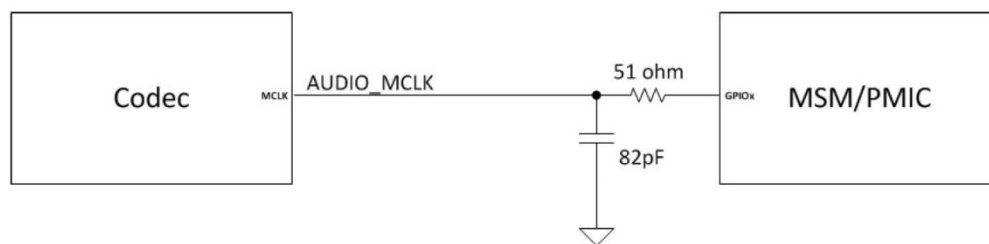


Figure 5-12 Codec RC filter

5.6.3.1 Codec charge-pump

The codec charge-pump design is **critical**.

- To prevent charge pump noise from propagating into the system, route the ground pad of the charge-pump bypass capacitor into the ground flood island through the dedicated via(s) before connecting to main reference GND. Use the dedicated via(s) to connect the charge-pump ground flood island to the main reference ground.
- Use the correct keep out area for the charge-pump transfer capacitor and its traces connecting to the codec chip.

5.6.3.2 Codec buck regulator

Use the following guidelines for the code buck regulator:

- The election of the components for the buck regulator of the codec must follow the recommendation in the corresponding codec reference design and codec design guidelines.
- The ground pads of all shunt components for the buck regulator must be connected to the island in the inner layer to isolate them from the other clean ground flood. Use the dedicated via(s) to connect the island to the main reference ground.
- To avoid noise issues from the codec charge pump and buck regulator, the corresponding codec layout guidelines must be followed.

5.6.3.3 Codec Checklist Summary

5.6.3.3.1 Traces from codec to headset

Use the following guidelines for the codec to headset traces:

- Place the 0 Ω resistor on HPH_REF right as close as possible to the codec HPH_REF pin.
- Place the ferrite beads (600 Ω @100 MHz) on AUDIO_L/R right as close as possible to the codec HPH_LP/RM pin.
- Isolate the AUDIO_L/R and HPH_REF traces from noisy signals, all the way from codec to the headset.
- AUDIO_L/R traces width must be 10 mil.
- The HPH_REF trace must be 20 mil.
- Ideally, the AUDIO_L/R needs to be less than 300 m Ω , inclusive of the ferrite bead for maximum audio volume. However, to improve FM antenna isolation, the ferrite bead impedance can be increased. This slightly reduces the maximum audio volume. Ferrite beads with a DC resistance of 0.1 Ω (for better maximum volume) to 0.4 Ω (for better FM isolation, [1000 Ω @ 100 MHz]) are acceptable.

5.6.3.3.2 Codec ground

Use the following guidelines for codec ground:

- Isolate codec analog GND and digital/charge-pump GND.
- Digital/charge-pump GND must be routed into the codec chip ground flood island before connecting to the main reference GND.

- Strongly connect codec chip ground flood to the main PCB ground, with multiple vias.
- Isolate the codec buck regulator GND.

5.6.3.3.3 Charge-pump capacitor recommendation

Charge-pump transfer capacitor: The requirement of the codec charge-pump transfer capacitor varies for different codec chips, and must follow the recommendations in the corresponding reference design and codec design guidelines.

Charge-pump voltage-output bypass capacitor(s): The requirement of the codec charge-pump voltage-output bypass capacitor varies for different codec chips, and must follow the recommendations in the corresponding reference design and codec design guidelines.

5.6.3.3.4 Noisy codec signals

Use the following guidelines to address noisy codec signals:

- I2S (I2S_Tx_sck and I2S_Rx_Sck), SLIMbus clock, and MCLK traces should be well isolated by routing on the stripline.
- The RC filter placeholder option must be included for the MCLK, if it is from the baseband (APQ8016E) device or the PMIC.
- Connect the ground of the loudspeaker and its shunt components to the ground flood island before connecting them to the main reference ground. Use the dedicated via(s) to connect the speaker ground flood island to the main reference ground.

Codec ground design example

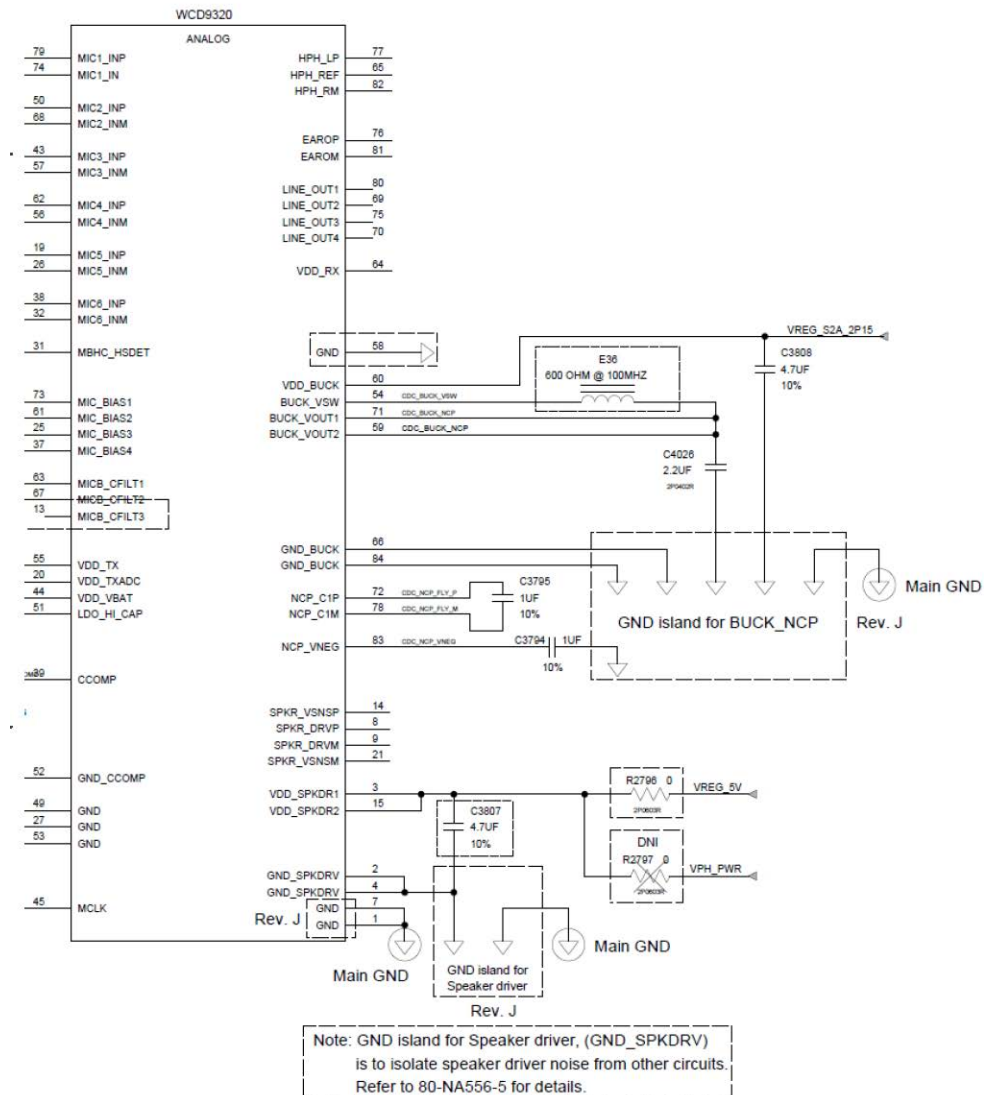


Figure 5-13 Codec ground design example

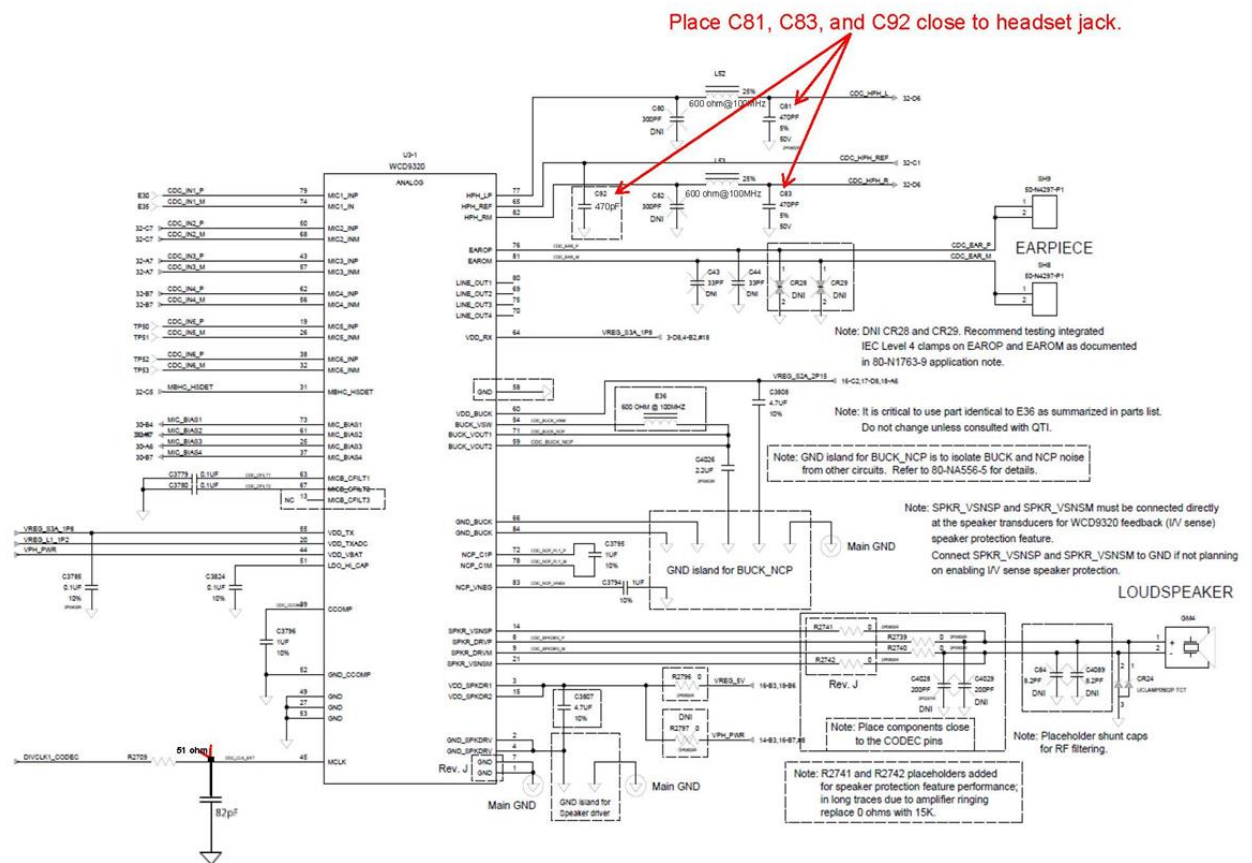


Figure 5-14 Codec Capacitor placement

5.6.4 Headset Jack Checklist

5.6.4.1 FM/audio inductor

Use the following guidelines for the FM/audio inductor:

- Refer to the next page for the FM/audio inductor target mini-specification.
- Ensure FM/audio inductor is well connected to main PCB ground with multiple vias, and is isolated from any possible noisy ground flood on the same layer.
- Do not connect FM/audio GND with codec digital/charge-pump ground before entering main reference ground.
- Place FM/audio inductor close to headset jack; **this is critical**.
- Isolate the return ground paths for the codec and the FM antenna.

5.6.4.2 Audio L/R and HPH REF

Use the following guidelines for the audio L/R and HPH REF:

- Place ferrite beads at AUDIO_L/R and HPH_REF close to headset jack; **this is critical.**

- It is recommended to connect shunt components (capacitors and ESD diodes) to the clean GND flood at the inner layer.
- Route AUDIO_L/R and HPH_REF in isolation all the way from the headset to codec.

5.6.4.3 Headset microphone

Use the following guidelines for the headset microphone:

- Place ferrite beads at the headset microphone, close to the headset jack.
- Try to use the minimum shunt capacitor at the headset microphone to reduce FM LNA load.
- It is recommended to connect shunt components (capacitors and ESD diode) to the clean GND flood at inner layer.
- It is recommended to route the headset microphone trace in isolation all the way from the headset jack to codec.

5.6.4.4 Headset detection

Use the following guidelines for headset detection:

- Place ferrite beads at the headset detection close to the headset jack.
- It is recommended to route the headset detection trace in isolation all the way from the headset jack to codec.

NOTE: Europe swaps the position of the audio ground pin and mic pin in the headset used by other countries.

5.6.5 FM/Audio Inductor Target Mini-Specification

The PCB design and headset type have a dependency on the FM choke inductor value. This inductor value must be optimized to get the maximum FM antenna gain through the headset.

See the following sections for headset schematic examples.

FM choke inductor (headset ground antenna) mini-specification example:

- 470 nH $\pm 10\%$
- 0402/0603 size
- Shielded: Shielding prevents FM receiver noise pickup and spurs when compared to a wire-wound inductor.
- 200 mA current rating
- DCR = 150 m Ω
- Q = 15 m Ω

FM choke inductor (headset audio antenna) mini-specification example

- 120 nH $\pm 10\%$
- 0402/0603 size

- Shielded: Shielding prevents FM receiver noise pickup and spurs when compared to a wire-wound inductor.
- 200 mA current rating
- DCR = 150 mΩ
- Q = 15 mΩ

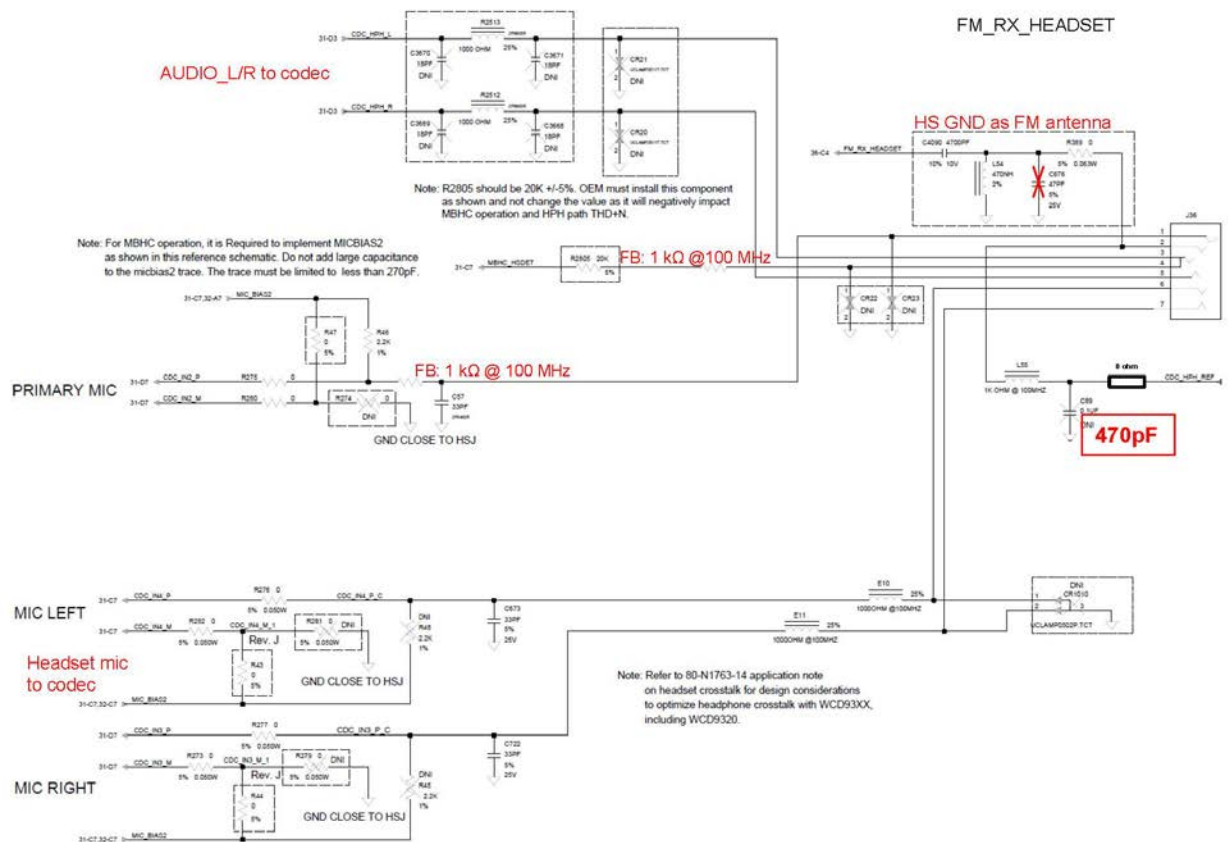


Figure 5-15 Headset design

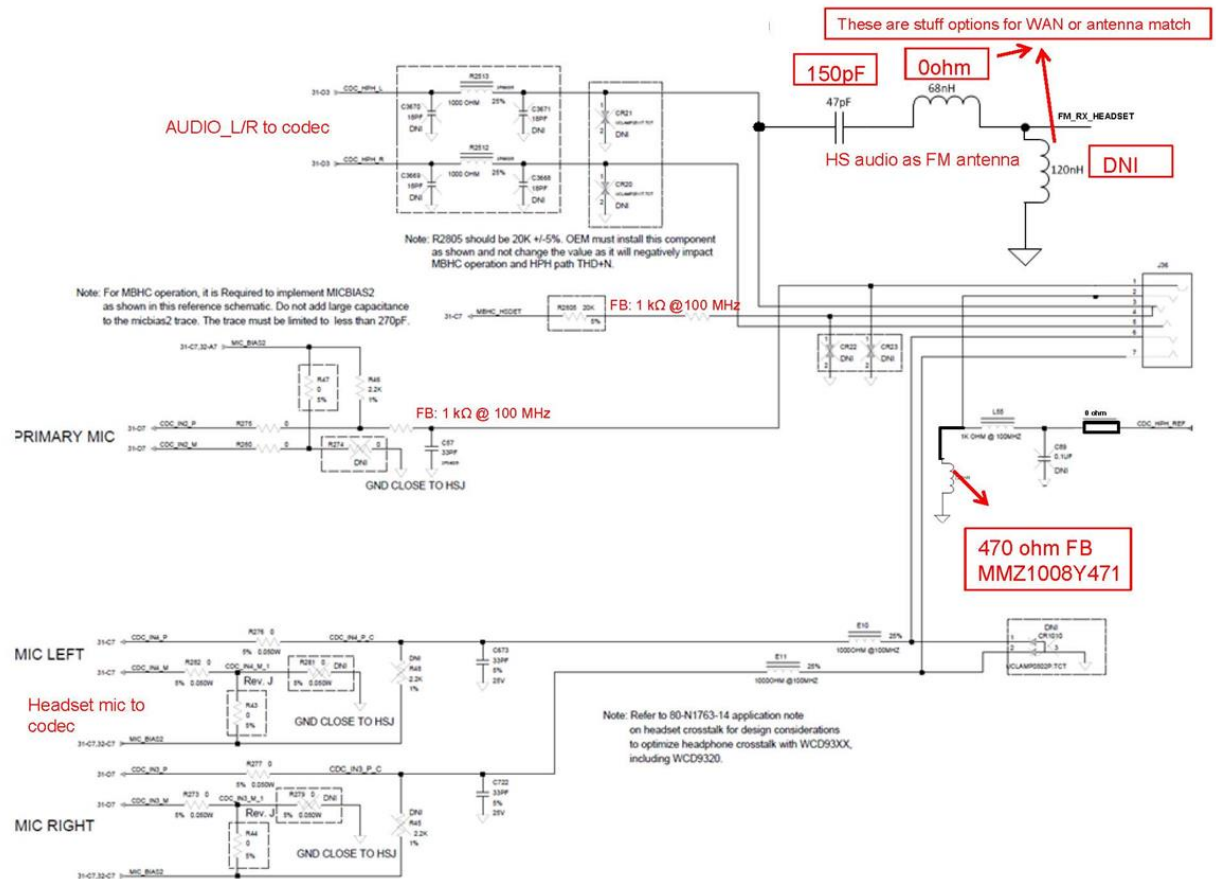


Figure 5-16 Headset design options for WAN or antenna match

5.6.6 Third-Party Loudspeaker Amplifier Checklist

Use the following guidelines for the third-party loudspeaker amplifier:

General

- Follow all of the vendor's recommendations (bypass cap value, component placement, etc.).
- Shielding must fully cover the loudspeaker amplifier.

Filtering

- Add a ferrite bead Pi network on the loudspeaker output, and place it right next to the pins.
- Add shunt capacitors at the speaker output, and place them close to the speaker.

Line-out traces

- Line-out traces to the loudspeaker amplifier can be long trace, and can even be on a flex cable; however, class-D amplifier outputs must be close to the loudspeaker – **this is critical**.
- Class-D trace outputs must be 25 mil.
- Route the speaker lines as a pair, and isolate from other traces.

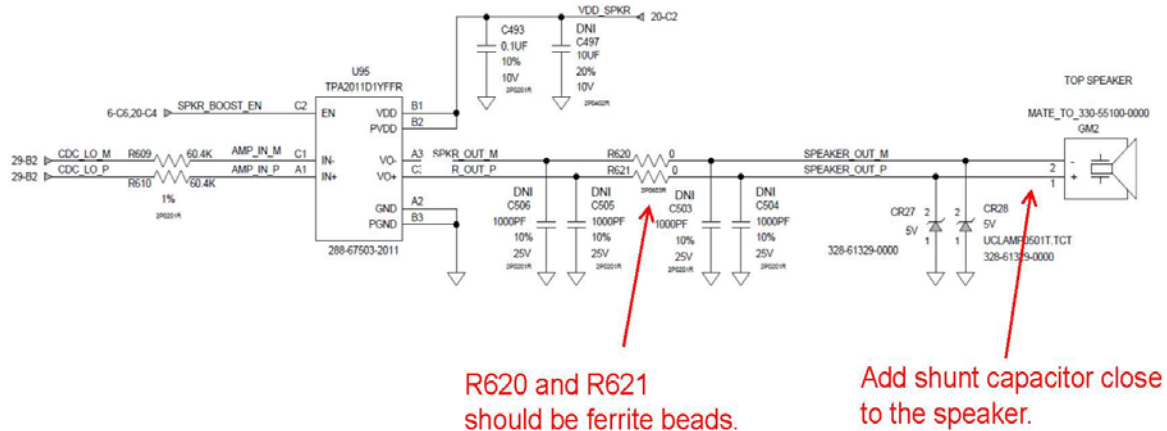


Figure 5-17 Loudspeaker design

5.6.7 Display Noise in FM Band Causing FM Concurrency Desense

Use the following guidelines to address display noise in the FM band:

Root cause

- Different displays have different levels of FM band electromagnetic interference (EMI).

Impact

- The FM radio misses weak stations during station search.
- Audio quality of weak stations is poor.

Display noise source:

Display usually contributes wideband noise and/or spur through:

Radiation from front panel and display flexible cable

Conducted coupling from MIPI interface

To identify noise/spur is from display, check if noise floor and spur reduce by turning off display. The noisiest circuits include:

- Unshielded flex cables
- Touch screen controller (TSC)
- Display controller circuitry

Design suggestions:

- Good flex and controller circuitry shielding are essential.
- Grounding of display and display component shields to the board is mandatory.
- Sensitive radio circuits, such as FM chip and audio jack, must be placed away from display aggressor components like flex, touch screen, display controllers, and MIPI interface.
- Good layout and shield for the *high-voltage display switcher supply* are mandatory.

5.6.8 Placement Recommendations

The following placement guidelines are recommended:

- Place the WCN device at the opposite PCB side of the PMIC and codec, as far away from them as possible.
- Place the headset jack at the opposite PCB side of the PMIC and codec, as far away from them as possible.
- Place the ferrite beads as close to the source or sink of the noise to limit interference.
- Place the bypass capacitors close to the source or sink of the noise to limit interference.
- Place the MCLK RF filter close to the source (APQ8016E/PMIC).

5.6.8.1 Shielding

Use the following guidelines for shielding the PMIC and the WTR/RTR must be shielded.

Shielding requirements include the following:

- Well-grounded shield walls
- No gap between the shield wall and PCB (that is, solder down the shield walls to the PCB)
- No gaps in the shield around the corners
- No holes on top of the shield
- No holes in the metal cavity of the phone, since it functions as a shield.
- **Basically, no holes or gaps in the shield, especially over the PMIC and the codec.**

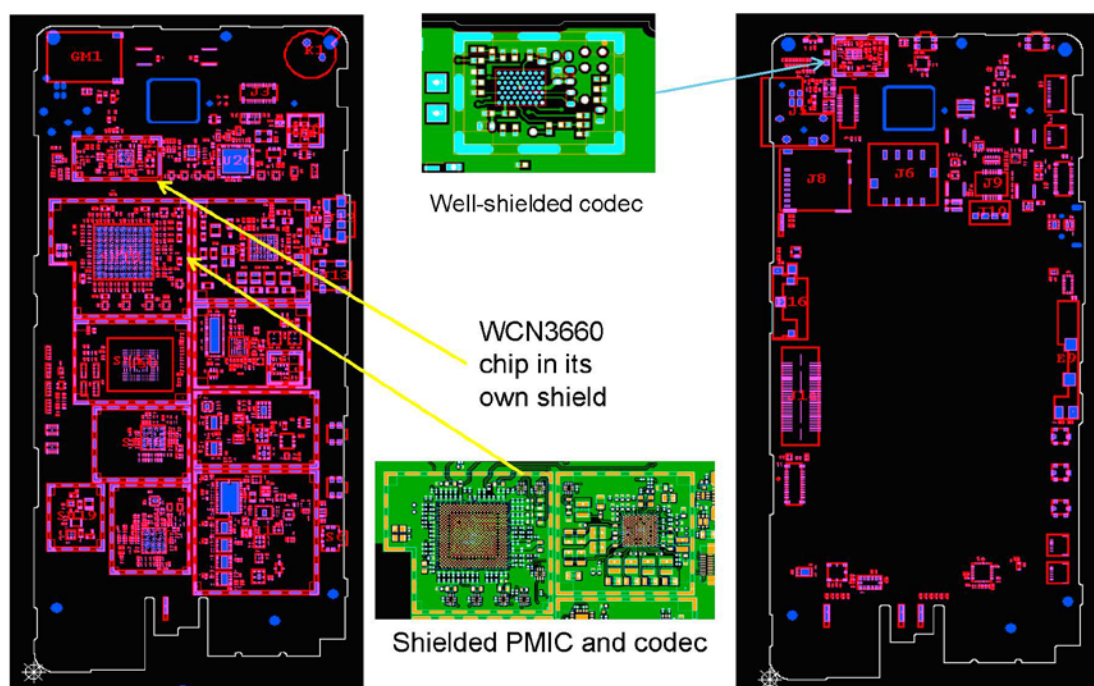


Figure 5-18 Shielding example

6 GPSA

The GPS guidelines are:

- The GPS LNA needs a separate shielding can.
- The GPS LNA input and output need to add the saw filter for out band isolation.
- The input saw filter need to be placed as close as possible to the GPS LNA; the output saw filter need to be placed as close as possible to the WGR GPS pin and matching network.
- The GPS RF trace must be placed away from clock XO and RF Tx traces.
- The power trace must be placed far away from clocks, e.g., 19.2 MHz, RTC, DDR clock, camera clock, etc.
- The GPS trace must have good isolation. There must be enough space and use fill for isolation.
- The input saw filter and the GPS LNA must be placed close to the GPS antenna.
- The GPS LNA and saw filter ground pads must have a good connection with ground and should be placed using a rough ground via, if possible.

EXHIBIT 1

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1.2 **Indemnification.** You agree to indemnify and hold harmless QTI and its officers, directors, employees and successors and assigns against any and all third party claims, demands, causes of action, losses, liabilities, damages, costs and expenses, incurred by QTI (including but not limited to costs of defense, investigation and reasonable attorney's fees) arising out of, resulting from or related to: (i) any breach of this Agreement by You; and (ii) your acts, omissions, products and services. If requested by QTI, You agree to defend QTI in connection with any third party claims, demands, or causes of action resulting from, arising out of or in connection with any of the foregoing.

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1.5 **LIMITATION OF LIABILITY.** IN NO EVENT SHALL QTI, QTI'S AFFILIATES OR ITS LICENSORS BE LIABLE TO YOU FOR ANY INCIDENTAL, CONSEQUENTIAL OR SPECIAL DAMAGES, INCLUDING BUT NOT LIMITED TO ANY LOST PROFITS, LOST SAVINGS, OR OTHER INCIDENTAL DAMAGES, ARISING OUT OF THE USE OR INABILITY TO USE, OR THE DELIVERY OR FAILURE TO DELIVER, ANY OF THE MATERIALS, OR ANY BREACH OF ANY OBLIGATION UNDER THIS AGREEMENT, EVEN IF QTI HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES. THE FOREGOING LIMITATION OF LIABILITY SHALL REMAIN IN FULL FORCE AND EFFECT REGARDLESS OF WHETHER YOUR REMEDIES HEREUNDER ARE DETERMINED TO HAVE FAILED OF THEIR ESSENTIAL PURPOSE. THE ENTIRE LIABILITY OF QTI, QTI'S AFFILIATES AND ITS LICENSORS, AND THE SOLE AND EXCLUSIVE REMEDY OF YOU, FOR ANY CLAIM OR CAUSE OF ACTION ARISING HEREUNDER (WHETHER IN CONTRACT, TORT, OR OTHERWISE) SHALL NOT EXCEED US\$10.

2. **COMPLIANCE WITH LAWS; APPLICABLE LAW.** You agree to comply with all applicable local, international and national laws and regulations and with U.S. Export Administration Regulations, as they apply to the subject matter of this Agreement. This Agreement is governed by the laws of the State of California, excluding California's choice of law rules.

3. **CONTRACTING PARTIES.** If the Materials are downloaded on any computer owned by a corporation or other legal entity, then this Agreement is formed by and between QTI and such entity. The individual accepting the terms of this Agreement represents and warrants to QTI that they have the authority to bind such entity to the terms and conditions of this Agreement.

4. **MISCELLANEOUS PROVISIONS.** This Agreement, together with all exhibits attached hereto, which are incorporated herein by this reference, constitutes the entire agreement between QTI and You and supersedes all prior negotiations, representations and agreements between the parties with respect to the subject matter hereof. No addition or modification of this Agreement shall be effective unless made in writing and signed by the respective representatives of QTI and You. The restrictions, limitations, exclusions and conditions set forth in this Agreement shall apply even if QTI or any of its affiliates becomes aware of or fails to act in a manner to address any violation or failure to comply therewith. You hereby acknowledge and agree that the restrictions, limitations, conditions and exclusions imposed in this Agreement on the rights granted in this Agreement are not a derogation of the benefits of such rights. You further acknowledges that, in the absence of such restrictions, limitations, conditions and exclusions, QTI would not have entered into this Agreement with You. Each party shall be responsible for and shall bear its own expenses in connection with this Agreement. If any of the provisions of this Agreement are determined to be invalid, illegal, or otherwise unenforceable, the remaining provisions shall remain in full force and effect. This Agreement is entered into solely in the English language, and if for any reason any other language version is prepared by any party, it shall be solely for convenience and the English version shall govern and control all aspects. If You are located in the province of Quebec, Canada, the following applies: The Parties hereby confirm they have requested this Agreement and all related documents be prepared in English.